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Student note to the lecture

Group theory in physics

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Motto, topics and literature

1 Motto

Numbers measure quantities, groups measure symmetries.

In physics: spatio-temporal (visible geometrical) symmetries and symmetries with respect to inner (abstract) spaces (isospin, color, flavor,...).

2 Topics

- finite, discrete groups
- continuous Lie-groups
- representations of groups and Lie-algebras

3 Literature

- *Groups and symmetries*, M. A. Armstrong, 1997, (Springer)
- *Lie Groups, Lie Algebras and Representations*, B. C. Hall, 2004, (Springer)
- *An Elementary Introduction to Groups and Representations*, B. C. Hall, 2000 (arXiv:math-ph/0005032v1)
- *Elementare Zahlentheorie*, R. Remmert and P. Ullrich, 2008, (Birkhäuser)
- *Gravitation*, C. W. Misner, K. S. Thorne and J. A. Wheeler, 1973, (Macmillan Education)

Chapter 1

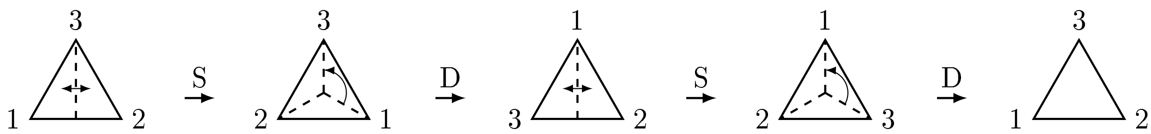
Finite groups

1.1 Definition and basic notations

Example: What are the symmetries of an equilateral triangle?

- Rotation D about $120°$: D, $D \cdot D = D^2$, $D^3 = id$
- Mirror reflection S with respect to an angle bisector[1] S, $S \cdot S = S^2 = id$

Figure 1.1: Rotation and mirror reflection



$$SDSD = id = DSDS$$

The symmetries of the equilateral triangle are *generated* by D and S. The following relations hold:

$$D^3 = S^2 = (DS)^2 = id\,.$$

Enumeration of elements: $id, D, D^2, S, DS, SD = D^2S$. These correspond to the six permutations of the three cornerpoints.

[1] dt. Winkelhalbierende

Dieder-group: D_3

Definition: An (abstract) group G is a non-empty set G in which to every ordered pair $(a, b) \in G \times G$ an element $a \circ b \in G$ is assigned such that the following axioms hold:

i) $a \circ (b \circ c) = (a \circ b) \circ c$ (associativity, brackets in products of $a, b, c \in G$ are not required)

ii) There exists a *neutral element* $e \in G$ such that $a \circ e = e \circ a = a$ for all $a \in G$

iii) For each element $a \in G$ there exists an *inverse element* $a' \in G$ such that $a \circ a' = a' \circ a = e$

In the following the notation "$\circ$" will be suppressed.

Corollary: The neutral element and the respective inverse element are *unique*.

Proof:

Let $e' \in G$ be another neutral element, then:

$$e' = ee' = e$$

Let $a'' \in G$ be another inverse element to $a \in G$, then:

$$a'' = ea'' = (a'a)a'' = a'(aa'') = a'e = a'$$

The unique inverse element of a is denoted by a^{-1}. One has

$$(ab)^{-1} = b^{-1}a^{-1},$$

since $b^{-1}a^{-1}ab = b^{-1}eb = b^{-1}b = e$. ✓

A group G is called *commutative* or *abelian*, if $ab = ba$ for all $a, b \in G$. The Dieder-group D_3 is *non-abelian*, since $SD \neq DS$.
The number of elements of a *finite* group G is called its *order* $|G|$, e.g. $|D_3| = 6$.

Examples of groups:

- The integer numbers $\mathbb{Z} = \{0, \pm 1, \pm 2, \dots\}$ form an abelian group with respect to the addition. $0 \in \mathbb{Z}$ is the neutral element and $-a \in \mathbb{Z}$ is the inverse element of $a \in \mathbb{Z}$.

- The set of numbers $\{0, 1, 2, \ldots, n-1\}$ forms a group with respect to the *addition modulo n*
$$a +_n b = \begin{cases} a+b, & 0 \le a+b < n \\ a+b-n, & a+b \ge n \end{cases}$$
0 is the neutral element and $n-a$ is the inverse element to $a \neq 0$. This abelian group is called $\mathbb{Z}_n$ with $|\mathbb{Z}_n| = n$, e.g. in $\mathbb{Z}_7$: $5+3=1$ and $-3=4$.
- Symmetrygroup of the regular n-gon: Let r be the rotations about $2\pi/n$ and let s be the mirror reflection with respect to the axis through one cornerpoint and the center. The following relations hold: $r^n = e$, $s^2 = e$ and $(rs)^2 = e$, since rs itself is a reflection.
The $2n$ different elements: $e, r, r^2, \ldots, r^{n-1}, s, rs, r^2s, \ldots, r^{n-1}s$ form the Dieder-group D_n which is non-abelian for $n \ge 3$. The following multiplication rules hold:
$$\left.\begin{aligned} r^a r^b &= r^k \\ r^a(r^b s) &= r^k s \end{aligned}\right\} k = a +_n b$$
From $rsrs = e$ it follows $sr = r^{-1}s$ and we get:
$$\left.\begin{aligned} r^a(sr^b) &= r^l s \\ (r^a s)(r^b s) &= r^l \end{aligned}\right\} l = a +_n (n-b)$$

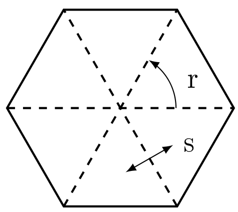


Figure 1.2: Illustration using a regular 6-gon.

Definition: A subset H of a group G that itself forms a group is called a *subgroup* $H \subset G$.

Criterion: For all $a, b \in H \subset G$, $ab^{-1} \in H$ must hold. Then the group axioms are all satisfied.
$$a = b \in H \Rightarrow e \in H$$
$$e, b \in H \Rightarrow b^{-1} \in H$$
$$a, b^{-1} \in H \Rightarrow a \cdot b \in H$$

Let G be a group and let $a \in G$ be an element. Then the set of powers $\{a^n,\ n \in \mathbb{Z}\}$ forms an abelian subgroup of G.

$$a^n a^m = a^{n+m}, \quad (a^n)^{-1} = a^{-n}$$

It is denoted by $\langle a \rangle$ and called the subgroup generated by a. If there exists an element $g \in G$ that generates the whole group G then G is called a *cyclic group*: $\langle g \rangle = G$.
$\mathbb{Z}_n$ is generated by 1. Thus $\mathbb{Z}_n$ is a cyclic group of order n.

Theorem of Lagrange: The order $|H|$ of a subgroup $H \subset G$ of a finite group G is a *divisor (factor)* of the group order $|G|$. $|G| : |H|$ is called the *index* of H in G.

Proof: Consider the following subsets of G: $gH = \{gh,\ h \in H\}$ where g runs over all $g \in G$. Each two of these subsets are either equal or disjoint. Let $g_1H \cap g_2H \neq \emptyset$ then $g_1h_1 = g_2h_2$, $g_1 = g_2 \underbrace{h_2h_1^{-1}}_{\in H}$, $g_1 = g_2H \Rightarrow g_1H \subset g_2H$.
Interchange g_1 and $g_2 \Rightarrow g_2H \subset g_1H$. Thus we find $g_1H = g_2H$.
We get a pairwise disjoint decomposition $G = g_1H \cup g_2H \cup \cdots \cup g_lH$
$\Rightarrow |G| = l|H|$. ✓

Corollary:

i) The order $|\langle g \rangle|$ of any element $g \in G$ is a divisor of the group order $|G|$. $H = \langle g \rangle = \{g^n,\ n \in \mathbb{Z}\}$ is a cyclic subgroup of G of order $|\langle g \rangle|$.

ii) For any element $x \in G$ one has $x^{|G|} = e$. Let $m_x = |\langle x \rangle|$ then $m_x \cdot k = |G|$ holds, $x^{m_x} = e$, $x^{m_x \cdot k} = (x^{m_x})^k = e^k = e = x^{|G|}$ for all $x \in G$.

iii) If the order of G is a *prime number* p then G is cyclic. Let $e \neq g \in G$ then $|\langle g \rangle| = p$ must hold $\Rightarrow \langle g \rangle = G$.

Example for a non-trivial cyclic group:
Let $p \in \mathcal{P}$ be a prime number. $\mathbb{Z}_p^* = \mathbb{Z}_p \backslash \{0\} = \{1, 2, \ldots, p-1\}$ forms a cyclic group of order $p-1$ with respect to the multiplication modulo p (C. F. Gauß).
Multiplication modulo p: $a \cdot_p b = a \cdot b - k \cdot p$ with k chosen such that $1 \leq ab - kp \leq p - 1$.
$1 \in \mathbb{Z}_p^*$ is the neutral element. If a and p are coprime[2], hence $\mathrm{hcf}(a,p) = 1$,[3] then there exist numbers b and k such that $ab + kp = 1$. This is obtained through the Euclidian division-algorithm applied to a, p).
Now b mod p is the multiplicative inverse of a, e.g.:

$$3^{-1} = 5 \text{ in } \mathbb{Z}_7^*, \text{ since } 3 \cdot 5 = 15 = 1 + 2 \cdot 7$$
$$5^{-1} = 9 \text{ in } \mathbb{Z}_{11}^*, \text{ since } 5 \cdot 9 = 45 = 1 + 4 \cdot 11$$

[2] dt. teilerfremd
[3] hcf = highest common factor, dt. ggT = größter gemeinsamer Teiler

Existence of a generating element in $\mathbb{Z}_p^*$ (called primitive root modulo p):
We know that $|\mathbb{Z}_p^*| = p-1$ and thus $a^{p-1} = \underbrace{a \cdot a \cdot \cdots \cdot a}_{p-1 \text{ times}} = 1$ for all $a \in \mathbb{Z}_p^*$ (Fermat's little theorem).
Assume there would be no generating element then we would have $a^m = 1$ for all $a \in \mathbb{Z}_p^*$ with $m < p-1$ being a divisor of $p-1$. Now $\mathbb{Z}_p$ is a *field*[4] with p elements. The polynomial $x^m - 1$ would have $p-1 > m$ roots.

But: In a field K a polynomial of degree m has at most m roots.

Proof by induction over m:
$f(X) = a_0 + a_1 X + \cdots + a_m X^m$ with $a_m \neq 0$. Let $c \in K$ be a root, then $f(X) = f(X) - f(c) = \sum_{\nu=1}^{m} a_\nu (X^\nu - c^\nu) = (X-c) \cdot g(x)$ with the degree of $g(x)$ being $m-1$. Note that $X^\nu - c^\nu$ is divisible by $X - c$ for all $\nu \in \mathbb{N}$.
Now we have $f(b) = (b-c) \cdot g(b)$ for all $b \in K$. $f(b) = 0$ can only hold for $b \neq c$ if $g(b) = 0$ (absence of zero-divisors is a field).
Further roots of $f(X)$ are roots of $g(X)$. ✓

Conclusion: Such $m < p-1$ can not exist. There exists $a \in \mathbb{Z}_p^*$ with $a^q = 1$ only for $q = p-1$.

$\mathbb{Z}_p^*$ is a cyclic group of order $p-1$

Comment: In general there exist *several* such primitive roots modulo p: There

Table 1.1: Smallest primitive root x_0 modulo p

p	2	3	5	7	11	13	17	19	23	29	31	37	41
x_0	1	2	2	3	2	2	3	2	5	2	3	2	6

are precisely $\varphi(p-1)$ primitive roots modulo p with $\varphi(p-1) =$ number of integers coprime to $p-1$ in $\{1, 2, \ldots, p-2\}$. E.g.: $\varphi(6) = 2$, $\varphi(12) = 4$, $\varphi(18) = 6$, etc. $\varphi(n)$ is called the *Euler totient function*.

1.2 Product groups

The *direct* product $G \times H$ of two groups G and H is constructed as follows: Its elements are ordered pairs (g, h) with $g \in G$ and $h \in H$. The composition is done componentwise, namely: $(g, h) \cdot (g', h') = (gg', hh')$.

[4] dt. Körper

(e, e) is the neutral element and (g^{-1}, h^{-1}) is the inverse element to (g, h).
The same construction applies to the direct product of finitely many groups $G_1 \times G_2 \times \cdots \times G_n$.

Examples:

i) $\mathbb{Z}_2 \times Z_3$ has six elements: $(0,0)$, $(1,0)$, $(0,1)$, $(1,1)$, $(0,2)$ and $(1,2)$. For each of these elements $(0,0)$, the sixth power is equal to the neutral element.
$\mathbb{Z}_2 \times Z_3$ is generated by $(1,1)$. It is an element of order six.
$2 \cdot (1,1) = (1,1) + (1,1) = (0,2)$, $3 \cdot (1,1) = (1,0)$, $4 \cdot (1,1) = (0,1)$, $5 \cdot (1,1) = (1,2)$ and $6 \cdot (1,1) = (0,0)$. We therefore have the group isomorphism

$$\mathbb{Z}_2 \times \mathbb{Z}_3 = \mathbb{Z}_6$$

ii) Klein's four-group $\mathbb{Z}_2 \times \mathbb{Z}_2$ consists of the elements $(0,0)$, $(0,1)$, $(1,0)$ and $(1,1)$. Each is of order two. $\mathbb{Z}_2 \times \mathbb{Z}_2 \neq \mathbb{Z}_4$ is not cyclic.

Theorem: $\mathbb{Z}_n \times \mathbb{Z}_m$ is cyclic if and only if n and m are coprime, hence $\mathrm{hfc}(n, m) = 1$.

Proof: Let k be the order of $(1,1)$ in $\mathbb{Z}_n \times \mathbb{Z}_m$, i.e. $k \bmod n = 0 = k \bmod m$. Therefore n divides k and m divides k.
If n and m are coprime, then the product nm must be a factor of $k \Rightarrow k = nm$. In this case $(1,1)$ generates the group $\mathbb{Z}_n \times \mathbb{Z}_m$ and we have a *cyclic* group.
Let $d > 1$ be the h.c.f. of n and m. Then we have integers $n' = n/d$ and $m' = m/d$. For each element $(x, y) \in \mathbb{Z}_n \times \mathbb{Z}_m$ one has $m'dn'(x, y) = (m' \underbrace{dn'}_{n} x \bmod n, \underbrace{m'd}_{m} n'y \bmod m) = (0,0)$, such that the order of (x, y) is at most $n'dm' = mn/d < mn$. Therefore $\mathbb{Z}_n \times \mathbb{Z}_m$ does not contain an element of order nm and thus it cannot be cyclic. ✓

Finitely generated abelian groups:

These commutative groups are generated by *finitely* many elements and can be *fully classified*.

Main theorem: Each finitely generated abelian group is isomorphic to a direct product of cyclic groups

$$\mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2} \times \cdots \times \mathbb{Z}_{m_k} \times \mathbb{Z}^s\,,$$

where m_i is a factor of m_{i+1} for $1 \leq i \leq k-1$, hence m_1 divides m_2, m_2 divides m_3,…,m_{k-1} divides m_k. The integer $s \in \mathbb{N}$ is called the *rank* of the abelian group and the integers $m_1, m_2, \ldots, m_k$ are the so-called *torsion coefficients.*

Corollaries: Each finite abelian group is isomorphic to the direct product of cyclic groups $\mathbb{Z}_{m_1} \times \cdots \times \mathbb{Z}_{m_k}$. m_1 divides m_2, m_2 divides m_3,..., m_{k-1} divides m_k. The rank must be zero, hence $s = 0$.
A finitely generated abelian group *without* an element $\neq 0$ of finite order is isomorphic to the direct product of *finitely* many copies of $\mathbb{Z}$,

$$G = \underbrace{\mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z}}_{s \text{ times}} = \mathbb{Z}^s\,, \quad \text{free abelian group of rank } s.$$

Examples:

i) $\mathbb{Z}_6 \times \mathbb{Z}_{10}$ does not seem to fit in this scheme as 6 does not divide 10. But one has $\mathbb{Z}_6 \times \mathbb{Z}_{10} = \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_{10} = \mathbb{Z}_2 \times \mathbb{Z}_{30}$ and 2 divides 30. We therefore have the torsion coefficients 2 and 30.

ii) Abelian groups of order 12 are isomorphic to $\mathbb{Z}_{12}$ and to $\mathbb{Z}_2 \times \mathbb{Z}_6$.

Proof of the main theorem: Let $x_1, \ldots, x_r$ be a minimal set of generating elements ($r-1$ elements no longer generate G). Since the group is abelian each element $g \in G$ can be written as

$$g = x_1^{n_1} \cdot x_2^{n_2} \cdots x_r^{n_r}\,, \quad \text{with } n_i \in \mathbb{Z} \text{ (integers)}$$

An expression of the form

$$e = x_1^{n_1} \cdot x_2^{n_2} \cdots x_r^{n_r}\,, \quad \text{with } n_1 \in \mathbb{Z} \backslash \{\pm 1\} \tag{$\star$}$$

is called a *relation* among the generators.
If there exists only the *trivial* relation ($\star$) with $n_i = 0$, then the expression $g = x_1^{n_1} \cdot x_2^{n_2} \cdots x_r^{n_r}$ is unique and we have the isomorphism

$$g \to (n_1, n_2, \ldots, n_r) \quad \Rightarrow G = \mathbb{Z}^r\,.$$

Now let there be non-trivial relations ($\star$). Let among all possible relations between minimal sets of generators be *one* with *smallest* exponent m_1. Without loss of generality, let it correspond to the generator x_1

$$e = x_1^{m_1} \cdot x_2^{n_2} \cdots x_r^{n_r}\,. \tag{$\star\star$}$$

We claim that m_1 divides n_2. Otherwise we would have $n_2 = qm_1 + u$ with $0 < u < m_1$. This implies

$$e = x_1^{m_1} \cdot x_2^{qm_1+u} \cdots x_r^{n_r} = (x_1 x_2^q)^{m_1} \cdot x_2^u \cdots x_r^{n_r} \tag{$\star\star\star$}$$

and $\{x_1 x_2^q, x_2, \dots, x_r\}$ is *also* a minimal set of generators.
Every $g \in G$ can be written as

$$g = x_1^{n_1} \cdot x_2^{n_2} \cdots x_r^{n_r} = (x_1 x_2^q)^{n_1} \cdot x_2^{n_2 - qn_1} \cdots x_r^{n_r} .$$

The above relation $(\star\star\star)$ contradicts the minimality of m_1. Therefore $u = 0 \Rightarrow m_1$ divides n_2. For the same reason m_1 divides $n_3, n_4, \dots, n_r$ with $n_i = m_1 q_i$ for $3 \leq i \leq r$.
Now we choose new generators $z_1, x_2, x_3, \dots, x_r$ with $z_1 = x_1 \cdot x_2^q \cdot x_3^{q_3} \cdots x_r^{q_r}$. Relation $(\star\star)$ turns into $e = z_1^{m_1}$.
$G = \mathbb{Z}_{m_1} \times G_1$ where G_1 is generated by $x_2, \dots, x_r$. $\mathbb{Z}_{m_1}$ is isomorphic to the subgroup generated by z_1 and thus $G = \langle z_1 \rangle \cdot G_1$. The previous procedure applied to G_1 yields $G_1 = \mathbb{Z}^{r-1}$ or $G_1 = \mathbb{Z}_{m_2} \times G_2$.
In the second case the exponent m_2 appears in the relation $e = y_2^{m_2} \cdot y_3^{n'_3} \cdots y_r^{n'_r}$ between the minimal set of generators of G_1.
But as $z_1, y_2, \dots, y_r$ form a minimal set of generators of G, the following relation holds

$$e = z_1^{m_1} \cdot y_2^{m_2} \cdot y_3^{n'_3} \cdots y_r^{n'_r}$$

and m_1 divides m_2.
This process of decomposition will eventually terminate, since we reduce the number of generators by one in each step.

Result: $G = \mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2} \times \cdots \times \mathbb{Z}_{m_k} \times \mathbb{Z}^s$ and m_1 divides m_2, m_2 divides m_3, …, m_{k-1} divides m_k. ✓

In the case of a finite abelian group we have $s = 0$. $G = \mathbb{Z}_{m_1} \times \cdots \times \mathbb{Z}_{m_k}$ and $|G| = m_1 \cdots m_k$. For each element $g \in G$ we get $g^{m_k} = e$, where m_k is the least common multiple[5] of the order of all group elements. m_k is the *exponent* $\epsilon(G)$ of G. It divides $|G|$. There exists an element $x \in G$ of order $\epsilon(G)$ namely $(0, 0, \dots, 0, 1)$. For *non-abelian* groups the latter statement is *wrong*.
E.g. D_3: order$(D) = 3$, order$(s) = 2$, $\epsilon(D_3) = 6$ but D_3 does not contain an element of order 6.

Remark: $G = \mathbb{Z}_{m_1} \times \cdots \times \mathbb{Z}_{m_k}$ with m_1 divides m_2,…is the *coarsest*[6] decomposition of the finite abelian group G. The primary decomposition $m_j = p_1^{\nu_{1j}} \cdots p_t^{\nu_{tj}}$ (coprime factors) gives $\mathbb{Z}_{m_j} = \mathbb{Z}_{p_1^{\nu_{1j}}} \times \cdots \times \mathbb{Z}_{p_t^{\nu_{tj}}}$ and leads to the *finest* decomposition of G.

[5]dt. kleinstes gemeinsames Vielfaches
[6]dt. gröbste

Little problem: Determine all abelian groups G of order $|G| = 360$:

$$\begin{array}{lll} G_1 = \mathbb{Z}_{360}\,, & G_2 = \mathbb{Z}_2 \times \mathbb{Z}_{180}\,, & G_3 = \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_{90} \\ G_4 = \mathbb{Z}_2 \times \mathbb{Z}_6 \times \mathbb{Z}_{30}\,, & G_5 = \mathbb{Z}_3 \times \mathbb{Z}_{120}\,, & G_6 = \mathbb{Z}_6 \times \mathbb{Z}_{60} \end{array}$$

The primary decomposition of $360 = 2^3 \cdot 3^2 \cdot 5$ leads to the non-isomorphic abelian groups:

$$\begin{aligned} \mathbb{Z}_8 \times \mathbb{Z}_9 \times \mathbb{Z}_5 = \mathbb{Z}_{360} &= G_1 \\ \mathbb{Z}_8 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5 = \mathbb{Z}_3 \times \mathbb{Z}_{120} &= G_5 \\ \mathbb{Z}_2 \times \mathbb{Z}_4 \times \mathbb{Z}_9 \times \mathbb{Z}_5 = \mathbb{Z}_2 \times \mathbb{Z}_{180} &= G_2 \\ \mathbb{Z}_2 \times \mathbb{Z}_4 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5 = \mathbb{Z}_6 \times \mathbb{Z}_{60} &= G_6 \\ \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_9 \times \mathbb{Z}_5 = \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_{90} &= G_3 \\ \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5 = \mathbb{Z}_2 \times \mathbb{Z}_6 \times \mathbb{Z}_{30} &= G_4 \end{aligned}$$

Explicit calculations with finitely generated abelian groups:
Now we write them additively. Let $x_1, \ldots, x_n$ be the n generators subject to the m relations:

$$\begin{array}{cccc} a_{11}x_1 + & \ldots & +a_{1n}x_n & = 0 \\ a_{21}x_1 + & \ldots & +a_{2n}x_n & = 0 \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1}x_1 + & \ldots & +a_{mn}x_n & = 0 \end{array} \quad , \quad a_{ij} \in \mathbb{Z}$$

Example: Three generators $x, y, z \in G$ and two relations:

$$\left.\begin{array}{rr} R_1: & 3x + 5y - 3z = 0 \\ R_2: & 4x + 2y = 0 \end{array}\right\} \quad \text{relations}$$

$$\begin{array}{rl} R_3 = R_2 - R_1: & x - 3y + 3z = 0 \\ R_4 = R_1 - 3R_3: & 14y - 12z = 0 = 2(y + 6(y - z)) \end{array}$$

Define new generators:

$$\begin{array}{lll} x' = x - 3y + 3z, & y' = y + 6(y - z), & z' = y - z \\ x = x' + 3z', & y = y' - 6z', & z = y' - 7z' \end{array}$$

such that the relations read:

$$\left.\begin{array}{rl} R_3': & x' = 0 \\ R_4': & 2y' = 0 \end{array}\right\} \Rightarrow G = \mathbb{Z}_2 \times \mathbb{Z}\,.$$

The canonical form of a finitely generated abelian group can be obtained by so called *elementary row and column transformations* of the coefficient matrix a_{ij}:

i) Interchange two rows or two columns

ii) Multiply a row or column by -1

iii) Add an *integer multiple* of a row/column to another row/column

Row transformations reversibly change the relations. Column transformations reversibly change the generators:

$$\begin{pmatrix} 3 & 5 & -3 \\ 4 & 2 & 0 \end{pmatrix} \xrightarrow{R_2-R_1} \begin{pmatrix} 3 & 5 & -3 \\ 1 & -3 & 3 \end{pmatrix} \xrightarrow{R_1\leftrightarrow R_2} \begin{pmatrix} 1 & -3 & 3 \\ 3 & 5 & -3 \end{pmatrix}$$

$$\xrightarrow{R_2-3R_1} \begin{pmatrix} 1 & -3 & 3 \\ 0 & 14 & -12 \end{pmatrix} \xrightarrow{C_2+3C_1} \begin{pmatrix} 1 & 0 & 3 \\ 0 & 14 & -12 \end{pmatrix}$$

$$\xrightarrow{C_3-3C_1} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 14 & -12 \end{pmatrix} \xrightarrow{C_2+C_3} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & -12 \end{pmatrix}$$

$$\xrightarrow{C_3+6C_2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix}$$

Theorem: A $m \times n$-matrix A with integer entries can be converted to a diagonal-matrix D by elementary row and column transformations such that $d_{ii} \geq 0$ where d_{11} divides d_{22}, d_{22} divides d_{33},...

Exercise: For five generators and four relations represented by the matrix

$$A = \begin{pmatrix} 3 & -4 & 5 & 3 & 7 \\ 3 & 2 & -1 & -3 & 1 \\ 8 & 2 & -2 & -8 & 2 \\ 11 & -8 & 9 & -5 & 9 \end{pmatrix}$$

show that $G = \mathbb{Z}_2 \times \mathbb{Z}_6 \times \mathbb{Z}_6 \times \mathbb{Z}$.

1.3 Homomorphisms

Definition: Let G and G' be two groups. A function $\varphi : G \to G'$ is called a *group-homomorphism* if it maps the group multiplication in G onto the one in G', i.e.

$$\varphi(x \cdot y) = \varphi(x) \cdot \varphi(y)\,, \quad \text{for all } x, y \in G$$

The *kernel* K of the group-homomorphism φ, defined as $K = \{x \in G | \varphi(x) = e\}$, forms an *invariant* subgroup (called *normal subgroup*[7]) of G:

$$xKx^{-1} = K, \quad \text{for all } x \in G\,.$$

In other words: $K \subset G$ is a normal subgroup if and only if left and right cosets[8] agree, i.e.

$$xK = Kx, \quad \text{for all } x \in G\,.$$

Confirmation: $x, y \in K = \text{kernel}\,(\varphi)$, $\varphi(x) = e = \varphi(y)$, $\varphi\big(xy^{-1}\big) = \varphi(x)\,\varphi\big(y^{-1}\big) = \varphi(x)\,\varphi(y)^{-1} = ee^{-1} = e \Rightarrow xy^{-1} \in K \Rightarrow K$ is a subgroup.
Now take an arbitrary $x \in G$ and $k \in K$: $\varphi\big(xkx^{-1}\big) = \varphi(x)\underbrace{\varphi(k)}_{=e}\varphi(x)^{-1} = \varphi(x)\,\varphi(x)^{-1} = e.$ $\Rightarrow xkx^{-1} \in K \Rightarrow xKx^{-1} \subset K$. Interchanging $x \leftrightarrow x^{-1}$ $\Rightarrow x^{-1}Kx \subset K$ or $K \subset xKx^{-1}$.
$\Rightarrow xKx^{-1} = K$ ✓

Definition: A *group-isomorphism* is an invertible group-homomorphism $\psi : G \to G'$, $\psi^{-1} : G' \to G$. The kernel of an isomorphism is trivial: $\text{kernel}\,(\psi) = \{e\}$.

Isomorphism-Theorem: Let $\varphi : G \to G'$ be a group-homomorphism with kernel K (normal subgroup in G). The mapping $\psi : xK \to \varphi(x)$ provides an isomorphism between the *quotient-group* G/K and the image of $\text{image}\,(\varphi)$ in G'

$$G/K = \text{image}\,(\varphi)$$

$\text{image}\,(\varphi)$ is a subgroup of G', since $\varphi(x)\,\varphi(y)^{-1} = \varphi\big(xy^{-1}\big) \in \text{image}\,(\varphi)$.

Quotient-group G/K: K is normal subgroup of G:

- Composition of left cosets: $(xK)(yK) = xKyK = xyKK = xyK$. Take representatives $k, k', k'', k''' \in K$: $xkyk' = xyk''k' = xyk'''$.
- K is the neutral element in G/K: $xKK = xK$, $KxK = xKK = xK$.
- $(xK)^{-1} = K^{-1}x^{-1} = x^{-1}K$ is the inverse element to xK in G/K: $x^{-1}KxK = x^{-1}xKK = eK = K$, $xKx^{-1}K = xx^{-1}KK = eK = K$.

[7] dt. Normalteiler
[8] dt. Nebenklassen

Back to the isomorphism-theorem: $\psi(xK) = \varphi(x)$, $\psi(xKyK) = \psi(xyK) = \varphi(xy) = \varphi(x)\,\varphi(y) = \psi(xK)\,\psi(yK)$.
$\varphi(x) = \varphi(y) \Rightarrow \varphi\big(xy^{-1}\big) = e \Rightarrow xy^{-1} \in K \Rightarrow x \in yK \Rightarrow xK = yK$.
Therefore $\psi : G/K \to G'$ is a well-defined group-homomorphism.

Examples:

i) $\mathbb{Z} \to \mathbb{Z}_n$ (additive abelian groups), $x \mapsto x \bmod n$. The kernel K consists of all integer multiples of n: $K = n\mathbb{Z}$.

$$\mathbb{Z}_n = \mathbb{Z}/n\mathbb{Z}$$

ii) $\mathbb{R} \to S^1$ (S^1 is the unit-circle in the complex plane $\mathbb{C}$), $x \mapsto e^{2\pi \mathrm{i} x}$. The kernel K consists of all the integers $\mathbb{Z}$: $K = \mathbb{Z}$.

$$S^1 = \mathbb{R}/\mathbb{Z}$$

Geometrically this corresponds to rolling up the line to a circle.

iii) $S^1 \to S^1$, $z \mapsto z^2$. The kernel K consists of $\{\pm 1\}$: $K = \{\pm 1\}$.

$$S^1 = S^1/\{\pm 1\}$$

iv) $S^1 \to S^1$, $z \mapsto z^n$. The kernel K consists of the n-th roots of unity. Geometrically this corresponds to winding up the circle n times.

Perspective: A group G is called *simple*, if it has no non-trivial normal subgroups N.

$$\{e\} \subset N \subset G \quad \Rightarrow N = \{e\} \text{ or } \{G\}$$

A major achievement of mathematics in the 20-th century: The finite, simple groups can be classified completely. This is the list:

- $\mathbb{Z}_p$: abelian cyclic groups of prime order
- A_n ($n \geq 5$): even permutations of $n \geq 5$ objects
- Lie-type: insert elements from a finite field into matrices

- 26 *sporadic* groups:
  - the largest of them is called "monster group" M with $|M| = 2^{46} \cdot 3^{20} \cdot 5^9 \cdot 7^6 \cdot 11^2 \cdot 13^3 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 31 \cdot 41 \cdot 47 \cdot 59 \cdot 71 \sim 8,08 \cdot 10^{53}$
  - the smallest sporadic group is called M_{11} with $|M_{11}| = 7920 = 8 \cdot 9 \cdot 10 \cdot 11$. There exists a series of these Mathieu groups: $M_{11}, M_{12}, M_{22}, M_{23}, M_{24}$ (discovered in 1861)
  - Janko groups J_1, J_2, J_3, J_4 (1964-1975)
  - Convay group Co_1, Co_2, Co_3 (1969)
  - Fischer groups F_{22}, F_{23}, F_{24} (1976)
  - Baby monster group (friendly giant) B with $|B| \sim 4,15 \cdot 10^{33}$

1.4 The permutation group

The group of permutations of n objects is called the *symmetric group* S_n. The order of S_n is $|S_n| = n! = n \cdot (n-1) \cdots 1$.
S_3 has the following 6 elements:

$$\underset{id}{\begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}} \quad \underset{1 \leftrightarrow 2}{\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}} \quad \underset{1 \leftrightarrow 3}{\begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}} \quad \underset{2 \leftrightarrow 3}{\begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}} \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \quad \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

For $\alpha, \beta \in S_n$ the product $\alpha \cdot \beta \in S_n$ means: Apply first β, then α.

Example:

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$
$$\begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

S_n is *non-abelian* for $n \geq 3$.
For larger n the notation $\begin{pmatrix} 1 & \dots & n \\ \alpha(1) & \dots & \alpha(n) \end{pmatrix}$ becomes too cumbersome[9].

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 5 & 4 & 3 & 6 & 1 & 2 \end{pmatrix} \in S_6$$

The same information is given by the *cycle-notation*: $\alpha = (15)(246)$:
Within a bracket a number is mapped to its (right) successor, this to its (right)

[9] dt. umständlich

successor and so on. The last number in the bracket is mapped to the first: $1 \to 5 \to 1$, $2 \to 4 \to 6 \to 2$.
The number 3 which is not transformed by α does not need to be mentioned.

Prescription: Open a bracket, write the smallest number that is transformed by α, then its image, then its image, and so on. Close the bracket once the starting number is reached. Open a new bracket and start with the smallest number not yet mentioned which is transformed by α. Repeat the process until you are done.

This shows: Each permutation $\alpha \in S_n$ can be written as a product of *disjoint cyclic permutations*:

$$(\dots)\underbrace{(\alpha_1\alpha_2\dots\alpha_k)}_{\text{cycle of length } k\geq 2}(\dots)$$

The decomposition is unique up to (irrelevant) ordering of the factors. Two-cycles are called *transpositions*.

Statement: The transpositions in S_n generate S_n.
Each $\alpha \in S_n$ is a product of cyclic permutations. A cyclic permutation can be written as a product of transpositions, because:

$$(\alpha_1\alpha_2\dots\alpha_k) = \underbrace{(\alpha_1\alpha_k)\dots(\alpha_1\alpha_3)(\alpha_1\alpha_2)}_{\begin{pmatrix}\alpha_1 & \alpha_2 & \dots & \alpha_{k-1} & \alpha_k\\ \alpha_2 & \alpha_3 & \dots & \alpha_k & \alpha_1\end{pmatrix}}$$

These transpositions are no longer disjoint. Therefore this decomposition is *not unique*.

Statement: The $n-1$ transpositions $(12), (13), \dots, (1n)$ generate S_n. Use the identity

$$(ab) = (1a)(1b)(1a)$$

and the previous statement.

Statement: The $n-1$ transpositions $(12), (23), \dots, ((n-1)n)$ generate S_n. Use the identity

$$(1k) = \underbrace{((k-1)k)\cdots(34)(32)(12)(23)(34)\cdots((k-1)k)}_{\begin{pmatrix}1 & 2 & 3 & \dots & k-1 & k\\ k & 2 & 3 & \dots & k-1 & 1\end{pmatrix}}$$

and the previous statement.

Statement: The transposition (12) and the long n-cycle $(12\ldots(n-1)n)$ generate S_n together.

We have to write the transpositions $(k(k+1))$ as a "word" in the "letters" (12) and $(12\ldots n)$:

$$(23) = (12\ldots n)(12)\underbrace{(12\ldots n)^{-1}}_{(n(n-1)\ldots 1)}$$

$$1 \to n \to 1, \quad 2 \to 1 \to 2 \to 3, \quad 3 \to 2 \to 2, \quad k \to k-1 \to k \text{ for } k \geq 4$$

$$\begin{aligned}(34) &= (12\ldots n)(23)(12\ldots n)^{-1}\\ &= (12\ldots n)^2(12)(12\ldots n)^{-2}\end{aligned}$$

Iteration gives us:

$$(k(k+1)) = (12\ldots n)^{k-1}(12)(12\ldots n)^{1-k} \checkmark$$

Each permutation $\alpha \in S_n$ can be written as a product of transpositions in different ways. The number of transpositions is however *always* either *even* or *odd*.
Consider the following polynomial with $\frac{n}{2}(n-1)$ factors:

$$\begin{aligned}P(x_1, x_2, \ldots, x_n) &= (x_1 - x_2)(x_1 - x_3)\cdots(x_1 - x_n)\cdot(x_2 - x_3)\cdots(x_{n-1} - x_n) =\\ &= \prod_{1\leq i<j\leq n}(x_i - x_j)\end{aligned}$$

and let $\alpha \in S_n$ operate on it:

$$\alpha P = \prod_{1\leq i<j\leq n}(x_{\alpha(i)} - x_{\alpha(j)}) = \pm P = sign(\alpha)\, P$$

Factors get interchanged and sometimes signs get introduced. The value ± 1 is the *signum* of the permutation α:

$$sign(\alpha) = \begin{cases} +1, & \text{even permutations}\\ -1, & \text{odd permutations}\end{cases}$$

One clearly has $sign(\alpha\beta) = sign(\alpha)\cdot sign(\beta)$. The sign-function is a group-homomorphism.
The transposition (12) is *odd* and since $(1a) = (2a)(12)(2a)$ one finds that $(1a)$ is

odd as well. Together with $(ab) = (1a)(1b)(1a)$ one finds that *any* transposition is odd.

$$\begin{aligned} sign : S_n &\to \mathbb{Z}_2 \\ sign(\alpha) &= (-1)^t \end{aligned}$$

with t being the number of transpositions. Therefore t is either even or odd.
The cyclic permutation

$$(\alpha_1\alpha_2\ldots\alpha_k) = \underbrace{(\alpha_1\alpha_k)\ldots(\alpha_1\alpha_3)(\alpha_1\alpha_2)}_{k-1 \text{ factors}}$$

is *even* if and only if the cycle-length k is *odd*.

Theorem: The even permutations in S_n form a *normal subgroup* of order $\frac{n!}{2}$, called the *alternating group* A_n.

Proof: id is even, even $\cdot$ even^{-1} = even. $\underbrace{(12)}_{\tau}$ even = odd. Note that A_n is the kernel of the sign-homomorphism, $sign : S_n \to \mathbb{Z}_2$, and therefore A_n is a normal subgroup in S_n.

$$S_n = \underbrace{A_n \cup \tau A_n = A_n \cup A_n\tau}_{\text{left coset quals right coset}}$$

Any subgroup of index 2 is necessarily normal. For $n \geq 3$ the alternating group A_n is generated by *3-cycles*. Each 3-cycle is even.
Now write $\alpha \in A_n$ as an even-numbered product of transpositions $(1a)$. Collect successive pairs and combine them to $(1a)(1b) = (1ba)$. ✓

Example:

- $A_3 = \mathbb{Z}_3$ since 3 is a prime number.
- The 12 elements of A_4 are:

$$\begin{array}{cccc} id & (12)(34) & (13)(24) & (14)(23) \\ (123) & (124) & (134) & (234) \\ (132) & (142) & (143) & (243) \end{array}$$

where one has e.g. $(13)(24) = (13)(12)(14)(12) = (123)(124)$.
$\{id, (12)(34), (13)(24), (14)(23)\}$ by itself forms a normal subgroup of A_4 that is isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$, since each of its elements is of order two.
One finds a chain of nested normal subgroups in S_4:

$$S_4 \supset A_4 \supset \mathbb{Z}_2 \times \mathbb{Z}_2 \supset \mathbb{Z}_2 \supset \{0\}$$

with abelian factor groups

$$\begin{aligned} S_4/A_4 &= \mathbb{Z}_2 \to \sqrt{\ } \\ A_4/(\mathbb{Z}_2 \times \mathbb{Z}_2) &= \mathbb{Z}_3 \to \sqrt[3]{\ } \\ (\mathbb{Z}_2 \times \mathbb{Z}_2)/\mathbb{Z}_2 &= Z_2 \to \sqrt{\ } \end{aligned}$$

This group structure is intimately[10] related to *Ferarri's formula* for the roots of a quartic polynomial.
For $n \geq 5$ one instead has the short chain: $S_n \supset A_n \supset \{e\}$.

1.5 Group operations

Definition: An *operation/action* of a group G on a set X is a homomorphism from G to S_X, the bijective maps from $X \to X$.

$$\varphi : G \to S_X, \quad \varphi(g \cdot h) = \varphi(g) \cdot \varphi(h)\,, \quad \varphi(e) = id_X$$

We simplify the notation as follows:

$$\varphi(g)(x) := g(x) \in X$$

This implies $(gh)(x) = g(h(x))$, $e(x) = x$.

Terminology: For fixed $x \in X$, the set of all $g(x)$, when g runs over the whole group G, is called the *orbit* of x, denoted by $G(x)$.
The decomposition of X in orbits is *disjoint* (two orbits never cross each other).
Let $y \in G(x)$, that means there exists a $g \in G$ with $y = g(x) \Rightarrow G(y) \subset G(x)$.
Similarly $x = g^{-1}(y) \Rightarrow G(x) \subset G(y)$. It follows $G(x) = G(y)$.
Let $x \in X$ be a point. The *subgroup* of G, which leaves x fixed under the group operation is called the *stabilizer* of x, denoted by $G_x := \{g \in G | g(x) = x\}$.

Example: Let G be a group and take the set X to be the group G itself. G operates on G by *conjugation*:

[10] dt. aufs Engste

$$g(x) = gxg^{-1}$$

$$\begin{aligned} (gh)(x) &= ghxh^{-1}g^{-1} = g\big(hxh^{-1}\big) = g(h(x)) \\ e(x) &= exe^{-1} = x \end{aligned}$$

The corresponding orbits are the *conjugation-classes*. A normal subgroup $N \subset G$ consists of conjugation-classes.
The stabilizer of $x \in G$ is built up by all elements $g \in G$ commuting with x:

$$G_x = \{g \in G | xg = gx\}$$

In the case $G = S_n$ one has: Permutations in the same conjugation-class have the same cycle structure. Let $\theta \in S_n$

$$\theta \ldots (a_1 a_2 \ldots a_k) \ldots \theta^{-1} = \ldots (\theta(a_1)\,\theta(a_2) \ldots \theta(a_k)) \ldots$$

$$\theta(a_j) \overset{\theta^{-1}}{\to} a_j \to a_{j+1} \overset{\theta}{\to} \theta(a_{j+1})$$

The conjugation-classes of S_n are in one to one correspondence to the partitions $(n_1, n_2, \cdots, n_r)$, with $n_j \geq 1$, of n, satisfying:

$$n = n_1 + n_2 + \cdots + n_r$$

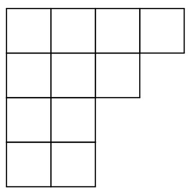
Examples:

- S_5 has 7 conjugation-classes:

$$\begin{array}{cccc} 5 = 1+1+1+1+1, & 5 = 1+1+1+2, & 5 = 1+1+3, & 5 = 1+4 \\ 5 = 1+2+2, & 5 = 2+3, & 5 = 5 & \end{array}$$

- S_{20} has 627 conjugation-classes.

This, in the end, leads to the *Young-tableaus*.



The *partition-function* $p(n)$ counts the number of partitions of n where $p(0) = 1$

$$\sum_{n=0}^{\infty} p(n)\, x^n = \prod_{n=1}^{\infty} (1 - x^n)^{-1}$$

By expanding the geometrical series $(1 - x^n)^{-1} = 1 + x^n + x^{2n} + x^{3n} + \ldots$, one sees that the prefactor of x^n in this product counts the number of ways to write

$$n = a_1 + 2a_2 + 3a_3 + \cdots$$

Also an asymptotic expression for the partition-function is known

$$p(n) \approx \frac{1}{4n\sqrt{3}} \exp\left\{\pi\sqrt{\frac{2n}{3}}\right\}$$

Now let us consider again a general operation of a group G on a set X. Points in the *same* orbit have *conjugated stabilizers*.
Let $y = g(x)$ then we show $gG_xg^{-1} = G_y$. Let $h \in G_x$, then

$$\begin{aligned} ghg^{-1}(y) &= ghg^{-1}g(x) = gh(x) = g(x) = y \\ &\Rightarrow G_y \supseteq gG_xg^{-1} \end{aligned}$$

Now interchange x and y

$$G_x \supseteq g^{-1}G_y g, \quad G_y \subseteq gG_xg^{-1}$$

Orbit-stabilizer-theorem: For any finite groups G the following holds

$$|G(x)| \cdot |G_x| = |G|$$

The *size* of any orbit is a *factor* of the *group order*.

Proof: Decompose G into disjoint cosets of G_x

$$\begin{array}{rcccc} & G_x\,, & g_1G_x\,, & \ldots\,, & g_nG_x \\ \text{group operation on } x & \updownarrow & \updownarrow & & \updownarrow \\ \text{orbit of } x & x\,, & g_1(x)\,, & \ldots\,, & g_n(x) \end{array}$$

Each coset generates *exactly one* point in the orbit $G(x)$.

$$g_i(x) = g_j(x) \Rightarrow g_i^{-1}g_j \in G_x \Rightarrow g_iG_x = g_jG_x$$

Theorem of Cauchy: Let $p \in P$ be a prime factor of the group order $|G|$ of the finite group G. Then there is an element $x \in G$ of order p.

Proof: Let X be the set of ordered p-tupels $(x_0, x_1, \ldots, x_{p-1})$ with $x_0 \cdots x_{p-1} = e$ and $x_j \in G$. Consider the natural *operation* of $\mathbb{Z}_p$ on X: $m \in \mathbb{Z}$ maps $(x_0, \ldots, x_{p-1})$ onto $(x_m, x_{m+1}, \cdots, x_{p-1}, x_0, \cdots, x_{m-1})$ shifting the index by m (count modulo p).

$$(x_m \cdots x_{p-1})(x_0 \cdots x_{m-1}) = e$$

From the previous theorem and $|\mathbb{Z}_p| = p$ we conclude that each orbit contains *either* 1 *or* p p-tupels. The total number of p-tupels in X is $|G|^{p-1}$ and is divisible by p.
The orbit of $(e, \cdots, e)$ consists of one point. *Not* all the other N orbits can consist of p points, since $Np + 1 = |G|^{p-1}$ would not be divisible by p. There must exist further orbits made up by one point. Such p-tupels are *invariant* under $\mathbb{Z}_p$-operations.
$\Rightarrow$ There exists $x_0 = x_1 = \cdots = x_{p-1}$ with $x_0^p = e$. ✓

Corollary from the orbit-stabilizer-theorem:
The number of elements in each *conjugation-class* of a finite group G is a *factor* of the group order $|G|$

$$|G| = 1 + c_1 + \cdots + c_r$$

where c_j divides $|G|$.

Example: Class equation of A_5

$$60 = 1 + 15 + 20 + 12 + 12$$

Theorem: Let $p \in P$ be a prime and let the order of the group G be *a power* of p, then G has a non-trivial *center* $Z(G)$

$$Z(G) = \{z \in G\,, gz = zg \quad \text{for all } g \in G\}$$

The center of a group consists of those group elements that commute with all group elements. It is an abelian, normal subgroup.

Proof: Consider the disjoint decomposition of G into conjugation-classes. Their size is 1 or a power of p.
By definition the center $Z(G)$ consists of the conjugation-classes with one element. If the center were trivial, then the group order $Np + 1 = |G|$ would be in contradiction to the assumption $|G| = p^k$. ✓

Theorem: A group G of order $|G| = p^2$ with $p \in P$ a prime number, is *either* cyclic or isomorphic to $\mathbb{Z}_p \times \mathbb{Z}_p$ (therefore abelian).

Proof: If G has an element of order p^2, then G is cyclic and $G = \mathbb{Z}_{p^2}$. Otherwise all elements of G, except e, have order p (theorem of Cauchy). Choose $e \neq x \in Z(G)$ and choose $y \notin \langle x \rangle$ outside from the generated set $\langle x \rangle$. The p^2 elements $x^i y^j$ with $1 \leq i, j \leq p$ are all different, since $\langle x \rangle$ and $\langle y \rangle$ share only e as a common element. Each element from $\langle x \rangle$ commutes with each element from $\langle y \rangle$, since $x \in Z(G)$.

$$G = \langle x \rangle \cdot \langle y \rangle\,, \quad \langle x \rangle \cap \langle y \rangle = e \quad \Rightarrow \quad G = \mathbb{Z}_p \times \mathbb{Z}_p$$

Remark: If $|G| = p^3$, there exist also non-abelian groups of this order, e.g. 3×3-matrices

$$M = \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}\,, \quad \text{with } a, b, c \in \mathbb{Z}_p$$

$$M_1 M_2 \neq M_2 M_1\,, \quad \text{since } M_1 M_2 - M_2 M_1 = \begin{pmatrix} 0 & 0 & a_1 c_2 - a_2 c_1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

The inverse is given by $M^{-1} = \begin{pmatrix} 1 & -a & ac - b \\ 0 & 1 & -c \\ 0 & 0 & 1 \end{pmatrix}$

Classification of the finite rotation groups in three dimensions:

A rotation is characterized by an axis (unit vector $\boldsymbol{e}$, eigenvector of the rotation matrix $g\boldsymbol{e} = \boldsymbol{e}$) and an angle $\theta = 2\pi/n$, $n \in \mathbb{N}$. The generated set $\langle g \rangle$ is clearly a sub-group of the finite rotation group.

Let G be a finite rotation group and X the set of all axes of rotation. G operates on X: $h \in G$ and $\boldsymbol{e} \in X$ with $g\boldsymbol{e} = \boldsymbol{e}$ for some $g \in G$, then it follows

$$(hgh^{-1})h\boldsymbol{e} = hg\boldsymbol{e} = h\boldsymbol{e}$$

$h\boldsymbol{e}$ is an eigenvector with eigenvalue 1 and thus is an axis of rotation.

X is decomposed into k orbits $B_1, \cdots, B_k$. If $\boldsymbol{e}$ is a n-fold axis, so is $h\boldsymbol{e}$:

$$g^n = id \quad \Rightarrow (hgh^{-1})^n = hg^n h^{-1} = id$$

We can assign natural numbers n_j to the orbits. B_j consists of n_j-fold axes. The following relation holds:

$$\sum_{j=1}^{k}\left(1-\frac{1}{n_j}\right)=2-\frac{2}{|G|}$$

Proof: Consider the set of pairs

$$P=\{(\boldsymbol{e},g)\,,\boldsymbol{e}\in X, id\neq g\in G \text{ such that } g\boldsymbol{e}=\boldsymbol{e}\}$$

For each $g\in G$ there are exactly two axes $\pm\boldsymbol{e}$, thus $|P|=2(|G|-1)$. On the other hand, for each $\boldsymbol{e}\in B_j$ there are n_j-1 group elements g with $g\boldsymbol{e}=\boldsymbol{e}$, thus

$$|P|=\sum_{j=1}^{k}|B_j|(n_j-1)$$

Now B_j is an orbit and the stabilizer of each element in B_j is isomorphic to $\mathbb{Z}_{n_j}$. This implies $|B_j|=|G|/n_j$. Then one gets

$$|P|=2(|G|-1)=\sum_{j=1}^{k}\frac{|G|}{n_j}(n_j-1)\Bigg|\cdot\frac{1}{|G|}$$

or finally

$$2-\frac{2}{|G|}=\sum_{j=1}^{k}\left(1-\frac{1}{n_j}\right)\ \checkmark$$

We have to find all possible integer solutions of this equation with $n_j\geq 2$, $|G|\geq 2$ and $n_j\leq |G|$.
The rational number $|2-\frac{2}{|G|}|$ lies in the interval $[1,2[$ and each summand $(1-\frac{1}{n_j})$ in the interval $[\frac{1}{2},1[$. This implies $k=2$ or $k=3$.

- $k=2$:

$$\frac{1}{n_1}+\frac{1}{n_2}=\frac{2}{|G|}\,,\quad n_1,n_2\leq |G|$$

Solution: $n_1=n_2=|G|$

$$G = \mathbb{Z}_{n_1}$$

This cyclic group describes rotations in a fixed plane.

- $k = 3$:
  After suitable relabeling one has $n_1 \leq n_2 \leq n_3$. If $n_1 \geq 3$, then each summand $(1-1/n_j) \geq 2/3$ and the sum would exceed 2 which is forbidden.

$$n_1 = 2$$

  If $n_2 \geq 4$ the sum has at least the value $1/2+3/4+3/4 = 2$, which is not possible.

$$n_2 = 2 \text{ or } n_2 = 3$$

  Consider the case $n_1 = n_2 = 2$, $n_3 \equiv n$.

$$2 - \frac{2}{|G|} = \frac{1}{2} + \frac{1}{2} + 1 - \frac{1}{n} \quad \Rightarrow |G| = 2n$$

  We identify the dieder group:

$$G = D_n$$

  Finally consider the case $n_1 = 2$, $n_2 = 3$, then $n_3 < 6$, since $1/2 + 2/3 + 5/6 = 2$. All remaining possibilities $n_3 \in \{3, 4, 5\}$ give integer values for $|G|$.

We can now list the symmetries of the *Platonic solids*:

k	n_1	n_2	n_3	$\|G\|$	group
2	n	n	$-$	n	$\mathbb{Z}_n$
3	2	2	n	$2n$	D_n
3	2	3	3	12	A_4 regular tetrahedron
3	2	3	4	24	S_4 cube/octahedron
3	2	3	5	60	A_5 icosahedron/dodecahedron

Including mirror reflections one gets

regular tetrahedron:	A_4,	S_4
cube/octahedron:	S_4,	$\mathbb{Z}_2 \times S_4$
icosahedron/dodecahedron:	A_5,	$\mathbb{Z}_2 \times A_5$

Chapter 2

Continuous groups

We now consider groups for which the group elements depend continuously (differentiablely) on a finite number of real parameters:

Lie-group

Definition: A Lie-group G is a differentiable manifold G that also is a group such that the group multiplication

$$\mu : G \times G \to G\,, \quad \mu(g,h) = g \cdot h$$

and the formation of the inverse elements

$$i : G \to G\,, \quad i(g) = g^{-1}$$

are differentiable maps.

This basically means

$$\left.\begin{array}{rl} g = f(\alpha_1, \cdots, \alpha_n)\;, & h = f(\beta_1, \cdots, \beta_n) \\ g \cdot h = f(\gamma_1, \cdots, \gamma_n)\;, & g^{-1} = f(\bar{\alpha}_1, \cdots, \bar{\alpha}_n) \end{array}\right\} \begin{array}{l} \text{prarameter representation} \\ \text{of group elements} \end{array}$$

The γ_j are differentiable functions of $\alpha_1, \cdots, \alpha_n, \beta_1 \cdots, \beta_n$. The $\bar{\alpha}_j$ are differentiable functions of $\alpha_1, \cdots, \alpha_n$.
In the following we restrict ourselves to matrix–Lie-groups (there are exceptions).

2.1 Linear and special linear groups

General linear group:

1) $\boldsymbol{Gl(n,\mathbb{R})}$: Group of the invertible $n \times n$-matrices with real-valued entries $a_{ij} \in \mathbb{R}$.
$A \in Gl(n,\mathbb{R}) \Leftrightarrow \det(A) \neq 0$, polynomial of degree n in a_{ij} that has to be different from zero. $\Rightarrow$ open subset of $\mathbb{R}^{n^2}$.

$$\dim\left(Gl(n,\mathbb{R})\right) = n^2$$

with n^2 real parameter.

2) $\boldsymbol{Gl(n,\mathbb{C})}$: Group of the invertible $n \times n$-matrices with complex-valued entries $a_{ij} \in \mathbb{C}$.

$$\dim\left(Gl(n,\mathbb{C})\right) = 2n^2$$

with $2n^2$ real parameter.

Special linear group:

1) $Sl(n,\mathbb{R}) = \{A \in Gl(n,\mathbb{R}), \det(A) = 1\}$ is a *closed subgroup* of $Gl(n,\mathbb{R})$:

$$\begin{aligned} \det(A) = 1 &= \det(B) \\ \det(A \cdot B) = \det(A) \cdot \det(B) &= 1 \cdot 1 = 1 \\ \det\left(A^{-1}\right) = \det(A)^{-1} &= 1 \end{aligned}$$

where closed means that the limit of a sequence in the subgroup lies in the subgroup (closure w.r.t. topology).

$$\dim\left(Sl(n,\mathbb{R})\right) = n^2 - 1$$

2) $Sl(n,\mathbb{C}) = \{A \in Gl(n,\mathbb{C}), \det(A) = 1\}$. The condition $\det(A) = 1$ eliminates one complex parameter (two real ones, respectively).

$$\dim\left(Sl(n,\mathbb{C})\right) = 2n^2 - 2$$

2.2 Orthogonal and special orthogonal groups

These linear transformations leave the (ordinary) scalar product on $\mathbb{R}^n$ invariant:

$$\langle x|\, y\rangle = \sum_{j=1}^{n} x_j y_j\,, \quad x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \in \mathbb{R}^n$$

$A \in O(n) \Leftrightarrow \langle Ax|\, Ay\rangle = \langle x|\, y\rangle$, $(Ax)_j = \sum_{k=1}^{n} A_{jk} x_k$, $\sum_{j,k,l=1}^{n} A_{jl} x_l A_{jk} y_k \stackrel{!}{=} \sum_{j=1}^{n} x_j y_j = \sum_{l,k=1}^{n} x_l y_k \delta_{lk}$.

$$\sum_{j=1}^{n} A_{jk} A_{jl} = \delta_{kl}$$

Written as a matrix equation:

$$(AB)_{kl} = \sum_{j=1}^{n} A_{kj} B_{jl}$$

For $A \in O(n)$ we have

$$A^t \cdot A = \mathbb{1}_{n\times n}$$

with $A^{-1} = A^t$ being the transposed matrix.

$$O(n) = \{A \in Gl(n, \mathbb{R})\ ,\ A^t A = \mathbb{1}_{n\times n}\}$$

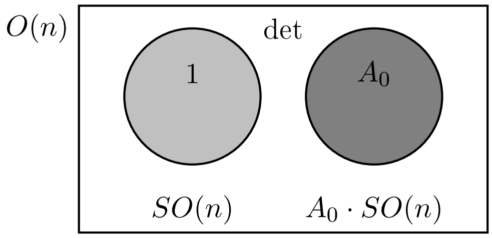
Since $\det\left(A^t\right) = \det(A)$ one has for orthogonal matrices

$$\det(A)^2 = 1 \quad \det(A) = \pm 1$$

The subgroup of $O(n)$ with determinant $+1$ is called *special orthogonal group*

$$SO(n) = \{A \in O(n)\ ,\ \det(A) = 1\}$$

$O(n)$ has two connected components



with $A_0 = \begin{pmatrix} -1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{pmatrix}$ and $\det(A_0) = -1$. The determinant as a continuous function of the matrix entries only takes the values ± 1. The conditional equations

$$\sum_{j=1}^{n} A_{jl} A_{jk} = \delta_{lk}$$

are identical for $l \leftrightarrow k$, therefore one can restrict oneself to $l \leq k$. There remain $\frac{n}{2}(n+1)$ independent conditional equations. The number of real parameters for $SO(n)$ is

$$n^2 - \frac{n}{2}(n+1) = \frac{n}{2}(n-1) = \binom{n}{2} \quad \text{(binomial coefficient)}$$

The row and column vectors of an orthogonal matrix form pairwise orthogonal unit vectors. E.g. $\sum\limits_{j=1}^{n} A_{jl}^2 = 1$, for $l = 1, \ldots, n$, (sum of positive squares where the range of the A_{jl} is bounded by $-1 \leq A_{jl} \leq 1$).

$SO(n)$ is a compact, connected manifold of dimension $\dim(SO(n)) = \frac{n}{2}(n-1)$.

Examples:

i) $O(2)$ consists of rotations

$$\begin{pmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{pmatrix} \in SO(2)$$

and mirror reflections

$$\begin{pmatrix} \cos\varphi & \sin\varphi \\ \sin\varphi & -\cos\varphi \end{pmatrix}$$

$SO(2) = S^1$ (1-dimensional circle) is abelian

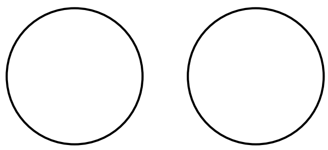
$$\begin{pmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{pmatrix} \begin{pmatrix} \cos\psi & -\sin\psi \\ \sin\psi & \cos\psi \end{pmatrix} = \begin{pmatrix} \cos(\varphi+\psi) & -\sin(\varphi+\psi) \\ \sin(\varphi+\psi) & \cos(\varphi+\psi) \end{pmatrix}$$

But $O(2) \neq SO(2) \times \mathbb{Z}_2$ as a group, only as a manifold, since $SO(2) \times \mathbb{Z}_2$ is abelian and $O(2)$ is not.

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$O(2)$ as a manifold consists of two disjoint circles



ii) Three dimensional rotations and mirror reflections

$$O(3) = SO(3) \times \mathbb{Z}_2$$

The pertinent group isomorphism is

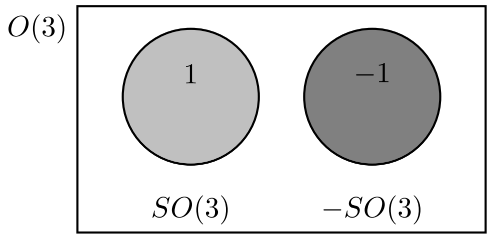
$$A \mapsto (sA, s) \quad \text{with } s = \det(A) = \pm 1$$

$$\det(sA) = s^3 \underbrace{\det(A)}_{s} = s^4 = 1$$

$$(R, s) \to sR$$

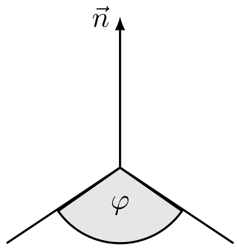
$$\det(sR) = s^3 \underbrace{\det(R)}_{1} = s^3 \cdot 1 = s$$

The crucial point for this isomorphism is the *odd* number of dimensions.

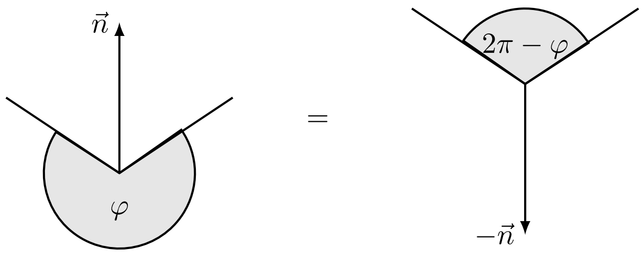


The rotations $R \in SO(3)$ can be characterized by giving an *axis of rotation* $\boldsymbol{n}$ (unit vector, $\boldsymbol{n}^2 = 1$) and an *angle of rotation* φ with $0 \leq \varphi \leq 2\pi$. $\boldsymbol{n}$ is the eigenvector of R with the eigenvalue 1:

$$R\boldsymbol{n} = \boldsymbol{n}$$



Observation: A rotation about an axis $\boldsymbol{n}$ with angle φ is equal to a rotation about $-\boldsymbol{n}$ with angle $2\pi - \varphi$.



Explicitly

$$R\boldsymbol{v} = \underbrace{\boldsymbol{n}(\boldsymbol{n} \cdot \boldsymbol{v})}_{\text{parallel component}} + \underbrace{\cos\varphi\ (\boldsymbol{v} - \boldsymbol{n}(\boldsymbol{n} \cdot \boldsymbol{v})) + \sin\varphi\ (\boldsymbol{n} \times \boldsymbol{v})}_{\text{perpendicular component}}$$

and $\cos(2\pi - \varphi) = \cos\varphi$ and $\sin(2\pi - \varphi) = -\sin\varphi$. Therefore we can restrict φ to $0 \leq \varphi \leq \pi$ and keep in mind that the rotation $(\boldsymbol{n}, \pi)$ is equivalent to $(-\boldsymbol{n}, \pi)$.

Result: The points in $SO(3)$ are in a one-to-one correspondence to the points in a 3-dimensional ball of radius π, where antipodal[1] points on the ball's surface are to be identified:

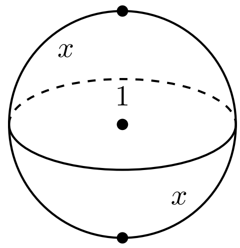


Figure 2.1: Antipodal points on a three dimensional ball

$SO(3)$ is *not simply-connected*. There exists a closed path in $SO(3)$ which *can not* be continuously contracted to a single point:

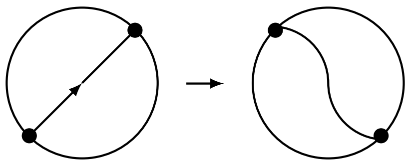


Figure 2.2: Opposing points can not be brought together.

However, if this closed path is traversed twice, this doubly traversed path can be continuously contracted to a single point:

[1]dt. antipodisch (aus dem Griechischen "auf der gegenüberliegenden Seite der Erdkugel")

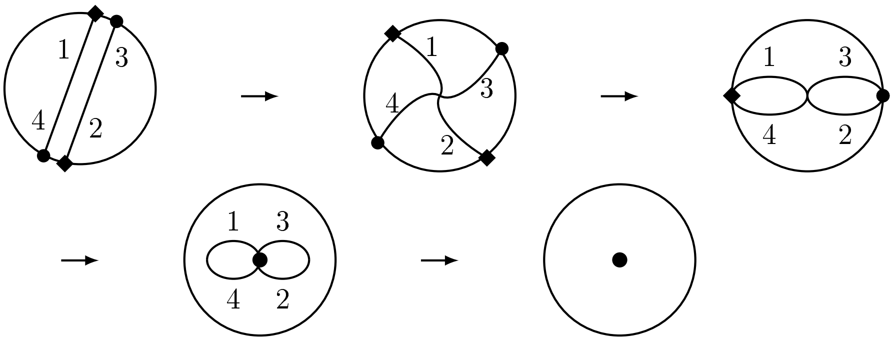


Figure 2.3: Continuous deformation of a doubly traversed, closed path.

Expressed in terms of the so-called *first homotopy-group*:

$$\pi_1(SO(3)) = \mathbb{Z}_2$$

Another representation of the manifold $SO(3)$ is given by

$$SO(3) = S^3/\sim\, = \mathbb{R}P^3$$

with $\sim$ being the equivalence relation: $x \sim y \Leftrightarrow x = \pm y$.
Visualization through an analogon in one dimension lower:

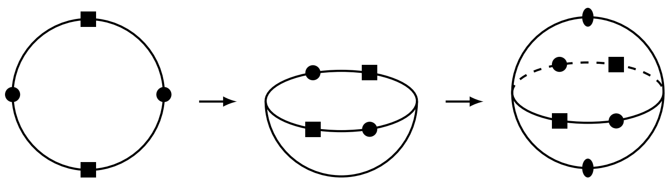


Figure 2.4: The points $\boldsymbol{r}$ and $-\boldsymbol{r}$ on the sphere S^2 get identified.

The points on the sphere S^2, identified with $\boldsymbol{r}$ and $-\boldsymbol{r}$, are in one-to-one correspondence with straight lines (without orientation) going through the origin in $\mathbb{R}^3$:

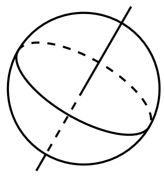


Figure 2.5: Two dimensional projective space $\mathbb{R}P^2$.

The three dimensional projective space $\mathbb{R}P^3$ is the space of all straight lines going through the origin in $\mathbb{R}^4$:

$$SO(3) = \mathbb{R}P^3$$

Euler's parametrization (1770)

$$\begin{pmatrix} x_1^2+x_2^2-x_3^2-x_4^2 & -2x_1x_4+2x_2x_3 & 2x_1x_3+2x_2x_4 \\ 2x_1x_4+2x_2x_3 & x_1^2-x_2^2+x_3^2-x_4^2 & -2x_1x_2+2x_3x_4 \\ -2x_1x_3+2x_2x_4 & 2x_1x_2+2x_3x_4 & x_1^2-x_2^2-x_3^2+x_4^2 \end{pmatrix} \cdot \frac{1}{x_1^2+x_2^2+x_3^2+x_4^2} \in SO(3)$$

with $(0,0,0,0) \neq (x_1,x_2,x_3,x_4) \in \mathbb{R}^4$ gives a rational parametrization of the rotation group in three dimensions.

The special orthogonal group $SO(n)$ "inherits" this topological property from $SO(3)$:

$$\pi_1(SO(n)) = \mathbb{Z}_2\,, \quad \text{for } n \geq 3$$

For comparison: $\pi_1(SO(2)) = \pi_1\big(S^1\big) = \mathbb{Z}$.
There arises the question about a simply-connected group (locally isomorphic to $SO(n)$), such that the $SO(n)$ emerges through the identification of pairs of points:

$$\text{spin-group: Spin}(n)$$

It is the *two-sheeted covering-group* of $SO(n)$ with

$$\pi_1(\mathrm{Spin}(n)) = 0$$

and

$$SO(n) = \mathrm{Spin}(n)/\mathbb{Z}_2$$

For $n = 3, 4, 5, 6$ the spin-group $\mathrm{Spin}(n)$ is given by certain "special" unitary groups. For $n \geq 7$ the spin-group $\mathrm{Spin}(n)$ can be constructed by means of *Clifford algebras*

$$\gamma_i\gamma_j + \gamma_j\gamma_i = -2\delta_{ij}\mathbb{1}$$

2.3 Unitary and special unitary groups

These linear transformations leave the *hermitian* scalar product on $\mathbb{C}^n$ invariant:

$$\langle z|\, w\rangle = \sum_{j=1}^{n} z_j^* w_j, \quad z = \begin{pmatrix} z_1 \\ \vdots \\ z_n \end{pmatrix},\ w = \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix} \in \mathbb{C}^n$$

The hermitian scalar product is complex-linear in the second argument and anti-linear in the first argument:

$$\begin{aligned} \langle z|\, \alpha w\rangle &= \alpha \langle z|\, w\rangle \ , \quad \text{with } \alpha \in \mathbb{C} \\ \langle \beta z|\, w\rangle &= \beta^* \langle z|\, w\rangle \ , \quad \text{with } \beta \in \mathbb{C} \end{aligned}$$

$\underbrace{A \in U(n)}_{\text{unitary matrix}} \Leftrightarrow \langle Az|\, Aw\rangle = \langle z|\, w\rangle.$

$$\sum_{j=1}^{n} A_{jk}^* A_{jl} = \delta_{kl}$$

For $A \in U(n)$ we have

$$A^{t*} \cdot A = \mathbb{1}_{n \times n}$$

with $A^{t*} = A^{\dagger}$ being the adjoint matrix. So we find

$$A^{\dagger} A = \mathbb{1} \Leftrightarrow A^{-1} = A^{\dagger}$$

Since $\det\left(A^{\dagger}\right) = (\det(A))^{*}$ one gets $|\det(A)|^{2} = 1$ and therefore

$$\det(A) = \zeta \quad \text{with } |\zeta| = 1$$

The subgroup of $U(n)$ with determinant equal to 1 is called the *special unitary group*

$$SU(n) = \{A \in U(n)\ , \det(A) = 1\}$$

The conditions for unitarity $\sum_{j=1}^{n} A_{jk}^{*} A_{jl} = \delta_{kl}$ are complex conjugated to each other for $l \leftrightarrow k$. Therefore consider only $l \leq k$.
The number of *real* parameters for $U(n)$ is

$$2n^{2} - \frac{n}{2}(n-1) \cdot 2 - n = n^{2}$$

$U(n)$ is a connected, *compact* manifold of *real* dimension $\dim(U(n)) = n^{2}$.

The condition $\det(A) = 1$ reduces the number of real parameters by one:

$SU(n)$ is a connected, *compact* manifold of *real* dimension $\dim(SU(n)) = n^2 - 1$.

Examples:

i) $U(1) = \{\zeta \in \mathbb{C}, |\zeta| = 1\} = S^1 = SO(2)$.

$$e^{i\varphi} \mapsto \begin{pmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{pmatrix}$$

ii) $A \in SU(2)$, $A^{-1} = A^\dagger$

$$A = \begin{pmatrix} z_1 & z_2 \\ z_3 & z_4 \end{pmatrix}, \quad A^\dagger = \begin{pmatrix} z_1^* & z_3^* \\ z_2^* & z_4^* \end{pmatrix}, \quad A^{-1} = \frac{1}{\underbrace{\det(A)}_{1}} \begin{pmatrix} z_4 & -z_2 \\ -z_3 & z_1 \end{pmatrix}$$

$\Rightarrow z_4 = z_1^*,\ z_3 = -z_2^*$ and therefore

$$A = \begin{pmatrix} z_1 & z_2 \\ -z_2^* & z_1^* \end{pmatrix}, \quad \det(A) = 1 = |z_1|^2 + |z_2|^2$$

Set $z_1 = x_1 + ix_2$ and $z_2 = x_3 + ix_4$ and the condition for the determinant reads $x_1^2 + x_2^2 + x_3^2 + x_4^2 = 1$. This defines points on the 3-sphere S^3 and therefore

$$SU(2) = S^3$$

S^3 is simply-connected and therefore $SU(2)$ is simply-connected. In fact all $SU(n)$ are simply-connected

$$\pi_1(SU(n)) = 0$$

Parametrization of $SU(2)$: Set $x_1 = \cos\varphi$ and $\begin{pmatrix} x_4 \\ x_3 \\ x_2 \end{pmatrix} = \boldsymbol{n} \cdot \sin\varphi$ with

$\boldsymbol{n}^2 = 1$ and $\boldsymbol{n} \in S^2$ being the unit vector.

$$\begin{aligned}
SU(2) \ni & \begin{pmatrix} \cos\varphi + \mathrm{i}n_3 \sin\varphi & (n_2 + \mathrm{i}n_1)\sin\varphi \\ (-n_2 + \mathrm{i}n_1)\sin\varphi & \cos\varphi - \mathrm{i}n_3 \sin\varphi \end{pmatrix} \\
&= \cos\varphi \cdot \mathbb{1} + \mathrm{i}\sin\varphi \Big\{ n_1 \underbrace{\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}}_{\sigma_1} + n_2 \underbrace{\begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}}_{\sigma_2} + n_3 \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}}_{\sigma_3} \Big\} \\
&= \cos\varphi + \mathrm{i}(\boldsymbol{\sigma} \cdot \boldsymbol{n}) \sin\varphi \\
&= \exp\{\mathrm{i}\boldsymbol{\sigma} \cdot \boldsymbol{n}\varphi\} \\
&= \sum_{j=0}^{\infty} \frac{(\mathrm{i}\boldsymbol{\sigma} \cdot \boldsymbol{n}\varphi)^j}{j!}
\end{aligned}$$

with $\sigma_1, \sigma_2, \sigma_3$ being the Pauli spin matrices.
Check this relation:

$$\begin{aligned}
(\boldsymbol{\sigma} \cdot \boldsymbol{n})^2 &= \begin{pmatrix} n_3 & n_1 - \mathrm{i}n_2 \\ n_1 + \mathrm{i}n_2 & -n_3 \end{pmatrix} \begin{pmatrix} n_3 & n_1 - \mathrm{i}n_2 \\ n_1 + \mathrm{i}n_2 & -n_3 \end{pmatrix} \\
&= \begin{pmatrix} n_1^2 + n_2^2 + n_3^2 & 0 \\ 0 & n_1^2 + n_2^2 + n_3^2 \end{pmatrix} = \mathbb{1} \\
\exp\left[\mathrm{i}\boldsymbol{\sigma} \cdot \boldsymbol{n}\varphi\right] &= \sum_{k=0}^{\infty} \left(\frac{\mathrm{i}^{2k}\varphi^{2k}}{(2k)!} \cdot \mathbb{1} + \frac{\mathrm{i}^{2k}\varphi^{2k+1}}{(2k+1)!} \cdot \mathrm{i}\boldsymbol{\sigma} \cdot \boldsymbol{n} \right) \\
&= \cos\varphi \cdot \mathbb{1} + \mathrm{i}\boldsymbol{\sigma} \cdot \boldsymbol{n} \sin\varphi
\end{aligned}$$

For compact Lie-groups the matrix exponential function is surjective. This for example is not true for the non-compact Lie-group $Sl(2, \mathbb{R})$.

2.4 Quaternions (Hamilton 1843)

Question: Is there a "vector product" on $\mathbb{R}^n$, such that each $0 \neq \boldsymbol{x} \in \mathbb{R}^n$ possesses a multiplicative inverse with respect to this vector product?

$\boldsymbol{n = 2}$**:** yes, the complex numbers $\mathbb{C} = \mathbb{R}^2$

$$\begin{aligned}
(x_1 + \mathrm{i}x_2)(y_1 + \mathrm{i}y_2) &= x_1 y_1 - x_2 y_2 + \mathrm{i}(x_1 y_2 + x_2 y_1) \\
\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \cdot \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} &= \begin{pmatrix} x_1 y_2 - x_2 y_2 \\ x_1 y_2 + x_2 y_1 \end{pmatrix} \\
(x_1 + \mathrm{i}x_2)^{-1} &= \frac{x_1 - \mathrm{i}x_2}{x_1^2 + x_2^2} \\
\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}^{-1} &= \frac{1}{x_1^2 + x_2^2} \begin{pmatrix} x_1 \\ -x_2 \end{pmatrix}
\end{aligned}$$

$\boldsymbol{n=3}$**:** no. After having chosen a basis $\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3$ of $\mathbb{R}^3$, the vector multiplication with $0 \neq \boldsymbol{x} \in \mathbb{R}^3$, $\boldsymbol{x} \neq c\boldsymbol{e}_1$ can be represented by a 3×3-matrix C

$$\boldsymbol{x} * \boldsymbol{e}_k = \sum_{l=1}^{3} C_{lk} \boldsymbol{e}_l \,, \quad C_{lk} \in \mathbb{R}$$

Since each cubic polynomial with real coefficients has at least one real root, the matrix C has at least one real eigenvalue λ and a corresponding eigenvector $0 \neq \boldsymbol{v} \in \mathbb{R}^3$.
Now one has $\boldsymbol{x}*\boldsymbol{v} = \lambda \boldsymbol{v}$ or $\underbrace{(\boldsymbol{x} - \lambda \boldsymbol{e}_1)}_{w \neq 0} * \underbrace{\boldsymbol{v}}_{\neq 0} = 0$. There are always zero-divisors.
Zero-divisors $\boldsymbol{v}$ cannot be invertible: $\boldsymbol{v} * \boldsymbol{a} = \boldsymbol{e}_1$, $\boldsymbol{w} * \boldsymbol{v} = 0 \Rightarrow \boldsymbol{w} = w * \boldsymbol{e}_1 = w * (\boldsymbol{v} * \boldsymbol{a}) = 0 \Rightarrow \boldsymbol{w} = 0 \lightning$.
This argument holds for *any odd* dimension.

$\boldsymbol{n=4}$**:** yes. This are Hamilton's quaternions. The multiplication on $\mathbb{R}^4$ is *associative*, but *no longer commutative*. One speaks of the skew field[2] $\mathbb{H}$. $q \in \mathbb{H}$, $q = x_1 + ix_2 + jx_3 + kx_4$ with $x_{1,2,3,4} \in \mathbb{R}$. The three "imaginary units" i, j, k obey the following multiplication rules

$$i^2 = j^2 = k^2 = -1$$

$$\begin{aligned} ij &= k, & jk &= i, & ki &= j \\ ji &= -k, & kj &= -i, & ik &= -j \end{aligned}$$

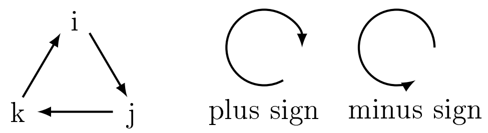


Figure 2.6: Multiplication rules for unit quaternions

$$\begin{aligned} qp &= (x_1 + ix_2 + jx_3 + kx_4)(y_1 + iy_2 + jy_3 + ky_3) \\ &= x_1y_2 - x_2y_2 - x_3y_3 - x_4y_4 + i(x_1y_2 + x_2y_1 + x_3y_4 - x_4y_3) \\ &\quad + j(x_1y_3 + x_3y_1 + x_4y_2 - x_2y_4) + k(x_1y_4 + x_4y_1 + x_2y_3 - x_3y_2) \neq pq \end{aligned}$$

[2] dt. Schiefkörper

The quaternion product contains the scalar dot-product and the cross-product. We define a *conjugated* quaternion $\bar{q} = x_1 - ix_2 - jx_3 - kx_4$ and get

$$q\bar{q} = x_1^2 + x_2^2 + x_3^2 + x_4^2 = \bar{q}q \equiv N(q) \in \mathbb{R}$$

$$q^{-1} = \frac{\bar{q}}{N(q)}$$

The quaternion conjugation fulfills $\bar{\bar{q}} = q$, $\overline{qp} = \bar{p}\bar{q}$.
The skew field $\mathbb{H}$ of the quaternions can be embedded into the complex 2×2-matrices

$$\mathbb{H} \to \mathbb{C}^{2\times 2}$$

$$x_1 + ix_2 + (x_3 + \underbrace{ix_4)j}_{kx_4} = z + wj \mapsto \begin{pmatrix} z & w \\ -w^* & z^* \end{pmatrix}, \quad z, w \in \mathbb{C}$$

$$1 \mapsto \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad i \mapsto \begin{pmatrix} \mathrm{i} & 0 \\ 0 & -\mathrm{i} \end{pmatrix} = \mathrm{i}\sigma_3$$

$$j \mapsto \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \mathrm{i}\sigma_2\,, \quad k \mapsto \begin{pmatrix} 0 & \mathrm{i} \\ \mathrm{i} & o \end{pmatrix} = \mathrm{i}\sigma_1$$

The imaginary basis quaternions correspond to i$\cdot$ Pauli-spin-matrices.
$\bar{q}$ corresponds to the 2×2-matrix $\begin{pmatrix} z^* & -w \\ w^* & z \end{pmatrix} = \begin{pmatrix} z & w \\ -w^* & z^* \end{pmatrix}^\dagger$.
The complex number $\mathbb{C}$ are contained in $\mathbb{H}$ as a subfield

$$\mathbb{C} \to \mathbb{H}\,, \quad z \mapsto \begin{pmatrix} z & 0 \\ 0 & z^* \end{pmatrix}$$

$\mathbb{H}$ is a 2-dimensional vector space of $\mathbb{C}$ with basis $1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ and $j = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ where $zj = jz^*$ for $z \in \mathbb{C}$ and $j^2 = -1$ holds.

$$\begin{aligned} N(q)\,\mathbb{1}_{2\times 2} &= \begin{pmatrix} z & w \\ -w^* & z^* \end{pmatrix} \begin{pmatrix} z^* & -w \\ w^* & z \end{pmatrix} \\ &= (zz^* + ww^*)\mathbb{1}_{2\times 2} \\ &= \det \begin{vmatrix} z & w \\ -w^* & z^* \end{vmatrix} \mathbb{1}_{2\times 2} \end{aligned}$$

$$N(pq) = pq\overline{pq} = p\underbrace{q\bar{q}}_{N(q)}\bar{p} = N(p)\,N(q) \quad \text{(leads to the four square formula)}$$

The unit-quaternions form the so-called *quaternion Lie-group* $Sp(1) = \{q \in \mathbb{H}\,, N(q) = 1\}$. Considered as complex 2×2-matrices, $Sp(1)$ consists of matrices

$$\begin{pmatrix} z & w \\ -w^* & z^* \end{pmatrix} \quad \text{with } zz^* + ww^* = 1$$

We find the group-isomorphism

$$Sp(1) = SU(2) = S^3$$

Remark: The *only* other real division algebra exists in $n = 8$ dimensions and is formed by the so-called *Cayley octonions*. The octonion multiplication is *neither commutative*, *nor associative* ("needle of topology in the flesh of algebra"). Complex division algebras do not exist at all.

Now we consider the purely imaginary quaternions $Im\mathbb{H} = \{ix_2+jx_3+kx_4\} = \mathbb{R}^3$ and the group operations of $Sp(1)$ on it

$$\begin{aligned} Im\mathbb{H} &\to Im\mathbb{H} \\ u &\mapsto qu\bar{q} \quad \text{with } q \in Sp(1) \end{aligned}$$

$u \in Im\mathbb{H} \Leftrightarrow \bar{u} = -u$, $q\bar{u}\bar{q} = \overline{qu\bar{q}} = -qu\bar{q} \in Im\mathbb{H}$, $N(u) = u\bar{u} = x_2^2 + x_3^2 + x_4^2$, $N(qu\bar{q}) = qu\underbrace{\bar{q}q}_{1}\bar{u}\bar{q} = qN(u)\,\bar{q} = N(u)\underbrace{q\bar{q}}_{1} = N(u)$. Each $q \in Sp(1)$ induces a 3-dimensional rotation, where $\pm q \in Sp(1)$ yield the *same* rotation. We have a group-homomorphism

$$Sp(1) \to SO(3) \quad \text{with kernel } \{\pm 1\}$$

and obtain the group-isomorphism

$$SO(3) = Sp(1)\,/\{\pm1\} = SU(2)\,/\{\pm1\}$$

$$\Rightarrow Spin(3) = SU(2) = Sp(1)$$

The same group operation written differently with $A \in SU(2)$

$$A\boldsymbol{\sigma} \cdot \boldsymbol{r} A^{\dagger} = \boldsymbol{\sigma} \cdot \boldsymbol{r}'$$

yields

$$\boldsymbol{r}' = R(A)\,\boldsymbol{r} \quad \text{with } R(A) \in SO(3)$$

$$\det(\boldsymbol{\sigma} \cdot \boldsymbol{r}) = \det \begin{vmatrix} z & x - \mathrm{i}y \\ x + \mathrm{i}y & -z \end{vmatrix} = -z^2 - x^2 - y^2 = -\boldsymbol{r}^2$$

$$\det\left(A\boldsymbol{\sigma} \cdot \boldsymbol{r} A^{\dagger}\right) = \underbrace{\det(A)}_{1} \det(\boldsymbol{\sigma} \cdot \boldsymbol{r}) \underbrace{\det\left(A^{\dagger}\right)}_{1} = -\boldsymbol{r}^2 = -\boldsymbol{r}'^2$$

Now we let $Sp(1) \times Sp(1)$ operate on $\mathbb{H}$ in the following way: $u = x_1 + ix_2 + jx_3 + kx_4 \in \mathbb{H}$ and

$$\begin{aligned} \psi(p,q) : \mathbb{H} &\to \mathbb{H} \\ u &\mapsto pu\bar{q} \quad \text{with } p, q \in Sp(1) \\ N(u) &= u\bar{u} = x_1^2 + x_2^2 + x_3^2 + x_4^2 \\ N(pu\bar{q}) &= pu \underbrace{\bar{q}q}_{1} \bar{u}\bar{p} = p \underbrace{u\bar{u}}_{N(u)} \bar{p} = N(u) \underbrace{p\bar{p}}_{1} = N(u) \end{aligned}$$

Each pair $(p,q) \in Sp(1) \times Sp(1)$ induces a four-dimensional rotation, where $\pm(p,q)$ yield the same result.
Calculate the kernel of ψ: $u = pu\bar{q}$ for all $u \in \mathbb{H}$ implies in particular for $u = 1$, that $p\bar{q} = 1$ and therefore $p = q$. Now $u = qu\bar{q}$ or $uq = qu \Rightarrow u = \pm 1$

$$\Rightarrow kernel(\psi) = \{\pm 1\}$$

We obtain the group-isomorphism

$$SO(4) = SU(2) \times SU(2) \,/ \{\pm(\mathbb{1}, \mathbb{1})\}$$

$$\Rightarrow Spin(4) = SU(2) \times SU(2)$$

Check the dimension: $\frac{4}{2}(4-1) = 6 = 3+3$ ✓

Supplement:

$$Spin(5) = Sp(2) = \{A \in SU(4)\ , A^t JA = J\}$$

with the antisymmetric matrix $J = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$.

Furthermore

$$Spin(6) = SU(4)$$

$$SO(6) = SU(4)\,/\{\pm 1\}$$

Reasoning: One obtains 6 antisymmetric tensors from 4 basis vectors.

2.5 The Lorentz group

The *Lorentz group* is of overwhelming importance in theoretical physics. It describes the symmetries of space-time according to special relativity. These linear transformations leave the indefinite Minkowski scalar-product on $\mathbb{R}^4$ invariant:

$$x^\mu = \begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix}, y^\mu = \begin{pmatrix} y^0 \\ y^1 \\ y^2 \\ y^3 \end{pmatrix}$$

with x^0, y^0 the time components and x^j, y^j the spatial components, where $j = 1, 2, 3$.

$$x \cdot y = x^0 y^0 - x^1 y^1 - x^2 y^2 - x^3 y^3 = \sum_{\mu,\nu=0}^{3} g_{\mu\nu} x^\mu y^\nu$$

with $g_{\mu\nu} = diag(1, -1, -1, -1)$ the *Minkowski metric tensor.*
Four-vectors $x^\mu \in \mathbb{R}$ are called

$$\left.\begin{array}{r}\text{timelike}\\ \text{lightlike}\\ \text{spacelike}\end{array}\right\} \text{ if } \left\{\begin{array}{ll} x \cdot x > 0 & \text{inside}\\ x \cdot x = 0 & \text{on}\\ x \cdot x < 0 & \text{outside}\end{array}\right\} \text{ the light cone}$$

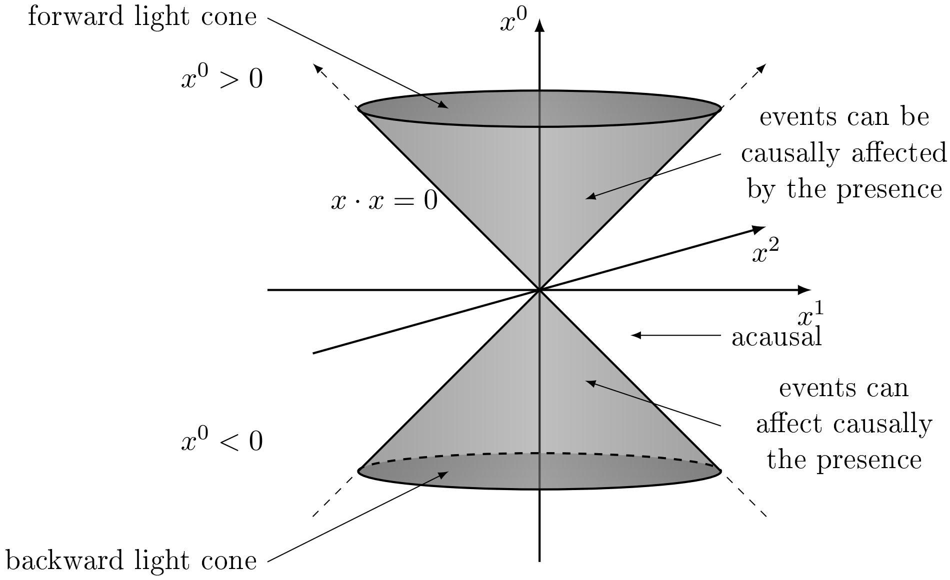


Figure 2.7: Light cone

$$x'^\mu = \Lambda^\mu_\alpha x^\alpha \, , y'^\mu = \Lambda^\mu_\beta y^\beta$$

Since $x' \cdot y' \stackrel{!}{=} x \cdot y$ we find the constraining equation

$$g_{\mu\nu} \Lambda^\mu_\alpha \Lambda^\nu_\beta = g_{\alpha\beta} \tag{2.1}$$

Only the equations with $\alpha \geq \beta$ are independent. For $\alpha \leftrightarrow \beta$ relabel the summation indices $\mu \leftrightarrow \nu$. The number of parameters is $16 - 4 - 6 = 6$.
Equation (2.1) can be written as a matrix-equation:

$$g = \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix} , \quad \Lambda = \begin{pmatrix} \Lambda^0_0 & \dots & \Lambda^0_3 \\ \vdots & & \vdots \\ \Lambda^3_0 & \dots & \Lambda^3_3 \end{pmatrix}$$

$$\Lambda^t g \Lambda = g$$

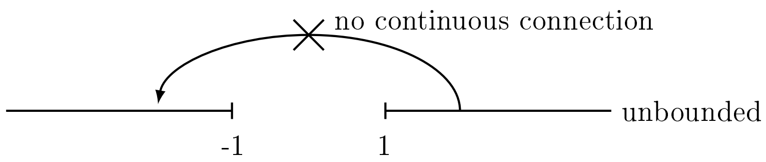
Taking the determinant, one finds $\det\left(\Lambda^t\right) \underbrace{\det(g)}_{-1} \det(\Lambda) = \underbrace{\det(g)}_{-1}$

$$\Rightarrow \det(\Lambda) = \pm 1$$

This means there are at least two connected components. Consider equation (2.1) for $\alpha = 0$

$$(\Lambda^0_0)^2 - (\Lambda^1_0)^2 - (\Lambda^2_0)^2 - (\Lambda^3_0)^2 = 1$$

$$\Rightarrow \Lambda^0_0 \geq 1 \quad \text{or} \quad \Lambda^0_0 \leq -1$$



$$\begin{array}{cccc}
\begin{pmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{pmatrix} &
\begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix} &
\begin{pmatrix} -1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{pmatrix} &
\begin{pmatrix} -1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix} \\
\text{identity} & \text{parity } P & \text{time-reversal } T & PT \\
det = 1, \Lambda^0_0 \geq 1 & det = -1, \Lambda^0_0 \geq 1 & det = -1, \Lambda^0_0 \leq -1 & det = 1, \Lambda^0_0 \leq -1
\end{array}$$

The Lorentz group $O(1,3)$ is a non-compact, six-dimensional manifold with four connected components.

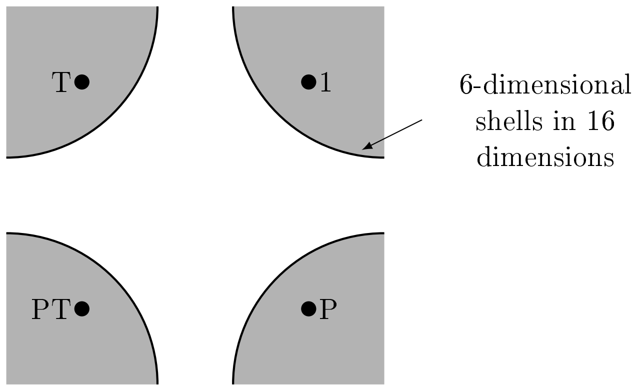


Figure 2.8: Connected components of the Lorentz group

The connected component of $\mathbb{1}$ is called the *proper orthochronous Lorentz group* $SO^+(1,3)$ ($\det(\Lambda) = 1$, $\Lambda^0_0 \geq 1$).

$$O(1,3) = \underbrace{SO^+(1,3) \cup P \cdot SO^+(1,3)}_{O^+(1,3) \text{ (orthochronous Lorentz group)}} \cup T \cdot SO^+(1,3) \cup P \cdot SO^+(1,3)$$

$$O(1,3)\,/SO^+(1,3) = \mathbb{Z}_2 \times \mathbb{Z}_2$$

The *spatial rotations* are contained in $SO^+(1,3)$ as a subgroup

$$\Lambda = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & & & \\ 0 & & R & \\ 0 & & & \end{pmatrix}, \quad \text{with } R \in SO(3)$$

as well as the *Lorentz boosts* along the three directions of space, e.g.

$$\Lambda = \begin{pmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

with $x'^0 = \gamma(x^0 - \beta x^1)$, $x'^1 = \gamma(x^1 - \beta x^0)$ and $(\Lambda^0_0)^2 - (\Lambda^1_0)^2 = 1 = \gamma^2(1-\beta^2)$, $\gamma = \frac{1}{\sqrt{1-\beta^2}}$, $\beta = \frac{v_x}{c} < 1$.
As a manifold, but not as a group, one has

$$SO^+(1,3) = SO(3) \times \mathbb{R}^3$$

As the rotation group the proper orthochronos Lorentz group $SO^+(1,3)$ is *not* simply connected

$$\pi_1\big(SO^+(1,3)\big) = \mathbb{Z}_2$$

With inclusion of space-time-translations one arrives at the so-called *Poincaré group*

$$x'^\mu = \Lambda^\mu_\nu x^\nu + a^\mu\,, \quad \text{with } a = \begin{pmatrix} a^0 \\ a^1 \\ a^2 \\ a^3 \end{pmatrix} \in \mathbb{R}^4$$

For pairs $(a, \Lambda) \in \mathbb{R}^4 \times O(1,3)$ we infer the group multiplication

$$\Lambda_1(\Lambda_2 x + a_2) + a_1 = (\Lambda_1\Lambda_2)x + (\Lambda_1 a_2 + a_1)$$

$$(a_1, \Lambda_1) \cdot (a_2, \Lambda_2) = (a_1 + \Lambda_1 a_2, \Lambda_1\Lambda_2)$$

The Poincaré group is a *semi-direct* product of the abelian translation group $\mathbb{R}^4$ and the non-abelian Lorentz group $O(1,3)$. There is a convenient representation in terms of 5×5-matrices

$$\begin{pmatrix} \Lambda^0_0 & \dots & \dots & \Lambda^0_3 & a^0 \\ \vdots & & & \vdots & \vdots \\ \Lambda^3_0 & \dots & \dots & \Lambda^3_3 & a^3 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \Lambda & a \\ 0 & 1 \end{pmatrix}$$

The group multiplication in the Poincaré group is now given by the ordinary matrix multiplication

$$\begin{pmatrix} \Lambda_1 & a_1 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} \Lambda_2 & a_2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \Lambda_1\Lambda_2 & \Lambda_1 a_2 + a_1 \\ 0 & 1 \end{pmatrix}$$

Relation of the Lorentz group to $Sl(2,\mathbb{C})$:

We construct another (equivalent) representation of $\mathbb{R}^4$ (space-time) with its indefinite Lorentz metric in the form of *hermitian* 2×2-matrices

$$X = x^0 \mathbb{1} + \boldsymbol{\sigma} \cdot \boldsymbol{x} = \begin{pmatrix} x^0 + x^3 & x^1 - \mathrm{i}x^2 \\ x^1 + \mathrm{i}x^2 & x^0 - x^3 \end{pmatrix} = X^\dagger = (X^*)^t$$

All hermitian 2×2-matrices are of this form, since

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = \begin{pmatrix} a_{11}^* & a_{21}^* \\ a_{12}^* & a_{22}^* \end{pmatrix}$$

requires $a_{11}, a_{22} \in \mathbb{R}$ and $a_{12} = a_{21}^*$.
One has $\det(X) = (x^0 + x^3)(x^0 - x^3) - (x^1 - \mathrm{i}x^2)(x^1 + \mathrm{i}x^2) = (x^0)^2 - (x^3)^2 - (x^1)^2 - (x^2)^2 = x \cdot x$. From the quadratic form $x \cdot x$ we obtain the scalar product (i.e. bilinear form) as follows

$$\begin{aligned} x \cdot y &= \frac{1}{2}((x+y)\cdot(x+y) - x \cdot x - y \cdot y) \\ &= \frac{1}{2}\big(\det(X+Y) - \det(X) - \det(Y)\big) \end{aligned}$$

The group $Sl(2,\mathbb{C})$ acts on hermitian 2×2-matrices

$$A \in Sl(2,\mathbb{C}) : X \mapsto AXA^\dagger$$

Check that the image is a hermitian 2×2-matrix: $(AXA^\dagger)^\dagger = A^{\dagger\dagger}X^\dagger A^\dagger = AXA^\dagger$. This again is hermitian ✓
The determinant $\det(X) = x \cdot x$ remains invariant

$$\det\left(AXA^\dagger\right) = \underbrace{\det(A)}_{1} \det(X) \underbrace{\det(A)^*}_{1} = \det(X)$$

Observation: Any $A \in Sl(2,\mathbb{C})$ induces a Lorentz transformation $\Lambda(A) \in SO^+(1,3)$.

$$\begin{aligned} AXA^\dagger = X' &= \begin{pmatrix} x'^0 + x'^3 & x'^1 - \mathrm{i}x'^2 \\ x'^1 + \mathrm{i}x'^2 & x'^0 - x'^3 \end{pmatrix} \\ x'^\mu &= \Lambda^\mu_\nu x^\nu \end{aligned}$$

and $\pm A \in Sl(2,\mathbb{C})$ lead to the same result.

Conclusion: We have a group homomorphism

$$\rho : Sl(2,\mathbb{C}) \to SO^+(1,3)$$
$$\mathbb{1}_{2\times 2} \mapsto \mathbb{1}_{4\times 4}$$

The kernel of ρ consists of $\{\pm \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\}$.

$$X = AXA^\dagger \quad \text{for all hermitian } 2\times 2\text{-matrices}$$

Take $X = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. This gives $A^\dagger = A^{-1}$ and thus $XA = AX$.

Setting $X = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ implies $a_{12} = a_{21} = 0$, since

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ -a_{21} & -a_{22} \end{pmatrix} = \begin{pmatrix} a_{11} & -a_{12} \\ a_{21} & -a_{22} \end{pmatrix}$$

And setting $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ implies $a_{11} = a_{22} = a$, since

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = \begin{pmatrix} 0 & a_{11} \\ a_{22} & 0 \end{pmatrix} \stackrel{!}{=} \begin{pmatrix} a_{11} & 0 \\ 0 & -a_{22} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

It follows that $A = \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}$ and with $\det(A) = 1$, one gets $a^2 = 1$ or $a = \pm 1$.

We do not show explicitly, that ρ is *surjective*, i.e. that it hits all elements of the proper, orthochronous Lorentz group $SO^+(1,3)$.

Result: We can derive the group isomorphism

$$SO^+(1,3) = Sl(2,\mathbb{C}) / \{\pm\mathbb{1}\}$$

$Sl(2,\mathbb{C}) = S^3 \times \mathbb{R}^3$ as a manifold is simply connected.
$Sl(2,\mathbb{C})$ is the two-sheeted covering-group of the proper, orthochronous Lorentz group $SO^+(1,3)$.

Physical application: The "smallest" objects that can be transformed linearly by the "Lorentz group" in a non-trivial way, are 2-component, complex vectors

$$\chi = \begin{pmatrix} \chi_1 \\ \chi_2 \end{pmatrix} \quad \text{with } \chi_{1,2} \in \mathbb{C}$$

These are called *Weyl-spinors*. They describe massless, relativistic spin-$\frac{1}{2}$ particles, e.g. neutrinos.
Massive, relativistic spin-$\frac{1}{2}$ particles, e.g. electrons, muons, τ-leptons or quarks, are described by 4-component *Dirac-spinors* (bispinors)

$$\psi_{\text{chiral}} = \begin{pmatrix} \chi \\ \phi \end{pmatrix}$$

They are transformed by $A \in Sl(2, \mathbb{C})$ as follows

$$\begin{aligned} \chi &\mapsto A\chi \\ \phi &\mapsto A^*\phi \end{aligned}$$

In standard representation Dirac-spinors are given by

$$\psi_{\text{Dirac}} = \frac{1}{\sqrt{2}} \begin{pmatrix} \mathbb{1} & \mathbb{1} \\ -\mathbb{1} & \mathbb{1} \end{pmatrix} \psi_{\text{chiral}}$$

Example:

$$\begin{aligned} A &= \begin{pmatrix} \zeta & 0 \\ 0 & \zeta^{-1} \end{pmatrix}, \quad \zeta = \left(\frac{1-\beta}{1+\beta}\right)^{1/4} \in \mathbb{R} \\ AXA^\dagger &= \begin{pmatrix} \zeta & 0 \\ 0 & \zeta^{-1} \end{pmatrix} \begin{pmatrix} x^0 + x^3 & x^1 - \mathrm{i}x^2 \\ x^1 + \mathrm{i}x^2 & x^0 - x^3 \end{pmatrix} \begin{pmatrix} \zeta & 0 \\ 0 & \zeta^{-1} \end{pmatrix} \\ &= \begin{pmatrix} \zeta^2(x^0 + x^3) & x^1 - \mathrm{i}x^2 \\ x^1 + \mathrm{i}x^2 & \zeta^{-2}(x^0 - x^3) \end{pmatrix} = \begin{pmatrix} x'^0 + x'^3 & x'^1 - \mathrm{i}x'^2 \\ x'^1 + \mathrm{i}x'^2 & x'^0 - x'^3 \end{pmatrix} \end{aligned}$$

This is a boost in z-direction.

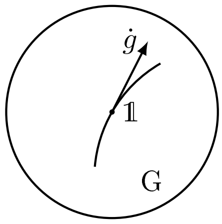
$$\begin{aligned} x'^0 &= \frac{1}{2}\big(\zeta^2(x^0 + x^3) + \zeta^{-2}(x^0 - x^3)\big) \\ &= \frac{\zeta^2 + \zeta^{-2}}{2} x^0 + \frac{\zeta^2 - \zeta^{-2}}{2} x^3 \\ &= \gamma(x^0 - \beta x^3) \\ x'^3 &= \frac{\zeta^2 - \zeta^{-2}}{2} x^0 + \frac{\zeta^2 + \zeta^{-2}}{2} x^3 \\ &= \gamma(x^3 - \beta x^0) \\ \beta &= -\frac{\zeta^2 - \zeta^{-2}}{\zeta^2 + \zeta^{-2}} = \frac{1 - \zeta^4}{1 + \zeta^4} = \frac{v}{c} \end{aligned}$$

Diagonal matrices $\begin{pmatrix} \zeta & 0 \\ 0 & \zeta^{-1} \end{pmatrix}$, $0 \neq \zeta \in \mathbb{R}$, generate boosts in the z-direction with velocity $\frac{v}{c} = \frac{1-\zeta^4}{1+\zeta^4}$.

2.6 Lie-algebras

We return to an arbitrary matrix–Lie-group G and consider differentiable *paths* through the *neutral element* e (unit matrix $\mathbb{1}_{n\times n}$)

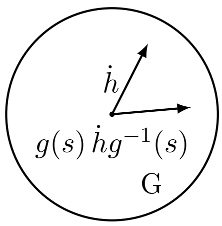
$$\begin{aligned}[-1,1] &\to G\\ t &\mapsto g(t) \quad \text{with } g(0)=\mathbb{1}\end{aligned}$$



The velocity vectors $\dot{g} = \frac{\mathrm{d}\,g(t)}{\mathrm{d}t}\Big|_{t=0}$ span the tangential space T_eG. $T_eG = LG$ is an n-dimensional, real vector space where $n = \dim(G)$.
Since G is a group (not just a manifold), LG possesses an additional *multiplicative structure*. Consider two paths $g(s)$ and $h(t)$ through the neutral element $\mathbb{1}$

$$\begin{aligned}\mathbb{1} &= g(s)\, g^{-1}(s) \Big| \left.\frac{\mathrm{d}}{\mathrm{d}s}\right|_{s=0}\\ 0 &= \mathbb{1}\dot{g}^{-1} + \dot{g}\mathbb{1}\\ \Rightarrow \dot{g}^{-1} &= -\dot{g} \in LG\end{aligned}$$

For any s, $g(s)\, h(t)\, g^{-1}(s)$ is a path through $\mathbb{1}$ with velocity vector $g(s)\,\dot{h}g^{-1}(s) \in LG$. This defines the *adjoint representation* of G on LG.



Next we differentiate with respect to s and evaluate at $s = 0$. The curve

$g(s)\,\dot{h}g^{-1}(s) \in LG$ has its velocity vector again in the vector space LG:

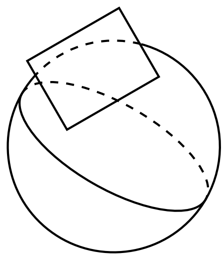
$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}s} g(s)\,\dot{h}g^{-1}(s)\bigg|_{s=0} &= \dot{g}\dot{h}\mathbb{1} + \mathbb{1}\dot{h}\dot{g}^{-1} \\ &= \dot{g}\dot{h} - \dot{h}\dot{g} = \left[\dot{g}, \dot{h}\right] \in LG\end{aligned}$$

The tangential bundle of a Lie-group is always trivial

$$TG = \bigcup_{g \in G} T_g G = G \times LG$$

whereas for the 2-sphere it's twisted with LG the Lie-algebra of G

$$TS^2 \neq S^2 \times \mathbb{R}^2$$



General definition:

A real/complex Lie-algebra $\mathcal{L}$ is a real/complex vector space with a bilinear product (*Lie-bracket*)

$$\begin{aligned}\mathcal{L} \times \mathcal{L} &\to \mathcal{L} \\ (X, Y) &\mapsto [X, Y]\end{aligned}$$

which has the following properties:

i) $[X, X] = 0$. This implies antisymmetry $[X, Y] = -[Y, X]$.

$$[X + Y, X + Y] = 0 = [X, X] + [X, Y] + [Y, X] + [Y, Y] \;\checkmark$$

ii) *Jacobi identity.* It serves as a substitute for the non-existing associativity

$$[[X, Y], Z] + [[Y, Z], X] + [[Z, X], Y] = 0$$

In an abstract (non-associative) Lie-algebra there exists *no* product $X \cdot Y$, such that $[X, Y] = X \cdot Y - Y \cdot X$.

But: One has the *theorem of I.D.Ado* which states:

> Any finite-dimensional (complex) Lie-algebra possesses a *faithful matrix representation.*

$$\begin{aligned}&\rho : \mathcal{L} \to \mathbb{C}^{N\times N}\\ &[X,Y] \mapsto \rho([X,Y]) = \underbrace{\rho(X)\cdot\rho(Y) - \rho(Y)\cdot\rho(X)}_{\text{matrix product}} = \underbrace{[\rho(X)\,,\rho(Y)]}_{\text{matrix commutator}}\end{aligned}$$

Simple example for a Lie-algebra: $\mathbb{R}^3$ with the cross product as the Lie-bracket $[\boldsymbol{a},\boldsymbol{b}] = \boldsymbol{a}\times\boldsymbol{b}$ defines a Lie-algebra. ✓

Most important result concerning Lie-algebras:

The so-called "simple" complex Lie-algebras can be completely classified into the four classical series (with associated Lie-groups) and into the five exceptions:

$$\begin{array}{ccccccccc} A_n & B_n & C_n & D_n & G_2 & F_4 & E_6 & E_7 & E_8\\ SU(n+1) & SO(2n+1) & Sp(n) & SO(2n) & & & & & \end{array}$$

A Lie-algebra is called *simple*, if it has no non-trivial invariant subalgebras (no trivial ideals). $I \subset \mathcal{L}$ is an ideal, if $[I,\mathcal{L}] \subset I$. The only ideals of a simple Lie-algebra $\mathcal{L}$ are the trivial ones, namely 0 and $\mathcal{L}$.

2.7 Lie-algebras of the classical groups

i) $\boldsymbol{u(n) = LU(n)}$:

$$\begin{aligned}&A(t)\,A^\dagger(t) = \mathbb{1} \overset{\text{differentiate}}{\longrightarrow} \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0}\\ &\dot{A} + \dot{A}^\dagger = 0\end{aligned}$$

$u(n)$ consists of *skew-hermitian matrices* $=$ i $\cdot$ hermitian matrices (if $B = B^\dagger$, then $\mathrm{i}B = \mathrm{i}B^\dagger = -(\mathrm{i}B)^\dagger$).

ii) $\boldsymbol{su(n) = LSU(n)}$: additional condition $\det(A(t)) = 1$

$$A(t) = \mathbb{1} + t\dot{A} + \mathcal{O}(t^2)$$

$$\begin{aligned}\det\left(\mathbb{1} + t\dot{A} + \dots\right) &= \det\begin{vmatrix} 1 + t\dot{a}_{11} & & t(*)\\ & \vdots & \\ t(*) & & 1 + t\dot{a}_{nn}\end{vmatrix} + \mathcal{O}(t^2)\\ &= (1 + t\dot{a}_{11})\cdots(1 + t\dot{a}_{nn}) + \mathcal{O}(t^2)\\ &= 1 + t(\dot{a}_{11} + \dots + \dot{a}_{nn}) + \mathcal{O}(t^2)\end{aligned}$$

Now $\frac{\mathrm{d}}{\mathrm{d}t}A(t)\big|_{t=0} = tr\left(\dot{A}\right) = 0$ (traceless).
$su(n)$ consists of traceless skew-hermitian $n\times n$-matrices $=$ i$\cdot$ traceless hermitian matrices.
A basis of the 3-dimensional real Lie-algebra $su(2)$ is given by

$$\begin{pmatrix} \mathrm{i} & 0 \\ 0 & -\mathrm{i} \end{pmatrix} = \mathrm{i}\sigma_3, \quad \begin{pmatrix} 0 & \mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix} = \mathrm{i}\sigma_1, \quad \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \mathrm{i}\sigma_2$$

The following commutator relations hold:

$$\begin{aligned} [\mathrm{i}\sigma_1, \mathrm{i}\sigma_2] &= -2\mathrm{i}\sigma_3 \\ [\mathrm{i}\sigma_2, \mathrm{i}\sigma_3] &= -2\mathrm{i}\sigma_1 \\ [\mathrm{i}\sigma_3, \mathrm{i}\sigma_1] &= -2\mathrm{i}\sigma_2 \\ [\mathrm{i}\sigma_i, \mathrm{i}\sigma_j] &= -2\sum_{k=1}^{3}\epsilon_{ijk}\mathrm{i}\sigma_k \end{aligned}$$

iii) $\boldsymbol{sl(2,\mathbb{C}) = LSl(2,\mathbb{C})}$: consists of traceless, complex 2×2-matrices $\dot{A}$:

$$\dot{A} = \frac{1}{2}\underbrace{(\dot{A}+\dot{A}^\dagger)}_{\text{hermitian}} + \frac{1}{2}\underbrace{(\dot{A}-\dot{A}^\dagger)}_{\text{skew-hermitian} = \mathrm{i}\cdot\text{ hermitian}}$$

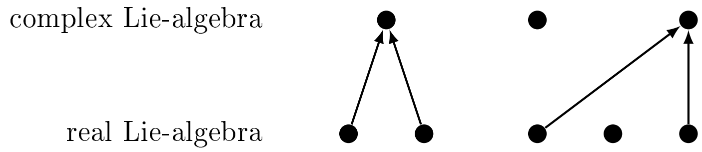
This is a complex linear combination of (skew-) hermitian 2×2-matrices.
By the means of complexification one finds

$$sl(2,\mathbb{C}) = \mathbb{C}\otimes su(2)$$

where $su(2)$ has a real basis $\mathrm{i}\sigma_1, \mathrm{i}\sigma_2, \mathrm{i}\sigma_3$.
Moreover

$$sl(2,\mathbb{C}) = \mathbb{C}\otimes sl(2,\mathbb{R})$$

where $sl(2,\mathbb{R})$ has a real basis $\sigma_1, \mathrm{i}\sigma_2, \sigma_3$ and $su(2) \neq sl(2,\mathbb{R})$ are non-isomorphic real Lie-algebras.



Observation: By the process of complexification information can get lost.

iv) $\boldsymbol{so(n) = LSO(n)}$:

$$A(t)\,A^t(t) = \mathbb{1} \xrightarrow{\text{differentiate}} \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0}$$

$$\dot{A} + \dot{A}^t = 0$$

$so(n)$ consists of skew-symmetric real $n \times n$-matrices.
A basis of $so(3)$ is given by:

$$L_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad L_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad L_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

and fulfills the commutator relations

$$[L_1, L_2] = L_3$$

$$[L_i, L_j] = \sum_{k=1}^{3} \epsilon_{ijk} L_k$$

The correspondence $L_j \leftrightarrow -\frac{\mathrm{i}}{2}\sigma_j$ gives an isomorphism of real Lie-algebras

$$so(3) = su(2)$$

but remember for the Lie-groups:

$$SO(3) = SU(2)\,/\mathbb{Z}_2$$

A basis of $so(4) = LSO(4)$ is given by:

$$L_1 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad L_2 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad L_3 = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

where we took the L_i from $so(3)$ and filled the 4th row and the 4th column with zeros. Furthermore we choose

$$M_1 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}, \quad M_3 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}$$

with entries $\neq 0$ only in the 4th row and 4th column.
The following commutation relations hold:

$$\begin{aligned} [L_j, L_k] &= \epsilon_{jkl} L_l \\ [M_j, M_k] &= \epsilon_{jkl} L_l \\ [M_j, L_k] &= \epsilon_{jkl} M_l \end{aligned}$$

Introduce half sums and differences:

$$J_j = \frac{1}{2}\left(L_j + M_j\right), \quad K_j = \frac{1}{2}\left(L_j - M_j\right)$$

$$\begin{aligned} [J_j, J_k] &= \frac{1}{4}\left[L_j + M_j, L_k + M_k\right] = \frac{1}{4}\epsilon_{jkl}\left(L_l + M_l + L_l + M_l\right) = \epsilon_{jkl} J_l \\ [K_j, K_k] &= \frac{1}{4}\left[L_j - M_j, L_k - M_k\right] = \frac{1}{4}\epsilon_{jkl}\left(L_l - M_l + L_l - M_l\right) = \epsilon_{jkl} K_l \\ [J_j, K_k] &= \frac{1}{4}\left[L_j + M_j, L_k - M_k\right] = \frac{1}{4}\epsilon_{jkl}\left(L_l + M_l - L_l - M_l\right) = 0 \end{aligned}$$

The two triples of basis elements $\{J_1, J_2, J_3\}$ and $\{K_1, K_2, K_3\}$ separately form $so(3)$ Lie-algebras. These commute with each other.

$$so(4) = so(3) \oplus so(3)$$

The Lie-algebra $so(4)$ is the *direct sum* of two $so(3)$ Lie-algebras.

Direct Sum of two Lie-algebras $\mathcal{L}_1$ and $\mathcal{L}_2$:

$$\left[\left(X_1, X_2\right), \left(Y_1, Y_2\right)\right] = (\underbrace{[X_1, Y_1]}_{\in \mathcal{L}_1}, \underbrace{[X_2, Y_2]}_{\in \mathcal{L}_2})$$

with $X_1, Y_1 \in \mathcal{L}_1$ and $X_2, Y_2 \in \mathcal{L}_2$.
Written in terms of sums:

$$[X_1 + X_2, Y_1 + Y_2] = [X_1, Y_1] + [X_2, Y_2]$$

i.e. all mixed commutators vanish: $[\mathcal{L}_1, \mathcal{L}_2] = 0$

Lie-algebra of the Lorentz group $so(1,3) = LSO^+(1,3)$:

$$\Lambda(t)^t \, g\Lambda(t) = g \stackrel{\text{differentiate}}{\longrightarrow} \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0}$$
$$\Rightarrow \dot{\Lambda}^t g + g\dot{\Lambda} = 0$$

$$\dot{\Lambda}^t = -g\dot{\Lambda}g$$

Basis of $so(1,3)$ consisting of 6 elements:

$$L_1 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad L_2 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}, \quad L_3 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$
$$B_1 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & & & \\ 0 & & 0 & \\ 0 & & & \end{pmatrix}, \quad B_2 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & & & \\ 1 & & 0 & \\ 0 & & & \end{pmatrix}, \quad B_3 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & & & \\ 0 & & 0 & \\ 0 & & & \end{pmatrix}$$

where the L_i are antisymmetric matrices with the 0th row and 0th column filled with zeros and the B_i are symmetric matrices.
Remember Lorentz boost in x-direction:

$$x'^0 = \gamma(x^0 - \beta x^1), \quad x'^1 = \gamma(x^1 - \beta x^0)$$

$$\frac{\partial}{\partial \beta} \left.\begin{pmatrix} \gamma & -\beta\gamma \\ -\beta\gamma & \gamma \end{pmatrix}\right|_{\beta=0} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$$

$$\gamma = \frac{1}{\sqrt{1-\beta^2}}$$

The following commutation relations hold:

$$\begin{aligned}[][L_j, L_k] &= \epsilon_{jkl} L_k \\ [L_j, B_k] &= \epsilon_{jkl} B_l \\ [B_j, B_k] &= -\epsilon_{jkl} L_l\end{aligned}$$

The minus sign in the last relation is the *only* difference to $so(4)$. It is sometimes discussed away by the (forbidden) identification $M_j = \mathrm{i}B_j$. But this is not allowed, because $\mathrm{i}B_j$ is *not* an element of the real Lie-algebra $so(1,3)$.
The commutation relations tell us that the B_js behave as $\mathrm{i}L_j$s.
$\Rightarrow$ The Lie-algebra of the Lorentz group is isomorphic to $so(3) \oplus \mathrm{i}so(3)$ which gives the *complexification* of $so(3)$:

$$\begin{aligned} so(1,3) &= so(3) \oplus \mathrm{i}so(3) = \mathbb{C} \otimes so(3) \\ &= \mathbb{C} \otimes su(2) = sl(2,\mathbb{C}) \end{aligned}$$

On the group level we had

$$SO^+(1,3) = Sl(2,\mathbb{C})\,/\mathbb{Z}_2$$

If we complexify $so(1,3)$, we find that it is the direct sum of two complexified $so(3)$s:

$$\mathbb{C} \otimes so(1,3) = \mathbb{C} \otimes so(3) \oplus \mathbb{C} \otimes so(3)$$

Of great importance in particle physics is the group $SU(3)$:

It describes flavor and color of quarks and the observed multiplet structure of hadrons.
The Lie-algebra $su(3) = LSU(3)$ consists of anti-hermitian, traceless 3×3-matrices. These are $\mathrm{i} \cdot$ hermitian, traceless matrices. A basis of $su(3)$ is given by: $\mathrm{i}\lambda_j$, with $j = 1, \ldots, 8$, the so-called *Gell-Mann-matrices*

$$\lambda_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \lambda_2 = \begin{pmatrix} 0 & -\mathrm{i} & 0 \\ \mathrm{i} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \lambda_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\lambda_4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \qquad \lambda_5 = \begin{pmatrix} 0 & 0 & -\mathrm{i} \\ 0 & 0 & 0 \\ \mathrm{i} & 0 & 0 \end{pmatrix}, \qquad \lambda_6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\lambda_7 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -\mathrm{i} \\ 0 & \mathrm{i} & 0 \end{pmatrix}, \qquad \lambda_8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}$$

The following commutation relations hold:

$$[\lambda_a, \lambda_b] = \mathrm{i} \sum_{c=1}^{8} f_{abc} \lambda_c$$

The real *structure constants* f_{abc} are totally antisymmetric:

$$f_{abc} = -f_{bac} = f_{bca}$$

Their non-vanishing values are:

abc	123	147	156	246	257	345	367	458	678
f_{abc}	2	1	-1	1	1	1	-1	$\sqrt{3}$	$\sqrt{3}$

Moreover the Gell-Mann-matrices are *orthonormal* with respect to the trace:

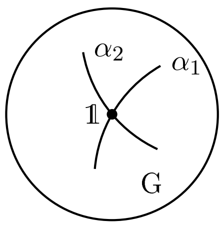
$$tr\,(\lambda_a \lambda_b) = 2\delta_{ab}, \quad a, b = 1, \ldots, 8$$

The matrix exponential function:

From the matrix Lie-algebra LG the matrix Lie group G can be reconstructed locally (in a neighborhood of the identity) through the *matrix exponential function*:

$$\exp\left(\sum_{j=1}^{n} \alpha_j X_j\right) = \sum_{\nu=0}^{\infty} \frac{1}{\nu!} \underbrace{\left(\sum_{j=1}^{n} \alpha_j X_j\right)^{\nu}}_{\text{ordinary matrix multiplication}}$$

where the X_j, with $j = 1, \ldots, n$, form a basis of LG and the $\alpha_j \in \mathbb{R}$, with $j = 1, \ldots, n$, are real parameters.



Consider the partial derivatives:

$$\frac{\partial}{\partial \alpha_k} \exp\left(\sum_{j=1}^{n} \alpha_j X_j\right)\Bigg|_{\alpha_j=0} = \frac{\partial}{\partial \alpha_k} \exp\left(\alpha_k X_k\right) = \frac{\partial}{\partial \alpha_k}\left(\mathbb{1} + \alpha_k X_k + \mathcal{O}(\alpha^2)\right) = X_k$$

Example: rotations in exponential representation

$$\begin{aligned}
\exp\left(\varphi L_3\right) &= \mathbb{1} + \sum_{\nu=1}^{\infty} \frac{1}{\nu!} \varphi^{\nu} L_3^{\nu} \\
&= \begin{pmatrix} \sum\limits_{k=0}^{\infty} \frac{(-1)^k \varphi^{2k}}{(2k)!} & -\sum\limits_{k=0}^{\infty} \frac{(-1)^k \varphi^{2k+1}}{(2k+1)!} & 0 \\ \sum\limits_{k=0}^{\infty} \frac{(-1)^k \varphi^{2k+1}}{(2k+1)!} & \sum\limits_{k=0}^{\infty} \frac{(-1)^k \varphi^{2k}}{(2k)!} & 0 \\ 0 & 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} \cos\varphi & -\sin\varphi & 0 \\ \sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{pmatrix}
\end{aligned}$$

where we have used

$$L_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad L_3^2 = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$
$$L_3^3 = -L_3, \quad L_3^4 = -L_3^2, \quad L_3^5 = L_3, \quad \ldots$$

and thus

$$L_3^{2k+1} = (-1)^k L_3, \qquad L_3^{2k} = (-1)^k \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad k \geq 1$$

Chapter 3

Representations

3.1 Representations of finite groups

Definition: A (complex) *representation* of a (finite) group G, is a homomorphism from G into the general linear group $Gl(n, \mathbb{C})$.

Representation D:

$$\begin{aligned} G &\to Gl(n, \mathbb{C}) \\ g &\mapsto D(g) \quad \text{(invertible } n \times n\text{-matrix)} \end{aligned}$$

$$\begin{aligned} D(g_1 g_2) &= D(g_1) \cdot D(g_2) \\ D(e) &= \mathbb{1} \\ D\big(g^{-1}\big) &= D(g)^{-1} \quad \text{(inverse matrix)} \end{aligned}$$

Through the representation D, the group G operates *linearly* on the vector space $V = \mathbb{C}^n$, where n the complex dimension of the representation D:

$$\begin{aligned} G \times V &\to V \\ (g, v) &\mapsto D(g)\, v \end{aligned}$$

Representations are of peculiar interest in *quantum mechanics*. If the Hamilton operator of a system possesses symmetries, then in general its energy eigenvalues are *degenerate*. In other words, there are several, linear independent, states (and thus a complete vector space of states) belonging to the same energy eigenvalue. The symmetry transformations transform these energetically degenerate states into each other. This gives rise to a representation of G.
The *representations* of G correspond to the possible *multiplets* of states, which *can be* degenerate due to the corresponding symmetry group G.
We want to classify the representations of G according to certain features. From a

given representation D, we immediately get a new representation $\tilde{D}$, by choosing a different set of basis vectors in $V = \mathbb{C}^n$:

$$|f_j\rangle = S\,|e_j\rangle$$

where S is an invertible matrix. Multiply

$$D(g)\,|e_j\rangle = \sum_{i=1}^{n} d_{ij}(g)\,|e_i\rangle$$

with S from the left and express the old basis $|e_j\rangle$ by the new one $|f_j\rangle$

$$\underbrace{SD(g)\,S^{-1}}_{\tilde{D}(g)}\,|f_j\rangle = \sum_{i=1}^{n} d_{ij}(g)\,|f_i\rangle$$

$$\tilde{D}(g) = SD(g)\,S^{-1}$$

Thus $\tilde{D}$ arises from D by a similarity transformation and therefore we will consider $D(g)$ and $\tilde{D}(g) = SD(g)\,S^{-1}$ as *equivalent* for *all* $S \in Gl(n, \mathbb{C})$.
We call a representation *reducible*, if there exists an *invariant subspace* U of $V = \mathbb{C}^n$.

$$D(g) = \left(\begin{array}{c:c} D_1(g) & * \\ \hdashline 0 & * \end{array}\right) \Big\updownarrow n \quad \updownarrow m \qquad \text{for all } g \in G$$

The subspace spanned by $\{|e_1\rangle, \ldots, |e_m\rangle\}$ is invariant.
If no such invariant subspace U exists (except for 0 and $V = \mathbb{C}^n$), the representation is called *irreducible*.
Furthermore a representation is called *fully reducible*, if it is equivalent to the direct sum of *irreducible* representations:

$$D(g) = \left(\begin{array}{c:c:c:c} D_1(g) & & & \\ \hdashline & D_2(g) & & \\ \hdashline & & \ddots & \\ \hdashline & & & D_r(g) \end{array}\right) \qquad \text{for all } g \in G$$

The matrix $D(g)$ possesses the *same* block diagonal form for all $g \in G$.
To obtain full reducibility one has to find an *invariant complement* U^C for each invariant subspace U:

$$V = U \oplus U^C$$

This is *always* possible for *finite, discrete* groups (and for compact Lie groups), but not in general.

Counterexample:

$$\mathbb{Z} \to Gl(2, \mathbb{C})$$
$$n \mapsto \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$$
$$\begin{pmatrix} 1 & n_1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & n_2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & n_1 + n_2 \\ 0 & 1 \end{pmatrix}$$

Nontrivial, invariant subspaces are 1-dimensional:

$$U = \mathbb{C} \begin{pmatrix} x \\ y \end{pmatrix}$$
$$\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$
$$\Rightarrow \frac{y}{x} = \frac{y}{x + ny}$$
$$\Rightarrow y = 0$$

$\Rightarrow$ only $U = \mathbb{C} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is an invariant subspace. U has no invariant complement.

Each representation of a *finite* group G is *equivalent* to a *unitary* representation.

Proof: Let $\langle \, | \, \rangle_0$ be some hermitian scalar product on $V = \mathbb{C}^n$. We construct a new one:

$$\langle v | \, w \rangle = \frac{1}{|G|} \sum_{h \in G} \langle \, D(h) \, v | \, D(h) \, w \rangle_0$$

The crucial property of *positivity* is conserved, since $\langle v | \, v \rangle_0 > 0$ implies $\langle v | \, v \rangle > 0$. Now one has

$$\begin{aligned}
\langle \, D(g) \, v | \, D(g) \, w \rangle &= \frac{1}{|G|} \sum_{h \in G} \langle \, D(h) \, D(g) \, v | \, D(h) \, D(g) \, w \rangle_0 \\
&= \frac{1}{|G|} \sum_{h \in G} \left\langle D(\underbrace{hg}_{h'}) \, v \middle| D(\underbrace{hg}_{h'}) \, w \right\rangle_0 \\
&= \frac{1}{|G|} \sum_{h' \in G} \left\langle \, D(h') \, v \middle| \, D(h') \, w \right\rangle_0 \\
&= \langle v | \, w \rangle
\end{aligned}$$

Where we have used that hg runs over the whole group G, if h does it.

Theorem: Each representation of a finite group G can be decomposed into a direct sum of unitary, irreducible representations. The orthogonal complement $U^\perp$ (with respect to the invariant scalar product $\langle\ |\ \rangle$) of any invariant subspace U is also invariant.

Proof: $\langle v|\, w\rangle = 0$ for $v \in U$, $w \in U^\perp$ and $D(g)\, v \in U$ for all $g \in G$

$$0 = \langle \underbrace{D(g)\, v}_{\in U} |w\rangle = \left\langle v \middle| D^\dagger(g)\, w \right\rangle = \langle v|\, D^{-1}(g)\, w\rangle = \langle v| \underbrace{D\big(g^{-1}\big)\, w}_{\in U^\perp} \rangle$$

$$\Rightarrow D\big(g^{-1}\big)\, w \in U^\perp$$

By the process of decomposing into invariant subspaced and their orthogonal complements, one arrives at full reducibility:

$$D(g) = \left(\begin{array}{c:c:c:c} D_1(g) & & & \\ \hdashline & D_2(g) & & \\ \hdashline & & \ddots & \\ \hdashline & & & D_r(g) \end{array}\right), \quad \text{with } D_j(g) \in U(n_j)$$

Goal: Identification of all unitary, irreducible and pairwise inequivalent representations (*irreps*) D_j, together with their dimensions n_j. All the following is a consequence of *Schur's lemma*:

i) Let D_1 and D_2 be two unitary, *inequivalent* irreps. of G. Suppose that for all $g \in G$ holds: $D_2(g)\, A = AD_1(g)$, then $A = 0$.

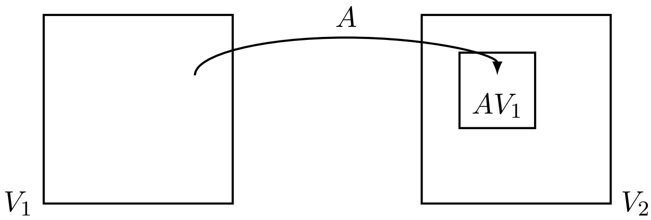
$$\begin{array}{ccc} V_1 & \xrightarrow{D_1(g)} & V_1 \\ A\downarrow & & \downarrow A \\ V_2 & \xrightarrow[D_2(g)]{} & V_2 \end{array}$$

ii) Assume $D(g)\, A = AD(g)$ holds for all $g \in G$, where $D(g)$ is an irrep., then $A = \lambda \cdot \mathbb{1}$, with A being a $n_2 \times n_1$ matrix.

Proof:

of i) Note that A is not necessarily a square-matrix. A describes a linear map from $V_1 = \mathbb{C}^{n_1}$ into $V_2 = \mathbb{C}^{n_2}$.

- $\boldsymbol{n_1 < n_2}$:

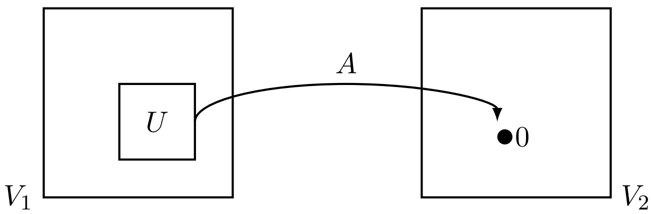


Since

$$\underbrace{D_2(g)\,A\,|v\rangle}_{\in AV_1} = A\,\underbrace{\underbrace{D_1(g)\,|v\rangle}_{\in V_1}}_{\in AV_1} \quad \text{with } |v\rangle \in V_1$$

one sees that AV_1 is an invariant subspace of V_2. Now the representation $D_2(g)$ is *irreducible*, therefore $AV_1 = 0$ or all of V_2. The latter alternative is excluded for dimensional reasons $\Rightarrow A = 0$ (zero map).

- $\boldsymbol{n_1 > n_2}$:



Consider the subspace $U \subset V_1$ with $AU = 0$ ($U = \text{kernel}\,(A)$) and $|u\rangle \in U$. Since

$$0 = D_2(g)\,\underbrace{A\,|u\rangle}_{0} = A\,\underbrace{D_1(g)\,|u\rangle}_{\in U}$$

one sees that U is an invariant subspace of V_1. Since $D_1(g)$ is an irrep., one has $U = V_1$ or 0. Again, it can't be the latter alternative for dimensional reasons $\Rightarrow A = 0$.

- $\boldsymbol{n_1 = n_2}$: The remaining alternatives are $AV_1 = V_2$ and $U = 0$. They imply that A is invertible. But then $D_1(g)$ and $D_2(g) = AD_1(g)\,A^{-1}$ would be equivalent to each other, in contradiction to our assumption $\Rightarrow A = 0$.

of ii) A has (at least) one eigenvector $|v\rangle \neq 0$, such that $A|v\rangle = \lambda|v\rangle$ (this holds, since we discuss everything with respect to the *complex numbers*, were the characteristic polynomial always has at least one root). Since

$$D(g)\,A\,|v\rangle = \lambda D(g)\,|v\rangle = AD(g)\,|v\rangle$$

the eigenspace, corresponding to the eigenvalue λ, is an invariant subspace.
However, since $D(g)$ is an irrep., this eigenspace has to be the whole vector space $V = \mathbb{C}^n$, on which $D(g)$ operates.

$$A = \lambda \cdot \mathbb{1}$$

Corollary: Every irrep. of a finite, abelian group G is one-dimensional, because for all $g, h \in G$:

$$\begin{aligned} gh &= hg \\ D(g)\,D(h) &= D(h)\,D(g) \\ \overset{\text{Schur's lemma}}{\Rightarrow} D(h) &= \lambda(h) \cdot \mathbb{1} \quad \text{with } \lambda(h) \in \mathbb{C} \\ \Rightarrow |\lambda(h)\,| &= 1 \quad \text{(because of unitarity)} \end{aligned}$$

Irreps. of $\mathbb{Z}_n = \{0, 1, \dots, n-1\}$:

$$D_r : k \bmod n \mapsto \exp\left\{\frac{2\pi\mathrm{i}}{n} k \cdot r\right\} = \zeta_n^{k\cdot r} \quad \text{for } r = 0, 1, \dots, n-1$$

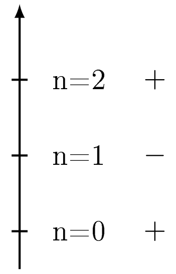
Here $\zeta_n = \exp\left\{\frac{2\pi\mathrm{i}}{n}\right\}$ is the primitive n-th root of unity and

$$\begin{aligned} D_r &= D_1 \cdots D_1 = D_1^r \\ D_n &= D_0 = 1 \end{aligned}$$

these representations themselves form a cyclic group of order n.
Consequently from *abelian* symmetry groups we cannot expect degenerate multiplets.

Example: Parity $\mathcal{P} = \mathbb{Z}_2$ for the harmonic oscillator

$$\begin{aligned} H\,|n\rangle &= \hbar\omega\left(n + \frac{1}{2}\right)|n\rangle \\ \mathcal{P}\,|n\rangle &= (-1)^n\,|n\rangle \end{aligned}$$



3.2 The fundamental orthogonality theorem

Let $D^{(\mu)}(g) : V_\mu \to V_\mu$ and $D^{(\nu)}(g) : V_\nu \to V_\nu$ be two inequivalent irreps. of the finite group G. Let A be an arbitrary linear map from V_ν to V_μ. Now we construct another one:

$$B = \sum_{g \in G} D^{(\mu)}(g)\, A D^{(\nu)}\big(g^{-1}\big)$$

Then for every $h \in G$ one has:

$$\begin{aligned}
D^{(\mu)}(h)\, B &= \sum_{g \in G} D^{(\mu)}(h)\, D^{(\mu)}(g)\, A D^{(\nu)}\big(g^{-1}\big) \\
&= \sum_{g \in G} D^{(\mu)} \underbrace{(hg)}_{g'}\, A\, \underbrace{D^{(\nu)}\big(g^{-1}\big)\, D^{(\nu)}\big(h^{-1}\big)}_{D^{(\nu)}((hg)^{-1})}\, D^{(\nu)}(h) \\
&= \sum_{g' \in G} D^{(\mu)}\big(g'\big)\, A D^{(\nu)}\big((g')^{-1}\big)\, D^{(\nu)}(h) \\
&= B D^{(\nu)}(h)
\end{aligned}$$

This means that B fulfills all assumptions of Schur's lemma. Hence, since $D^{(\mu)}(g)$ and $D^{(\nu)}(g)$ are inequivalent, one gets:

$$B = \sum_{g \in G} D^{(\mu)}(g)\, A D^{(\nu)}\big(g^{-1}\big) = \lambda_A^{(\mu)} \delta^{\mu\nu} \cdot \mathbb{1} \qquad (\star)$$

We choose the matrix A such that all of its entries are zero, except for one: $A_{rs} = 1$. Then the ij-th matrix element of equation $(\star)$ reads:

$$\sum_{g \in G} D^{(\mu)}_{ir}(g)\, D^{(\nu)}_{sj}\big(g^{-1}\big) = \lambda^{(\mu)}_{rs} \delta_{\mu\nu} \delta_{ij}$$

In order to determine the constant $\lambda_{rs}^{(\mu)}$, we set $\mu = \nu$ and take the trace $\sum\limits_{i=j}$:

$$\sum_{g\in G} \big(\underbrace{D^{(\nu)}\big(g^{-1}\big)\, D^{(\mu)}(g)}_{D^{(\mu)}(e)=\mathbb{1}}\big)_{rs} = \underbrace{n_\mu}_{\dim(V^\mu)} \lambda_{rs}^{(\mu)} = |G|\delta_{rs}$$

$$\Rightarrow \lambda_{rs}^{(\mu)} = \frac{|G|\delta_{rs}}{n_\mu}$$

Furthermore, these representations are unitary:

$$D^{(\nu)}\big(g^{-1}\big) = D^{(\nu)}(g)^{-1} = \Big(D^{(\nu)}(g)\Big)^{t*}$$

Finally, the fundamental orthogonality relation states

$$\sum_{g\in G} D_{ir}^{(\mu)}(g)\, D_{js}^{(\nu)*}(g) = \frac{|G|}{n_\mu}\delta^{\mu\nu}\delta_{ij}\delta_{rs} \tag{3.1}$$

where n_μ is the dimension of the corresponding representation $D^{(\mu)}(g)$.
The matrix entries of irreps. form an *orthogonal set* of functions on the finite group G. All these functions together form the vector space $\mathbb{C}^{|G|}$.
One deduces the inequality

$$\sum_\mu n_\mu^2 \le |G|$$

and equality will be proven later.

3.3 Characters

Definition: Let $D(g)$ be a representation of the finite group G. The function

$$\chi(g) = tr\,(D(g))$$

is called its *character*.

Characters of equivalent representations are equal

$$tr\left(SD(g)\, S^{-1}\right) = tr\,(D(g)) = \chi(g)$$

Characters are constant on conjugation classes

$$\begin{aligned}\chi\big(hgh^{-1}\big) &= tr\left(D\big(hgh^{-1}\big)\right) = tr\left(D(h)\, D(g)\, D(h)^{-1}\right)\\ &= tr\,(D(g)) = \chi(g)\end{aligned}$$

Theorem:

The characters of irreps. form a (complete) orthogonal set of functions on the conjugation classes of the group G.

Proof: Taking the sums $\sum\limits_{i=r=1}^{n_\mu}$ and $\sum\limits_{j=s=1}^{n_\mu}$ in the fundamental orthogonality relation (3.1), using $\sum\limits_{r=s=1}^{n_\mu} \delta_{rs} = n_\mu$, yields

$$\frac{1}{|G|} \sum_{g \in G} \chi^{(\mu)}(g)\, \chi^{(\nu)*}(g) = \delta^{\mu\nu}$$

or

$$\frac{1}{|G|} \sum_i k_i \chi_i^{(\mu)} \chi_i^{(\nu)*} = \delta^{\mu\nu}$$

where there are k_i elements in the i-th conjugation class of G.

Corollary: The number of the inequivalent irreps. of a finite group G is smaller or *equal* to the number of the conjugation classes.

Now we can decompose an arbitrary representation $D(g)$ into the direct sum of irreducible ones:

$$D(g) = \bigoplus_\mu D^{(\mu)}(g)$$

where some μ may occur several times.
By taking the trace, one finds for the characters:

$$\chi(g) = tr\,(D(g)) = \sum_\mu a_\mu \chi^{(\mu)}(g)$$

where the sum only runs over the irreps. and $a_\mu \in \mathbb{N}_0$ is the multiplicity of $D^{(\mu)}(g)$ in $D(g)$. Multiplying this expression with $\frac{1}{|G|} \sum\limits_{g \in G} \chi^{(\nu)*}(g)$ and relabeling $\nu \to \mu$ afterwards, yields

$$a_\mu = \frac{1}{|G|} \sum_{g \in G} \chi(g)\, \chi^{(\mu)*}(g)$$

A representation is uniquely determined by its characters.
A representation is *irreducible* if, and only if

$$\frac{1}{|G|}\sum_{g\in G}|\chi(g)|^2 = 1$$

holds.

Proof:

$$\begin{aligned}\frac{1}{|G|}\sum_{g\in G}|\chi(g)|^2 &= \sum_{\mu,\nu} a_\mu a_\nu \underbrace{\frac{1}{|G|}\sum_{g\in G}\chi^{(\mu)}(g)\,\chi^{(\nu)*}(g)}_{\delta_{\mu\nu}} \\ &= \sum_{\mu} a_\mu^2\end{aligned}$$

This equals 1 if $a_\mu = 1$ for one particular μ and 0 else.

The decomposition into irreps. is now applied to the so-called *regular* representations. The underlying vector space V_R is spanned by basis vectors that are labeled by the group elements $h \in G$:

$$R(g)\,|h\rangle = |gh\rangle\,, \quad \dim\,(V_R) = |G|$$

$$R(e) = \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{pmatrix} \qquad R(g\neq e) = \begin{pmatrix} 0 & & & & 1 \\ 1 & 0 & & & 0 \\ \vdots & & 1 & \ddots & \vdots \\ & & & & 0 \end{pmatrix}$$

Here, the latter one is a permutation matrix with 0 on the diagonal and exactly one 1 in each row and column.
The character of the regular representation is

$$\chi_R(g) = \begin{cases} |G|, & g = e \\ 0, & g \neq e \end{cases}$$

The multiplicities are given by

$$\begin{aligned} a_\mu^{(R)} &= \frac{1}{|G|}\sum_{g\in G}\chi_R(g)\,\chi^{(\mu)*}(g) = \chi^{(\mu)*}(e) = n_\mu \\ \dim\,(V_R) &= \sum_\mu a_\mu^{(R)} n_\mu \end{aligned}$$

$$|G| = \sum_{\mu} n_{\mu}^2$$

Finally, we show that the number of inequivalent irreps. is *equal* to the number of conjugation classes. Let $\alpha : G \mapsto G$ be a class-function, that is orthogonal on *all* irreducible characters. $\sum_{g \in G} \alpha(g)\, \chi^{(\nu)*}(g) = 0$ for all ν. We show that $\alpha(g) = 0$ for all $g \in G$.
Let $D^{(\nu)}(g)$ be an irrep. of G and consider

$$\begin{aligned}
\left(\sum_{g \in G} \alpha^*(g)\, D^{(\nu)}(g) \right) D^{(\nu)}(h) &= \sum_{g' \in G} \alpha^*\left(hg'h^{-1}\right) D^{(\nu)}(h)\, D^{(\nu)}\left(g'\right) \\
&= D(h) \sum_{g' \in G} \alpha^*\left(g'\right) D^{(\nu)}\left(g'\right)
\end{aligned}$$

where we have set $g = hg'h^{-1}$.
According to Schur's lemma (ii):

$$\sum_{g \in G} \alpha^*(g)\, D^{(\nu)}(g) = \lambda^{(\nu)} \cdot \mathbb{1}$$

Taking the trace yields

$$\lambda^{(\nu)} n_{\nu} = \sum_{g \in G} \alpha^*(g)\, \chi^{(\nu)}(g) = 0$$

$$\lambda^{(\nu)} = 0$$

One finds for all irreps. that

$$\sum_{g \in G} \alpha^*(g)\, D^{(\nu)}(g) = 0$$

and indeed this holds for *all representations*:

$$\sum_{g \in G} \alpha^*(g)\, D(g) = 0$$

In the case of the regular representation, the matrices $D_R(g)$ are however linearly independent and one concludes

$$\alpha^*(g) = 0 = \alpha(g)\,, \quad \text{for all } g \in G$$

Besides the irreducible characters there exist no further orthogonal class-functions different from zero.

Summary: A finite group G has exactly as many inequivalent irreps. $D^{(\mu)}(g)$ as it has conjugation classes. For their dimensions, the relation

$$\sum_\mu n_\mu^2 = |G|$$

holds and each dimension n_μ divides $|G|/|Z(G)|$.[1]

Character tables: In a character table the *columns* are labeled by the conjugation classes of the group G and the *rows* are labeled by the irreps., hence names for the corresponding vector spaces. The *entries* are the values of the respective character on a conjugation class. This number is a sum of *N-th roots of unity*

$$g^N = e \quad \Rightarrow D(g)^N = \mathbb{1}$$

where $D(g)$ is a unitary matrix and thus can be diagonalized.

The character table of S_3 is given by:

N S_3	1 id	3 (12)	2 (123)
trivial U dim $(U) = 1$	1	1	1
alt. U' dim $(U') = 1$	1	-1	1
V dim $(V) = 2$	2	0	-1

N gives the number of elements in a conjugation class.

$$\text{trivial repr.: } D(\sigma) = 1, \quad \sigma \in S_3$$
$$\text{alternating repr.: } D(\sigma) = sign(\sigma)$$

The 3-dimensional permutation representation

$$D\left(\sigma^{-1}\right)\begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix} = \begin{pmatrix} z_{\sigma(1)} \\ z_{\sigma(2)} \\ z_{\sigma(3)} \end{pmatrix}$$

[1] This theorem by Burnside and Tate will not be proven.

has an invariant subspace $\mathbb{C}\begin{pmatrix}1\\1\\1\end{pmatrix} = U$

$$\begin{aligned}\mathbb{C}^3 &= V \oplus U\\ \chi_{\mathbb{C}^3} &= (3,1,0)\\ \chi_V &= \chi_{\mathbb{C}^3} - \chi_U\\ &= (3,1,0) - (1,1,1) = (2,0,-1)\end{aligned}$$

We check the orthonormality relations

$$\begin{aligned}\langle \chi_U | \chi_U \rangle &= \frac{1}{6}(1+3+2) = 1\\ \langle \chi_V | \chi_V \rangle &= \frac{1}{6}(4+0+2\cdot 1^2) = 1\\ \langle \chi_U | \chi_{U'} \rangle &= \frac{1}{6}(1-3+2) = 0\\ \langle \chi_U | \chi_V \rangle &= \frac{1}{6}(2+0-2) = 0 = \langle \chi_{U'} | \chi_V \rangle\end{aligned}$$

The character table of S_4 is given by:

N $\lvert S_4\rvert = 24$	1 id	6 (12)	8 (123)	6 (1234)	3 (12)(34)
trivial U	1	1	1	1	1
alt. U'	1	-1	1	-1	1
stand. V	3	1	0	-1	-1
$V' = V \otimes U'$	3	-1	0	1	-1
myst. W	2	$a = 0$	$b = -1$	$c = 0$	$d = 2$

With

$$\begin{aligned}\mathbb{C}^4 &= V \oplus U\\ \chi_{\mathbb{C}^4} &= (4,2,1,0,0)\\ \chi_V &= (3,1,0,-1,-1)\end{aligned}$$

Using the sum rule for the dimensions gives $1^2 + 1^2 + 3^2 = 11$. For the remaining two irreps. of S_4 one thus has $n_4^2 + n_5^2 = 13$ with the only solution $n_4 = 3$, $n_5 = 2$. This gives us the representation $V' = V \otimes U'$. It is indeed irreducible, since its character is normalized to one

$$\langle \chi_{V'} | \chi_{V'} \rangle = \frac{1}{24}(9+6+0+6+3) = 1$$

To finally determine the character of the 2-dim. representation W, we exploit the orthogonality relations for the entries $(2, a, b, c, d)$. The determining equations are

$$\begin{aligned}
0 &= 2 + 6a + 8b + 6c + 3d \\
0 &= 2 - 6a + 8b - 6c + 3d \\
0 &= 6 + 6a - 6c - 3d \\
0 &= 6 - 6a + 6c - 3d
\end{aligned}$$

From the first two equations we find $a = -c$ and $8b + 3d + 2 = 0$. From the last two equations we find that that $a = c$ and $d = 2$. Thus $a = c = 0$, $d = 2$ and $b = -1$. We check the norm square:

$$\langle \chi_W | \chi_W \rangle = \frac{1}{24}(4 + 8 + 12) = 1$$

and find that W is irreducible as well. Effectively W is a representation of the factor group $S_3 = S_4/N_4$.
Note that S_4 is the rotational symmetry of the 3-dim. cube.

Comment:

There are 5 "angular momenta" on a cube.

Reduction of tensor products (Clebsch-Gordon decomposition):

Let the two irreps. $D_1(g) : V_1 \to V_1$ and $D_2(g) : V_2 \to V_2$ be given. From these one obtains a (in general reducible) representation on the *tensor product space* $V_1 \otimes V_2$ of dimension $\dim(V_1 \otimes V_2) = \dim(V_1) \cdot \dim(V_2)$

$$\begin{aligned}
D_1(g)\,|e_j\rangle &= \sum_i D_{1ij}(g)\,|e_i\rangle \quad (|e_i\rangle \text{ is a basis of } V_1) \\
D_2(g)\,|f_l\rangle &= \sum_k D_{1kl}(g)\,|f_k\rangle \quad (|f_k\rangle \text{ is a basis of } V_2)
\end{aligned}$$

$$D_1 \otimes D_2(g)\,(|e_j\rangle \otimes |f_l\rangle) = \sum_{i,k} (D_{1ij}(g)\, D_{2kl}(g))\,|e_i\rangle \otimes |f_k\rangle$$

where the tensor product "$\otimes$" is associative and bilinear.
For the characters of tensor product representations one finds

$$\chi_{D_1 \otimes D_2}(g) = \sum_{\substack{i=j \\ k=l}} D_{1ij}(g)\, D_{2kl}(g) = \chi_{D_1}(g) \cdot \chi_{D_2}(g)$$

The decomposition of $V_1 \otimes V_2$ into invariant and irreducible subspaces is the reduction of the tensor product, also called the *Clebsch-Gordon decomposition*

$$V_1 \otimes V_2 = \bigoplus_{\mu} V_\mu$$

This decomposition is done most easily with the help of characters

$$\chi_{D_1} \cdot \chi_{D_2} = \sum_{\mu} a_\mu \chi_\mu$$

Examples for the group S_3:

$$V \otimes U' = V$$

since $(2, 0, -1) \cdot (1, 1, 1) = (2, 0, -1)$

$$V \otimes V = U \oplus U' \oplus V$$

since $(2, 0, -1) \cdot (2, 0, -1) = (4, 0, 1) = (1, 1, 1) + (1, -1, 1) + (2, 0, -1)$

Representations of the product group $G \times H$:

Let V_μ be the irreps. of the finite group G and W_ν the irreps. of the finite group H. The irreps. of the direct product group $G \times H$ are the tensor products $V_\mu \otimes W_\nu$.

Proof: The character of the tensor product representation obeys

$$\chi_{V_\mu \otimes W_\nu}(g, h) = \chi^{(\mu)}(g) \cdot \chi^{(\nu)}(h)$$

and we calculate its norm square as:

$$\begin{aligned} \frac{1}{|G \times H|} \sum_{(g,h) \in G \times H} |\chi_{V_\mu \otimes W_\nu}(g, h)|^2 &= \frac{1}{|G||H|} \sum_{g \in G} \sum_{h \in H} |\chi^{(\mu)}(g)|^2 |\chi^{(\nu)}(h)|^2 \\ &= \frac{1}{|G|} \sum_{g \in G} |\chi^{(\mu)}(g)|^2 \cdot \frac{1}{|H|} \sum_{h \in H} |\chi^{(\nu)}(h)|^2 \\ &= 1 \cdot 1 = 1 \end{aligned}$$

Thus $V_\mu \otimes W_\nu$ is irreducible.

Moreover, the characters $\chi^{(\mu)}(g) \cdot \chi^{(\nu)}(h)$ form a complete set of orthonormal class-functions on $G \times H$, since

$$\frac{1}{|G \times H|} \sum_{(g,h) \in G \times H} \chi^{(\mu)}(g) \cdot \chi^{(\nu)}(h) \cdot \chi^{(\mu')}(g) \cdot \chi^{(\nu')}(h) = \delta^{\mu\mu'} \delta^{\nu\nu'}$$

Corollary: A finite abelian group

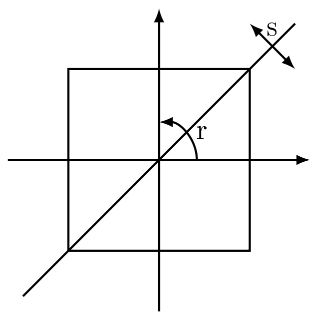
$$G = \mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2} \times \cdots \times \mathbb{Z}_{m_s}$$

where m_1 divides m_2 divides m_3 ... has the following irreducible 1-dim. representation

$$(k_1 \bmod m_1, \ldots, k_s \bmod m_s) \mapsto \exp\left(2\pi\mathrm{i} \sum_{j=1}^{s} \frac{k_j \cdot r_j}{m_j}\right)$$

where $0 \leq r_j \leq m_j - 1$.

Character table of the dieder group D_4: (symmetry group of the square)



$$D_4 = \{e, r, r^2, r^3, s, rs, r^2 s, r^3 s\}$$

There are eight elements and the following relations hold

$$r^4 = e, \quad s^2 = e, \quad sr = r^3 s = r^{-1} s$$

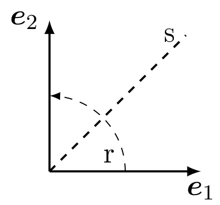
The conjugation classes are given by

$$\{e\}, \quad \{r, r^a srsr^{-a} = r^3\}, \quad \{r^2, r^a sr^2 sr^{-a} = r^6 = r^2\}$$
$$\{s, r^a sr^{-a} = r^{2a} s = r^2 s\}, \quad \{rs, r^a rsr^{-a} = r^{2a+1} s = r^3 s\}$$

N D_4	1 $\{e\}$	1 $\{r^2\}$	2 $\{r, r^3\}$	2 $\{s, r^2s\}$	2 $\{rs, r^3s\}$
trivial U	1	1	1	1	1
stand. V $\dim(V) = 2$	2	-2	0	0	0
U'	1	1	-1	1	-1
U''	1	1	1	-1	-1
$U''' = U' \otimes U''$	1	1	-1	-1	1

The 2-dim. standard representation is given by

$$r\boldsymbol{e}_1 = \boldsymbol{e}_2, \qquad s\boldsymbol{e}_1 = \boldsymbol{e}_2$$
$$r\boldsymbol{e}_2 = -\boldsymbol{e}_1, \qquad s\boldsymbol{e}_2 = \boldsymbol{e}_1$$



The corresponding matrices read

$$r \mapsto \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad r^2 \mapsto \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$
$$s \mapsto \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad rs \mapsto \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

Since

$$\langle \chi_V | \chi_V \rangle = \frac{1}{8}(2^2 + (-2)^2) = 1$$

V is irreducible and since

$$8 = 1^2 + 2^2 + 1^2 + 1^2 + 1^2$$

there are three remaining 1-dim. irreps.
Furthermore, since $s^2 = e$ and $r^4 = e$, one will assign to s and r the values ± 1. Note that r and r^3 are in the same conjugation class and thus one must assign

the *same* value to both of them which excludes the assignments $r \mapsto \pm \mathrm{i}$. Finally we find the tensorproduct decomposition

$$V \otimes V = U \oplus U' \oplus U'' \oplus U'''$$

where all the summands U, U', U'', U''' are one dimensional.

Character table of A_4: (rotations of the tetrahedron)

N A_4	1 id	4 (123)	4 (132)	3 (12)(34)
trivial U	1	1	1	1
stand. V	3	0	0	-1
U' $\dim(U') = 1$	1	$a = \frac{-1+\mathrm{i}\sqrt{3}}{2}$	$b = \frac{-1-\mathrm{i}\sqrt{3}}{2}$	$c = 1$
U'' $\dim(U'') = 1$	1	$a = \frac{-1-\mathrm{i}\sqrt{3}}{2}$	$b = \frac{-1+\mathrm{i}\sqrt{3}}{2}$	1

$$\begin{aligned} \mathbb{C}^4 &= V \oplus U \\ \chi_{\mathbb{C}^4} &= (4, 1, 1, 0) \\ \chi_V &= (3, 0, 0, -1) \end{aligned}$$

We check the norm square

$$\langle \chi_V | \, \chi_V \rangle = \frac{1}{12}(9 + 3 \cdot (-1)^2) = 1$$

and find that V is irreducible. We use orthogonality and find the determining equations for the characters of the remaining 1-dim. irreps. U' and U''

$$\begin{aligned} 0 &= 1 + 4a + 4b + 3c \\ 0 &= 3 - 3c \end{aligned}$$

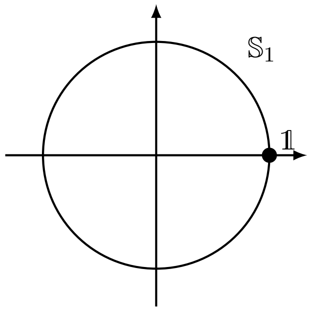
from which we read off $c = 1$ and $a + b = -1$. Moreover $|a| = 1 = |b|$, since the representations are unitary and 1-dim. From

$$(123)^3 = id = (132)^3$$

we conclude that a and b are *cubic roots of unity*. Furthermore, U' and U'' are complex conjugated to each other.

3.4 Representations of $U(1)$ or $SO(2)$

The unitary group $U(1) = \{z \in \mathbb{C},\ |z| = 1\}$ is *abelian*. Therefore all its *complex irreps.* are *one-dimensional* due to Schur's lemma II.



$$z = e^{\mathrm{i}\varphi} \mapsto z^n = e^{\mathrm{i}n\varphi}$$

where $n \in \mathbb{Z}$ is an *integer* (quantized charge).
The irreducible characters

$$\chi_n(z) = z^n = e^{\mathrm{i}n\varphi}$$

form a complete and orthonormal set of functions on the unit circle $S^1 = U(1)$.

$$\langle \chi_n | \chi_m \rangle = \underbrace{\frac{1}{2\pi} \int_0^{2\pi} d\varphi}_{\text{invariant integral for } U(1)} \underbrace{e^{\mathrm{i}(m-n)\varphi}}_{\chi_n^* \chi_m} = \delta_{mn}$$

3.5 Representations of $SU(2)$

We already had the Lie-algebra of $SU(2)$ in the form of the i-fold Pauli matrices:

$$\mathrm{i}\sigma_1 = \begin{pmatrix} 0 & \mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \quad \mathrm{i}\sigma_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \mathrm{i}\sigma_3 = \begin{pmatrix} \mathrm{i} & 0 \\ 0 & -\mathrm{i} \end{pmatrix}$$

together with their commutation relations:

$$[\mathrm{i}\sigma_j, \mathrm{i}\sigma_k] = -2 \sum_{l=1}^{3} \epsilon_{jkl} \mathrm{i}\sigma_l$$

It is easier to find matrix representations of the Lie-algebra $su(2)$. Using the matrix exponential, we obtain a representation of the Lie-group $SU(2)$. We are searching for finite dimensional and linear operators (i.e. matrices):

$$L_j = D\left(-\frac{\mathrm{i}}{2}\sigma_j\right)$$

which fulfill the following commutation relations:

$$[L_j, L_k] = L_j L_k - L_k L_j = \sum_{l=1}^{3} \epsilon_{jkl} L_l$$

Consider

$$\mathrm{ad}\ L_3(L_j) = [L_3, L_j]$$

which has the following 3×3 matrix representation:

$$\mathrm{ad}\ L_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

with eigenvalues $0, \pm\mathrm{i}$. In order to exploit the possibilities of diagonalization etc., it is beneficial to go over to the *complexified* Lie-algebra $\mathbb{C} \otimes su(2) = sl(2, \mathbb{C})$.
We now choose a *complex basis* of $sl(2, \mathbb{C})$:

$$J_3 = \mathrm{i}L_3, \quad J_\pm = \mathrm{i}L_1 \mp L_2$$

and obtain the following simpler commutation relations:

$$\begin{aligned} [J_3, J_\pm] &= \mathrm{i}\left(\mathrm{i}L_2 \pm L_1\right) = \pm\left(\mathrm{i}L_1 \mp L_2\right) = \pm J_\pm \\ [J_+, J_-] &= [\mathrm{i}L_1 - L_2, \mathrm{i}L_1 + L_2] = \mathrm{i}L_3 + \mathrm{i}L_3 = 2J_3 \end{aligned}$$

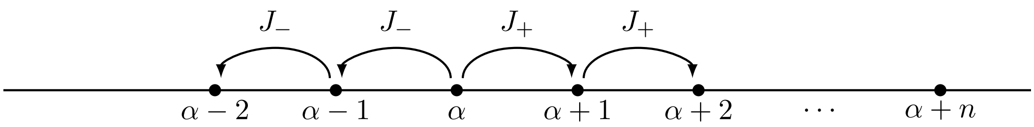
$$[J_3, J_\pm] = \pm J_\pm, \quad [J_+, J_-] = 2J_3$$

J_3 and $J_\pm$ represent the linear operators of an irrep. of $sl(2, \mathbb{C})$. Since we are working with complex numbers, J_3 has at least one eigenvector $|u\rangle$ with eigenvalue α:

$$J_3 |u\rangle = \alpha |u\rangle$$

Important observation: $J_\pm |u\rangle$ is either an eigenvector of J_3 with eigenvalue $\alpha \pm 1$ or the zero vector:

$$J_3 J_\pm |u\rangle = (J_\pm J_3 \pm J_\pm) |u\rangle = (\alpha \pm 1)\, J_\pm |u\rangle$$



Since the representation of $sl(2,\mathbb{C})$ has to be finite dimensional, there must exist an integer $n \in \mathbb{N}_0$ with $(J_+)^{n+1} |u\rangle = 0$, and $(J_+)^n \neq 0$.
We define:

$$|u_\lambda\rangle = (J_+)^n |u\rangle \neq 0$$

and set $\lambda = \alpha + n$. We find:

$$\begin{aligned} J_3 |u_\lambda\rangle &= \lambda |u_\lambda\rangle\,, & J_+ |u_\lambda\rangle &= 0 \\ (J_-)^k |u_\lambda\rangle &= |u_{\lambda-k}\rangle\,, & J_3 |u_{\lambda-k}\rangle &= (\lambda - k) |u_{\lambda-k}\rangle \end{aligned}$$

Not all $|u_{\lambda-k}\rangle$ can be different from zero:

$$(J_-)^{N+1} |u_\lambda\rangle = 0 \quad \text{for some } N \in \mathbb{N}_0$$

Furthermore, one finds:

$$\begin{aligned} J_+ |u_{\lambda-k}\rangle &= J_+ J_- J_-^{k-1} |u_\lambda\rangle = (J_- J_+ + 2J_3) J_-^{k-1} |u_\lambda\rangle \\ &= 2(\lambda + 1 - k) |u_{\lambda+k-1}\rangle + J_- \underbrace{J_+ J_- J_-^{k-2} |u_\lambda\rangle}_{\text{same as above with } k \to k-1} \\ &= 2 \sum_{\nu=1}^{k} (\lambda + \nu - k) |u_{\lambda+1-k}\rangle = k(2\lambda + 1 - k) |u_{\lambda+1-k}\rangle \end{aligned}$$

where in the last step we have used the arithmetic series $\sum\limits_{\nu=1}^{k} \nu = \frac{k}{2}(k+1)$.
We apply the latter relation for the case $k = N + 1$ and obtain:

$$0 = (N+1)(2\lambda - N) \underbrace{|u_{\lambda-N}\rangle}_{\neq 0} = J_+ \underbrace{|u_{\lambda-N-1}\rangle}_{=0} \quad \Rightarrow \lambda = \frac{N}{2} \equiv j$$

with j being either an integer or a half-integer.
The possible eigenvalues of J_3 in an irrep. are:

$$-j, 1-j, 2-j, \dots, j-2, j-1, j \quad \text{with } j \in \left\{0, \frac{1}{2}, 1, \frac{3}{2}, 2, \dots\right\}$$

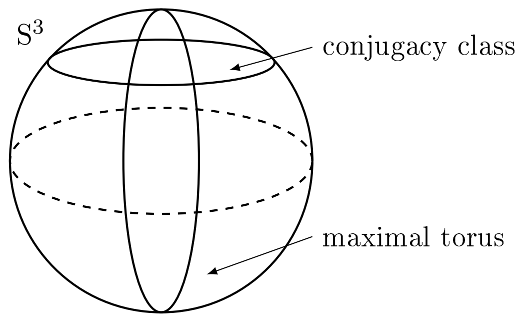
And therefore J_3 is given by the diagonal matrix

$$J_3 = \begin{pmatrix} j & & & & \\ & j-1 & & & \\ & & \ddots & & \\ & & & 1-j & \\ & & & & -j \end{pmatrix}, \quad (J_-)^{2j+1} |u_j\rangle = 0$$

Note that the spectrum of eigenvalues of J_3 is mirror symmetric (including $\pm m$), equidistant (distance 1) and specified by one single, non-negative integer $2j \in \mathbb{N}_0$.
Now we use the correspondence $J_3 = \mathrm{i}L_3 = \mathrm{i}D\big(-\frac{\mathrm{i}}{2}\sigma_3\big) = D\big(\frac{\sigma_3}{2}\big)$ and then obtain the form of diagonal $SU(2)$-matrices in the representation D.
The diagonal matrices

$$\begin{pmatrix} \mathrm{e}^{\mathrm{i}\frac{\varphi}{2}} & 0 \\ 0 & \mathrm{e}^{-\mathrm{i}\frac{\varphi}{2}} \end{pmatrix} = \exp\left(\frac{\mathrm{i}\varphi}{2}\sigma_3\right) \in SU(2)$$

form an abelian subgroup of $SU(2)$, isomorphic to $U(1) = S^1$, called the *maximal torus* of $SU(2)$.



In general the maximal torus is the largest abelian closed subgroup of a Lie group. It is well known that any unitary matrix can be diagonalized with complex eigenvalues of magnitude 1:

$$U = U' \begin{pmatrix} \zeta & 0 \\ 0 & \zeta^{-1} \end{pmatrix} U'^{-1} \quad \text{with } U, U' \in SU(2) \text{ and } |\zeta| = 1$$

Thus the maximal torus contains a representative of each conjugation class of $SU(2)$. Because of the relation

$$\begin{pmatrix} \mathrm{e}^{\mathrm{i}\frac{\varphi}{2}} & 0 \\ 0 & \mathrm{e}^{-\mathrm{i}\frac{\varphi}{2}} \end{pmatrix} = \exp\left(\frac{\mathrm{i}}{2}\sigma_3\varphi\right)$$

the irreducible representation of the maximal torus is given by

$$D_j\left(\exp\left(\frac{\mathrm{i}}{2}\sigma_3\varphi\right)\right) = \exp\left(\mathrm{i}J_3\varphi\right) = \begin{pmatrix} \mathrm{e}^{\mathrm{i}j\varphi} & & & & \\ & \mathrm{e}^{\mathrm{i}(j-1)\varphi} & & & \\ & & \ddots & & \\ & & & \mathrm{e}^{\mathrm{i}(1-j)\varphi} & \\ & & & & \mathrm{e}^{-\mathrm{i}j\varphi} \end{pmatrix}$$

Therefore, the corresponding irreducible character can be read of immediately:

$$\begin{aligned} \chi_j(\varphi) &= \sum_{m=-j}^{j} \mathrm{e}^{\mathrm{i}m\varphi} = \mathrm{e}^{-\mathrm{i}j\varphi} \underbrace{\sum_{m=0}^{2j} \mathrm{e}^{\mathrm{i}m\varphi}}_{\text{geometrical series}} \\ &= \mathrm{e}^{-\mathrm{i}j\varphi} \frac{1 - \mathrm{e}^{\mathrm{i}(2j+1)\varphi}}{1 - \mathrm{e}^{\mathrm{i}\varphi}} = \frac{\mathrm{e}^{\mathrm{i}(j+1)\varphi} - \mathrm{e}^{-\mathrm{i}j\varphi}}{\mathrm{e}^{\mathrm{i}\varphi} - 1} \frac{\mathrm{e}^{-\mathrm{i}\frac{\varphi}{2}}}{\mathrm{e}^{-\mathrm{i}\frac{\varphi}{2}}} \end{aligned}$$

and we find

$$\chi_j(\varphi) = \frac{\sin\left((2j+1)\frac{\varphi}{2}\right)}{\sin\left(\frac{\varphi}{2}\right)}$$

with $0 < \varphi < 4\pi$, but the interval $0 < \varphi < 2\pi$ is already sufficient.

We list some of the irreducible characters:

$$\begin{aligned}
\chi_0(\varphi) &= 1 \quad \text{trivial 1-dim. rep.} \\
\chi_{\frac{1}{2}}(\varphi) &= tr\left(\begin{pmatrix} e^{i\frac{\varphi}{2}} & 0 \\ 0 & e^{-i\frac{\varphi}{2}} \end{pmatrix}\right) = e^{i\frac{\varphi}{2}} + e^{-i\frac{\varphi}{2}} \\
&= 2\cos\left(\frac{\varphi}{2}\right) \quad \text{(defining 2-dim. rep.)} \\
\chi_1(\varphi) &= tr\left(\begin{pmatrix} \cos\varphi & -\sin\varphi & 0 \\ \sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{pmatrix}\right) \\
&= 1 + 2\cos\varphi \quad \text{(3-dim. vector rep.)}
\end{aligned}$$

Furthermore, these irreducible characters form a complete and orthonormal set of *class functions* with respect to the following scalar product:

$$\langle f | g \rangle = \underbrace{\frac{1}{\pi}\int_0^{2\pi} d\varphi \sin^2\frac{\varphi}{2}}_{\substack{\text{invariant group integral for class functions on } SU(2) \\ \text{replacing } \frac{1}{|G|}\sum\limits_{g\in G} \text{ in the case of finite groups}}} f(\varphi)^* \, g(\varphi)$$

Indeed, one finds

$$\begin{aligned}
\langle \chi_j | \chi_{j'} \rangle &= \frac{1}{\pi}\int_0^{2\pi} d\varphi \sin\left((2j+1)\frac{\varphi}{2}\right) \cdot \sin\left((2j'+1)\frac{\varphi}{2}\right) \\
&= \frac{1}{2\pi}\int_0^{2\pi} d\varphi \left[\cos\left((j-j')\varphi\right) - \cos\left((j+j'+1)\varphi\right)\right] = \delta_{jj'}
\end{aligned}$$

since $\int\limits_0^{2\pi} d\varphi \cos(l\varphi) = 2\pi\delta_{l0}$.

With the knowledge about the irreducible characters, we can write down the Clebsch-Gordon decomposition for $SU(2)$. Let $V(j_1)$ and $V(j_2)$ be the representation spaces of dimensions $\dim\left(V(j_1)\right) = 2j_1 + 1$ and $\dim\left(V(j_2)\right) = 2j_2 + 1$ respectively. Their tensor product, $V(j_1) \otimes V(j_2)$, is decomposed into reducible subspaces in the following way:

$$V(j_1) \otimes V(j_2) = \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} V(j)$$

Proof: Set $z = \mathrm{e}^{\mathrm{i}\varphi}$

$$\begin{aligned}\chi_{V(j_1)\otimes V(j_2)}(\varphi) = \chi_{j_1}(\varphi) \cdot \chi_{j_2}(\varphi) &= \sum_{\mu=-j_1}^{j_1} z^\mu \cdot \sum_{\nu=-j_2}^{j_2} z^\nu \\ &= \sum_{j=|j_1-j_2|}^{j_1+j_2} \sum_{m=-j}^{j} z^m = \sum_{j=|j_1-j_2|}^{j_1+j_2} \chi_j(\varphi)\end{aligned}$$

See also Figure 3.1.

Those representations of $SU(2)$ for which $-\mathbb{1} \in SU(2)$ is mapped onto the $(2j+1) \times (2j+1)$-unit matrix are representations of the factor group $SU(2)/\{\pm\mathbb{1}\} = SO(3)$.

$$D_j(-\mathbb{1}) = D_j(\exp(\mathrm{i}\pi\sigma_3)) = \exp(2\pi\mathrm{i}J_3) = \begin{pmatrix} \mathrm{e}^{2\pi\mathrm{i}j} & & \\ & \ddots & \\ & & \mathrm{e}^{-2\pi\mathrm{i}j} \end{pmatrix} = \begin{pmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{pmatrix} \quad \Rightarrow j \in \mathbb{N}_0$$

The irreducible representations of $SO(3)$ have odd dimension $2l+1$, with $l \in \mathbb{N}_0$.
A concrete realization of all basis vectors of $\bigoplus_{l=0}^{\infty} V(l)$ is given in terms of spherical harmonics

$$|l, m\rangle = Y_{lm}(\vartheta, \varphi)$$

where $l \in \mathbb{N}_0$ and $m \in \{-l, \ldots, l\}$.
The generators L_i, $i = 1, 2, 3$, are represented by the "angular momentum" operator: $\boldsymbol{\mathcal{L}} = -\boldsymbol{r} \times \boldsymbol{\nabla}$ and thus $\boldsymbol{J} = \mathrm{i}\boldsymbol{\mathcal{L}} = -\mathrm{i}(\boldsymbol{r} \times \boldsymbol{\nabla})$. In spherical coordinates one gets differential operators:

$$\begin{aligned} J_3 &= \mathrm{i}\mathcal{L}_3 = -\mathrm{i}\frac{\partial}{\partial\varphi} \\ J_\pm &= \mathrm{i}(\mathcal{L}_1 \pm \mathrm{i}\mathcal{L}_2) = \mathrm{e}^{\pm\mathrm{i}\varphi}\left(\pm\frac{\partial}{\partial\vartheta} + \mathrm{i}\cot\vartheta\frac{\partial}{\partial\varphi}\right) \end{aligned}$$

and furthermore one has the relations:

$$\begin{aligned} J_3\,|l, m\rangle &= m\,|l, m\rangle \\ J_\pm\,|l, m\rangle &= \sqrt{(l \mp m)(l \pm m + 1)}\,|l, m \pm 1\rangle \end{aligned}$$

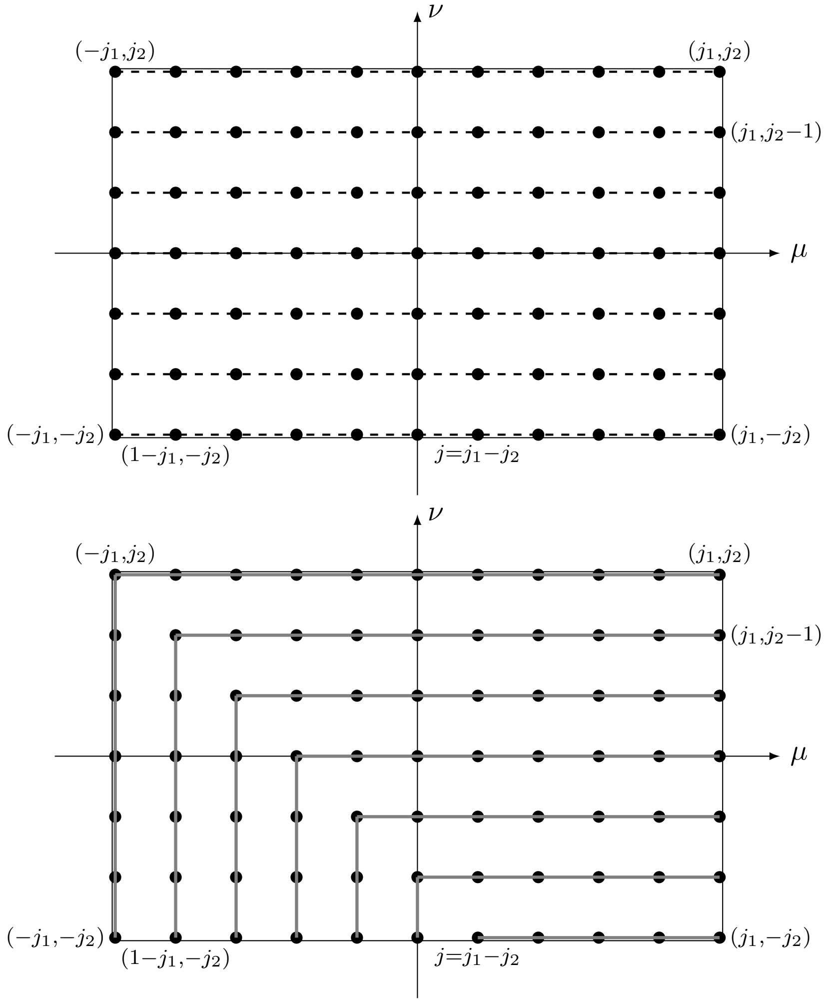
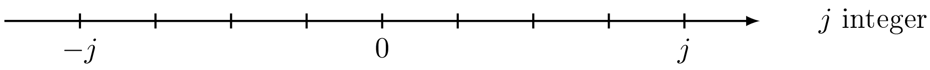
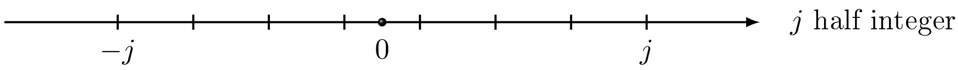


Figure 3.1: Equality of the different summations over lattice points in a rectangle.

Addition of weight diagrams:

To each irrep. of the complexified Lie-algebra $sl(2,\mathbb{C})$ one can associate a weight diagram:





This corresponds to an arrangement of eigenvalues (and eigenvectors) of J_3

$$D_j(J_3)\,|jm\rangle = m\,|jm\rangle$$

where m is the weight and the weight vectors $|jm\rangle$ form a basis of the $2j+1$-dim. representation space $V(j)$.
Consider the tensor product of two $SU(2)$ representations

$$U \in SU(2) : |v_1\rangle \otimes |v_2\rangle \mapsto D_1(U)\,|v_1\rangle \otimes D_2(U)\,|v_2\rangle$$

This provides a representation of the Lie-algebra $su(2)$

$$\dot{U} \in su(2) : |v_1\rangle \otimes |v_2\rangle \mapsto D_1(\dot{U})\,|v_1\rangle \otimes \mathbb{1}\,|v_2\rangle + \mathbb{1}\,|v_1\rangle \otimes D_2(\dot{U})\,|v_2\rangle$$

This implies that e.g. J_3 maps $|v_1\rangle \otimes |v_2\rangle$ onto $D_1(J_3)\,|v_1\rangle \otimes |v_2\rangle + |v_1\rangle \otimes D_2(J_3)\,|v_2\rangle$.
For tensor products of weight vectors

$$\begin{aligned}|j_1,m_1\rangle \otimes |j_2,m_2\rangle \mapsto\ & m_1\,|j_1,m_1\rangle \otimes |j_2,m_2\rangle + |j_1,m_1\rangle \otimes m_2\,|j_2,m_2\rangle \\ =\ & (m_1+m_2)\,|j_1,m_1\rangle \otimes |j_2,m_2\rangle\end{aligned}$$

the weights are added:

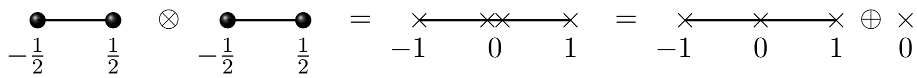
$$m = m_1 + m_2$$

We add weight diagrams by positioning the center of one on all the weights of the other. The set of resulting weights is decomposed (regrouped) into multiplets of weights corresponding to irreps.

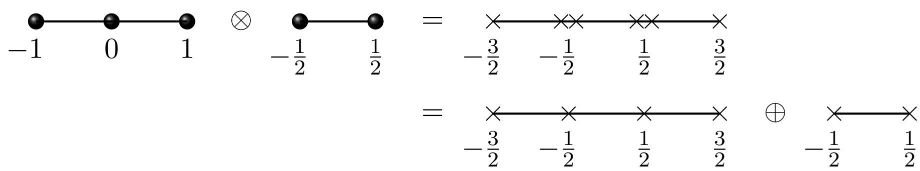
$$V(j_1) \otimes V(j_2) = \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} V(j)$$

Examples:

- $\boldsymbol{V(\frac{1}{2}) \otimes V(\frac{1}{2}) = V(1) \oplus V(0)}$:



- $\boldsymbol{V(1) \otimes V(\frac{1}{2}) = V(\frac{3}{2}) \oplus V(\frac{1}{2})}$:



3.6 Representations of $sl(3,\mathbb{C})$ resp. $SU(3)$

The complexified Lie-algebra of $SU(3)$ is $sl(3,\mathbb{C})$, consisting of traceless, complex 3×3 matrices. We use the following complex basis of $sl(3,\mathbb{C})$ (a 8-dim. complex vector space):

$$H_1 = \frac{1}{2}\begin{pmatrix}1 & 0 & 0\\ 0 & -1 & 0\\ 0 & 0 & 0\end{pmatrix}, \qquad H_2 = \frac{1}{2\sqrt{3}}\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & -2\end{pmatrix}$$

$$X_+ = \begin{pmatrix}0 & 1 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{pmatrix}, \qquad X_- = \begin{pmatrix}0 & 0 & 0\\ 1 & 0 & 0\\ 0 & 0 & 0\end{pmatrix}$$

$$Y_+ = \begin{pmatrix}0 & 0 & 0\\ 0 & 0 & 1\\ 0 & 0 & 0\end{pmatrix}, \qquad Y_- = \begin{pmatrix}0 & 0 & 0\\ 0 & 0 & 0\\ 0 & 1 & 0\end{pmatrix}$$

$$Z_+ = \begin{pmatrix}0 & 0 & 1\\ 0 & 0 & 0\\ 0 & 0 & 0\end{pmatrix}, \qquad Z_- = \begin{pmatrix}0 & 0 & 0\\ 0 & 0 & 0\\ 1 & 0 & 0\end{pmatrix}$$

The following commutation relations hold:

$$[(H_1,H_2),X_\pm] = \pm(1,0)X_\pm \quad \text{root } \alpha$$
$$[(H_1,H_2),Y_\pm] = \pm\left(-\frac{1}{2},\frac{\sqrt{3}}{2}\right)Y_\pm \quad \text{root } \beta$$
$$[(H_1,H_2),Z_\pm] = \pm\left(\frac{1}{2},\frac{\sqrt{3}}{2}\right)Z_\pm \quad \text{root } \gamma = \alpha+\beta$$

Root diagram of $sl(3,\mathbb{C})$:

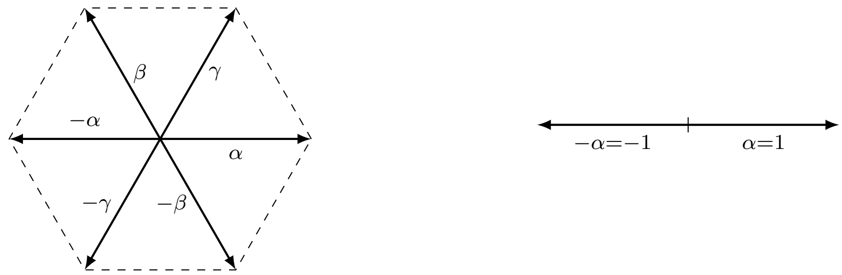


Figure 3.2: Root diagram of $sl(3,\mathbb{C})$ (left) and $sl(2,\mathbb{C})$ (right)

There are further 2-dim. root diagrams:

Figure 3.3: Root diagram of $so(5,\mathbb{C})$ (left) and G_2 (right)

We have obtained a decomposition of the Lie-algebra $sl(3,\mathbb{C})$ into the direct sum of a maximal abelian subalgebra $h = \mathbb{C}H_1 \oplus \mathbb{C}H_2$ spanned by H_1 and H_2 and eigenspaces $\mathcal{E}_\rho$, indexed by the six roots $\rho \in \{\pm\alpha,\pm\beta,\pm\gamma\}$. h is called the *Cartan-subalgebra* and we have the *root space decomposition*

$$sl(3,\mathbb{C}) = h \oplus \bigoplus_{\rho} \mathcal{E}_{\rho}$$

The commutation relation takes the form of an eigenvalue equation:

$$[H, X_{\rho}] = \rho(H)\, X_{\rho}$$

where

$$H = x_1 H_1 + x_2 H_2$$

and

$$\rho(H) = x_1 \rho_1 + x_2 \rho_2$$

Here $\rho(H)$ is a *linear form* on h, and the roots $\rho \in \{\pm\alpha, \pm\beta, \pm\gamma\} \subset h^*$ are infact elements of the *dual space* h^* to the Cartan-subalgebra h.
How do elements from different eigenspaces $\mathcal{E}_{\rho}$ commute with each other? A simple calculation using the Jacobi identity gives:

$$\begin{aligned} [H,[X_\alpha,X_\beta]] &= [X_\alpha,[H,X_\beta]] + [X_\beta,[X_\alpha,H]] = [X_\alpha,\beta(H)\,X_\beta] - [X_\beta,\alpha(H)\,X_\alpha] \\ &= (\alpha(H)+\beta(H))\,[X_\alpha,X_\beta] = (\alpha+\beta)(H)\,[X_\alpha,X_\beta] \end{aligned}$$

and thus we obtain

$$\left[\mathcal{E}_{\rho}, \mathcal{E}_{\rho'}\right] = \mathcal{E}_{\rho+\rho'}$$

if $\rho + \rho'$ is a root, otherwise $\left[\mathcal{E}_{\rho}, \mathcal{E}_{\rho'}\right]$ is zero. We also see that

$$[\mathcal{E}_{\rho}, \mathcal{E}_{-\rho}] \subset h$$

To each pair of root $\pm\rho$ corresponds a $sl(2,\mathbb{C})$ subalgebra

$$\begin{aligned} \pm\alpha = \pm(1,0): &\quad \left\{X_+, X_-, H_1 = \frac{1}{2}\,[X_+, X_-]\right\} \\ \pm\beta = \pm\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right): &\quad \left\{Y_+, Y_-, \frac{1}{2}\,[Y_+, Y_-] = -\frac{1}{2}H_1 + \frac{\sqrt{3}}{2}H_2\right\} \\ \pm\gamma = \pm\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right): &\quad \left\{Z_+, Z_-, \frac{1}{2}\,[Z_+, Z_-] = \frac{1}{2}H_1 + \frac{\sqrt{3}}{2}H_2\right\} \end{aligned}$$

The diagonal matrices $H_1, \pm\frac{1}{2}H_1 + \frac{\sqrt{3}}{2}H_2$ are linear combinations following the components of the roots α, β, γ.
We can now construct the irreps. of $sl(3,\mathbb{C})$ in analogy to those of $sl(2,\mathbb{C})$. Since H_1 and H_2 commute with each other, their representative linear operators can be diagonalized simultaneously:

$$\left.\begin{aligned} H_1 \left|v\right\rangle &= \mu_1 \left|v\right\rangle \\ H_2 \left|v\right\rangle &= \mu_2 \left|v\right\rangle \end{aligned}\right\} H\left|v\right\rangle = \mu(H)\left|v\right\rangle$$

where

$$\mu(H) = x_1\mu_1 + x_2\mu_2 \quad \text{and} \quad H = x_1 H_1 + x_2 H_2$$

$\mu = (\mu_1, \mu_2)$ is called a *weight* and $\left|v\right\rangle$ the corresponding *weight vector*.
By applying the ladder operators X_ρ, $\rho \in \{\pm\alpha, \pm\beta, \pm\gamma\}$ we obtain new weights and new weight vectors

$$\begin{aligned} H X_\rho \left|v\right\rangle &= ([H, X_\rho] + X_\rho H)\left|v\right\rangle = (\rho(H)\, X_\rho + \mu(H)\, X_\rho)\left|v\right\rangle \\ &= (\rho + \mu)(H)\, X_\rho \left|v\right\rangle \end{aligned}$$

When μ is a weight, we obtain the further possible weights

$$\mu + n_1\alpha + n_2\beta + n_3\gamma = \mu + n_1'\alpha + n_2'\beta$$

with $n_{1,2,3} \in \mathbb{Z}$.
Since the representation should be finite dimensional, almost all of the new weight vectors $\ldots X_{\rho''} X_{\rho'} X_\rho \left|v\right\rangle$ must vanish.

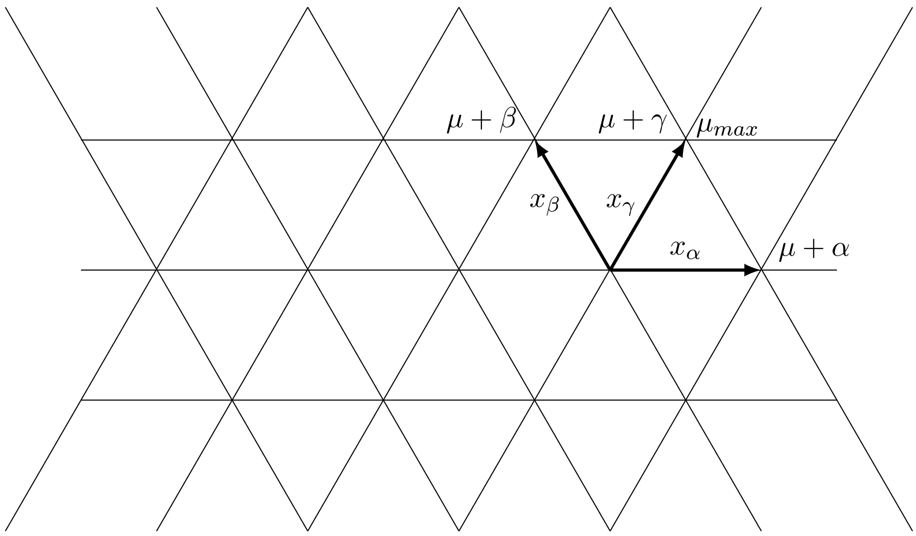


Figure 3.4: The weights are points on a regular, triangular lattice.

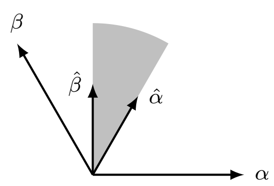
The weights are points on a regular, triangular lattice. The operators X_ρ act as shifting-operators. There must exist an extremal weight vector, which gets annihilated by X_α and X_β

$$X_\alpha |v_{\max}\rangle = 0 = X_\beta |v_{\max}\rangle$$

This implies $X_\gamma |v_{\max}\rangle = [X_\alpha, X_\beta] |v_{\max}\rangle = (X_\alpha X_\beta - X_\beta X_\alpha) |v_{\max}\rangle = 0$.
The weight corresponding to $|v_{\max}\rangle$ is called the *dominant weight* $\mu_{\max}$ of the representation of $sl(3, \mathbb{C})$.
In the case of $sl(2, \mathbb{C})$, the weight corresponding to $|j, j\rangle$ has been the extremal weight vector and $j \in \frac{1}{2}\mathbb{N}_0$ has been the dominant weight.
Along the straight lines in direction of α, β and γ, the weight vectors form irreps. of $sl(2, \mathbb{C})$. The eigenvalues of the corresponding J_3-type operators $H_1, -\frac{1}{2}H_1 + \frac{\sqrt{3}}{2}H_2$ and $\frac{1}{2}H_1 + \frac{\sqrt{3}}{2}H_2$ are known to be integer or half integer and non-negative:

$$\mu_{\max} \cdot \alpha \in \frac{1}{2}\mathbb{N}_0 \quad \text{and} \quad \mu_{\max} \cdot \beta \in \frac{1}{2}\mathbb{N}_0$$

The condition $\mu_{\max} \cdot \gamma \in \frac{1}{2}\mathbb{N}_0$ is redundant, since $\gamma = \alpha + \beta$, where the plus sign is crucial.
With this knowledge we can determine all dominant weights (keyword: *reciprocal lattice*).



Define vectors $\hat{\alpha}$ and $\hat{\beta}$ which satisfy:

$$\hat{\alpha} \cdot \alpha = \frac{1}{2} \qquad \hat{\beta} \cdot \alpha = 0$$
$$\hat{\alpha} \cdot \beta = 0 \qquad \hat{\beta} \cdot \beta = \frac{1}{2}$$

This linear equations have the solution:

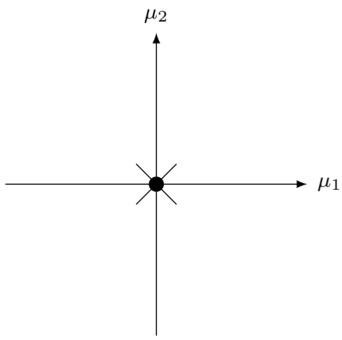
$$\hat{\alpha} = \left(\frac{1}{2}, \frac{\sqrt{3}}{6}\right) = \frac{1}{3}(2\alpha + \beta)$$
$$\hat{\beta} = \left(0, \frac{\sqrt{3}}{3}\right) = \frac{1}{3}(\alpha + 2\beta)$$

Each dominant weight is now determined by a pair of non-negative integers (p, q), such that

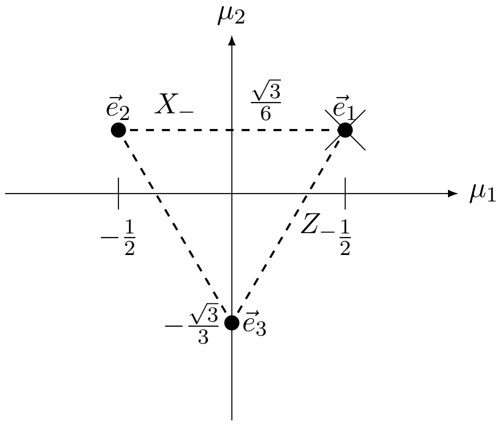
$$\mu_{\max}(p, q) = p\hat{\alpha} + q\hat{\beta} \quad \text{with } p, q \in \mathbb{N}_0$$

Examples of $sl(3,\mathbb{C})$-representations:

$p = q = 0$**:** $\mu_{\max} = (0,0)$, trivial 1-dim. representation: **1** (singlet)

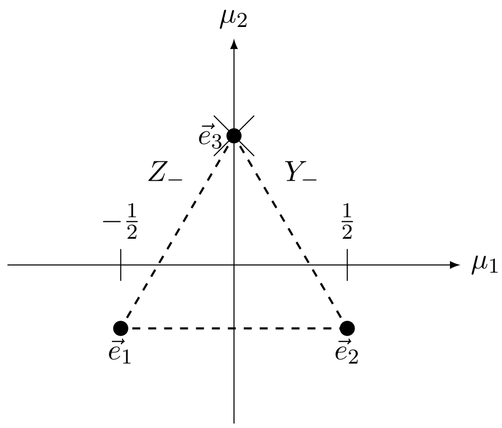


$p = 1, q = 0$**:** $\mu_{\max} = \hat{\alpha} = \left(\frac{1}{2}, \frac{\sqrt{3}}{6}\right)$, defining 3-dim. representation: **3** (triplet)



$$H_1 = \mathrm{diag}\left(\frac{1}{2}, -\frac{1}{2}, 0\right), \quad H_2 = \mathrm{diag}\left(\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{3}\right) = \frac{1}{2\sqrt{3}}\mathrm{diag}(1,1,-2)$$

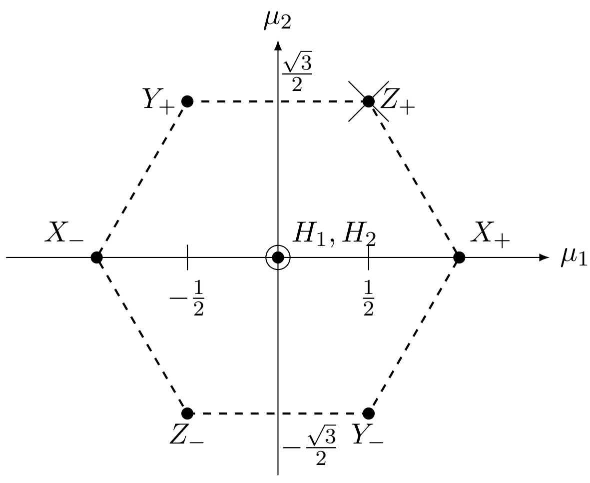
$\boldsymbol{p=0, q=1}$**:** $\mu_{\max} = \hat{\beta} = \left(0, \frac{\sqrt{3}}{3}\right)$, complex conjugated 3-dim. representation: $\bar{\mathbf{3}}$ (anti-triplet)



$$\bar{H}_1 = \operatorname{diag}\left(-\frac{1}{2}, \frac{1}{2}, 0\right) = -H_1, \quad \bar{H}_2 = \operatorname{diag}\left(-\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{3}\right) = -H_2$$

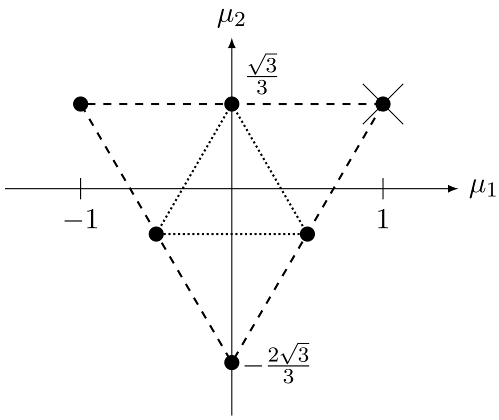
$$\left(\exp\left(\mathrm{i}\theta_1 H_1 + \mathrm{i}\theta_2 H_2\right)\right)^* = \exp\left(-\mathrm{i}\theta_1 H_1 - \mathrm{i}\theta_2 H_2\right) = \exp\left(\mathrm{i}\theta_1 \bar{H}_1 + \mathrm{i}\theta_2 \bar{H}_2\right)$$

$\boldsymbol{p=1, q=1}$**:** $\mu_{\max} = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$, adjoint representation: $\mathbf{8} = \bar{\mathbf{8}}$ (octet)

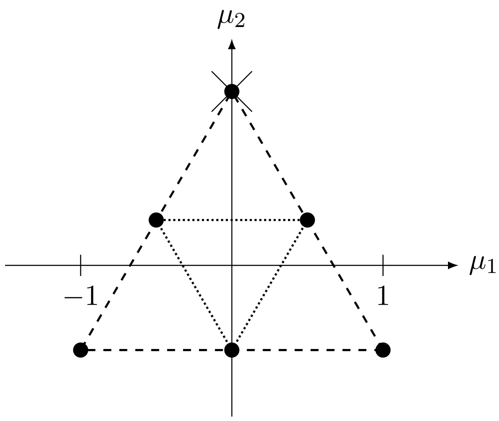


The representation space is the Lie-algebra $sl(3, \mathbb{C})$ itself and the weight diagram is equal to the root diagram.

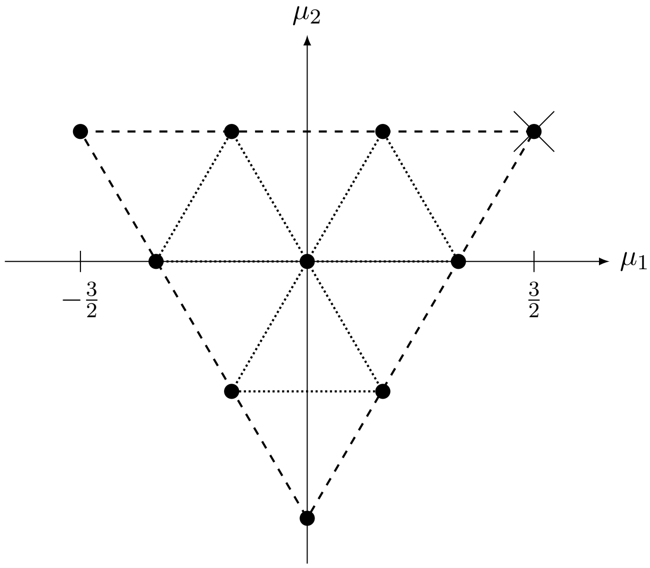
$\boldsymbol{p = 2, q = 0}$**:** $\mu_{\text{max}} = \left(1, \frac{\sqrt{3}}{2}\right)$, 6-dim. representation: **6** (sixtet)



$\boldsymbol{p = 0, q = 2}$**:** $\mu_{\text{max}} = \left(0, \frac{2\sqrt{2}}{3}\right)$, complex conjugated 6-dim. representation: $\bar{\mathbf{6}}$ (anti-sixtet)



$\boldsymbol{p=3, q=0}$**:** $\mu_{\max} = \left(\frac{3}{2}, \frac{\sqrt{3}}{2}\right)$, 10-dim. representation: **10** (decuplet)



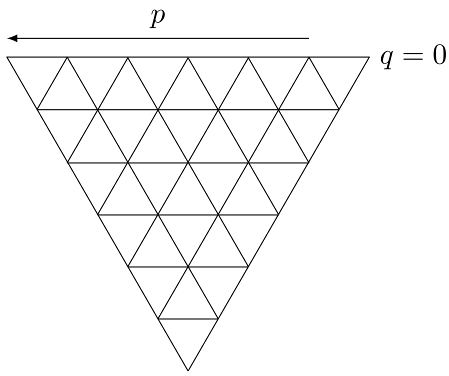
General rules for $sl(3,\mathbb{C})$ weight diagrams:

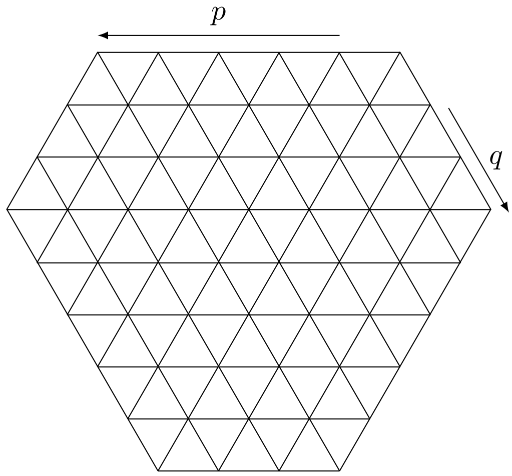
- p is the number of allowed steps with $X_- = X_{-\alpha}$
- q is the number of allowed steps with $Y_- = X_{-\beta}$
- The representative $V(q,p)$ has the negative weight with respect to the representative $V(p,q)$. $V(p,q)$ and $V(q,p)$ are complex conjugated to each other.
- The dimension of the representation $V(p,q)$ is given by

$$\dim\left(V(p,q)\right) = \frac{1}{2}(p+1)(q+1)(p+q+2), \quad \text{with } p,q \in \mathbb{N}_0$$

- The circumference (boundary) of a weight diagram is either a *equilateral triangle* (if $p=0$ or $q=0$) or a *half-regular hexagon* with two different side lengths p and q.

$$\mu_{\max} \cdot \left\{ \begin{array}{c} \alpha \\ \beta \\ \gamma \end{array} \right\} = \left\{ \begin{array}{l} p/2 \\ q/2 \\ (p+q)/2 \end{array} \right.$$





For $p = 0$ or $q = 0$ (triangular shape), each weight is simply occupied, i.e. the corresponding eigenspace is 1-dimensional.

$$\begin{aligned}\dim\left(V(p,0)\right) &= (p+1) + p + (p-1) + (p-2) + \cdots + 2 + 1 \\ &= \sum_{\nu=1}^{p+1} \nu = \frac{1}{2}(p+1)(p+2) \checkmark\end{aligned}$$

In the case where neither p, nor q vanishes, the weights are arranged in concentric hexagonal or triangular rings. The multiplicity (dimension of the eigenspace) on the outermost ring is 1. The multiplicity increases successively with each step inwards by one unit until the triangular ring is reached. From there on the multiplicity stays constant.

With this knowledge we can compute the dimension of $V(p,q)$ (assume without loss of generality $p \geq q$):

- outermost ring ($\nu = 1$): $3(p+q)\cdot 1$
- next inner ring: $3(p-1+q-1)\cdot 2$
- ...
- last hexagonal ring ($\nu = q$): $3(p-q+1+q-q+1)\cdot q$
- Summing up these numbers gives

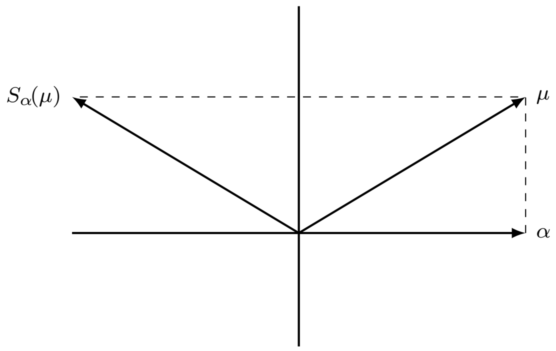
$$\begin{aligned}3\sum_{\nu=1}^{q} \nu(p+q+2-2\nu) &= \frac{3}{2}q(q+1)(p+q+2) - 6\frac{q}{6}(q+1)(2q+1) \\ &= \frac{q}{2}(q+1)(3p+3q+6-4q-2) \\ &= \frac{9}{2}(q+1)(3p-q+4)\end{aligned}$$

– The triangle with side length $p-q$ has the multiplicity $q+1$. This gives the contribution $(q+1)\frac{1}{2}(p-q+1)(p-q+2)$ to the dimension.

– In total this sums up to

$$\begin{aligned}\dim\left(V(p,q)\right) &= \frac{q+1}{2}(3pq-q^2+4q+p^2-2pq+q^2+3p-3q+2)\\ &= \frac{1}{2}(q+1)(p^2+pq+3p+q+2)\\ &= \frac{1}{2}(p+1)(q+1)(p+q+2)\ \checkmark\end{aligned}$$

The symmetries of weight diagrams of irreducible $sl(3,\mathbb{C})$-representations are generated by reflections about lines orthogonal to the (positive) roots $\{\alpha,\beta,\gamma=\alpha+\beta\}$



$$S_\alpha(\mu) = \underbrace{(\mu-\mu\cdot\alpha\alpha)}_{\text{orthogonal}} - \underbrace{\mu\cdot\alpha\alpha}_{\text{parallel}} = \mu - \underbrace{2\mu\cdot\alpha}_{\in\mathbb{Z}}\,\alpha$$

where $\alpha\cdot\alpha=1$. This finite discrete symmetry group is called the *Weyl-group*. The Weyl-group of $sl(3,\mathbb{C})$ or $SU(3)$ is

$$S_3 = D_3$$

Casimir operators:

If we have a matrix representation of a Lie-algebra, we can also consider the ordinary products of the generators (basis elements) and not just their Lie-brackets (commutators).

Example for $sl(2,\mathbb{C})$:

$\boldsymbol{J}^2 = J_1^2 + J_2^2 + J_3^2$ commutes with all the J_k:

$$
\begin{aligned}
\left[\boldsymbol{J}^2, J_k\right] &= \sum_{j=1}^{3}\left[J_j^2, J_k\right] = \sum_{j=1}^{3}(J_j\left[J_j, J_k\right] + \left[J_j, J_k\right]J_j) \\
&= \sum_{j,l=1}^{3} \mathrm{i}(J_j\epsilon_{jkl}J_l + \epsilon_{jkl}J_lJ_j) \\
&= \mathrm{i}\sum_{j,l=1}^{3} \underbrace{\epsilon_{jkl}}_{\text{antisym.}} \underbrace{(J_jJ_l + J_lJ_j)}_{\text{sym.}} = 0
\end{aligned}
$$

$$
\boldsymbol{J}^2 J_k = J_k \boldsymbol{J}^2
$$

According to Schur's lemma II, $\boldsymbol{J}^2$ is proportional to the unit matrix in an irreducible representation. We calculate the eigenvalue of $\boldsymbol{J}^2$ by applying the operator $\boldsymbol{J}^2$ to the highest weight vector:

$$
\begin{aligned}
\boldsymbol{J}^2\left|j,j\right\rangle &= \left[\underbrace{\frac{1}{4}(J_+ + J_-)^2}_{J_1^2} - \underbrace{\frac{1}{4}(J_+ - J_-)^2}_{J_2^2} + J_3^2\right]\left|j,j\right\rangle \\
&= \left[\underbrace{\frac{1}{2}J_+J_- + \frac{1}{2}J_-J_+}_{\text{use } \frac{1}{2}[J_+,J_-]=J_3} + J_3^2\right]\left|j,j\right\rangle \\
&= (\underbrace{J_-J_+}_{0} + \underbrace{J_3}_{j} + \underbrace{J_3^2}_{j^2})\left|j,j\right\rangle
\end{aligned}
$$

$$
\boldsymbol{J}^2\Big|_{V(j)} = j(j+1)
$$

In applications one often needs the eigenvalue of the (quadratic) $SU(3)$-Casimir operator:

$$C_2 = \frac{1}{4}\sum_{a=1}^{8}\lambda_a^2 \quad \text{(with generators } \lambda_a \text{ in the irrep. } V(p,q))$$

$$\begin{aligned}[C_2,\lambda_b] &= \frac{1}{4}\sum_{a=1}^{8}(\lambda_a\,[\lambda_a,\lambda_b] + [\lambda_a,\lambda_b]\,\lambda_a) \\ &= \frac{\mathrm{i}}{4}\sum_{a,c=1}^{8}\underbrace{f_{abc}}_{\text{antisym.}}\ \underbrace{(\lambda_a\lambda_c+\lambda_c\lambda_a)}_{\text{sym.}} = 0\end{aligned}$$

We calculate the eigenvalue of C_2 by applying this operator on the dominant weight vector $|v_{\max}\rangle$:

$$\begin{aligned}C_2 = &\underbrace{H_1^2}_{\left(\frac{\lambda_3}{2}\right)^2} + \underbrace{H_2^2}_{\left(\frac{\lambda_8}{2}\right)^2} + \underbrace{\frac{1}{4}(X_+ + X_-)^2}_{\left(\frac{\lambda_1}{2}\right)^2} - \underbrace{\frac{1}{4}(X_+ - X_-)^2}_{\left(\frac{\lambda_2}{2}\right)^2} \\ &+ \underbrace{\frac{1}{4}(Y_+ + Y_-)^2}_{\left(\frac{\lambda_6}{2}\right)^2} - \underbrace{\frac{1}{4}(Y_+ - Y_-)^2}_{\left(\frac{\lambda_7}{2}\right)^2} + \underbrace{\frac{1}{4}(Z_+ + Z_-)^2}_{\left(\frac{\lambda_4}{2}\right)^2} - \underbrace{\frac{1}{4}(Z_+ - Z_-)^2}_{\left(\frac{\lambda_5}{2}\right)^2} \\ &= H_1^2 + H_2^2 + \frac{1}{2}(X_+X_- + X_-X_+ + Y_+Y_- + Y_-Y_+ + Z_+Z_- + Z_-Z_+) \\ &= H_1^2 + H_2^2 + \underbrace{X_-X_+ + Y_-Y_+ + Z_-Z_+}_{\text{vanishes when applied to } |v_{\max}\rangle} + H_1 - \frac{1}{2}H_1 + \frac{\sqrt{3}}{2}H_2 + \frac{1}{2}H_1 + \frac{\sqrt{3}}{2}H_2\end{aligned}$$

The highest weight is then given by

$$\mu_{\max} = p\hat{\alpha} + q\hat{\beta} = \Bigg(\underbrace{\frac{p}{2}}_{\text{eigenvalue of } H_1}\ ,\ \underbrace{\frac{\sqrt{3}}{6}(p+2q)}_{\text{eigenvalue of } H_2}\Bigg)$$

$$\begin{aligned}C_2\,|v_{\max}\rangle &= |v_{\max}\rangle\left(\frac{p^2}{4} + \frac{1}{12}(p+2q)^2 + \frac{p}{2} + \frac{p+2q}{2}\right) \\ &= |v_{\max}\rangle\left(\frac{p^2}{3} + \frac{pq}{3} + \frac{q^2}{3} + p + q\right)\end{aligned}$$

$$C_2|_{V(p,q)} = \frac{1}{3}(p^2 + pq + q^2) + p + q$$

Nota that the eigenvalue of C_2 is even under the exchange of p and q.

dim	1	3	6	8	10	15	15	21	24	27
C_2	0	$\frac{4}{3}$	$\frac{10}{3}$	3	6	$\frac{16}{3}$	$\frac{28}{3}$	$\frac{40}{3}$	$\frac{25}{3}$	8
p, q $(p \geq q)$	$0, 0$	$1, 0$	$2, 0$	$1, 1$	$3, 0$	$2, 1$	$4, 0$	$5, 0$	$3, 1$	$2, 2$

We observe that there are 4 inequivalent irreps. of $sl(3, \mathbb{C})$ of dimension $N = 15$. The often used notation N and N^* is not unique for $N > 10$.

3.7 Representations of the Lie group $SU(3)$

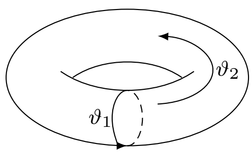
We now restrict the irrep. of the complex Lie-algebra to the real Lie-algebra $su(3)$. Using the matrix exponential we obtain an irrep. of the real Lie-group $SU(3)$:

$$D_{p,q}\left(\underbrace{\exp\left(\mathrm{i}\sum_{a=1}^{8} \lambda_a \vartheta_a\right)}_{\in SU(3)}\right) = \exp\left(\mathrm{i}\sum_{a=1}^{8} \underbrace{D_{p,q}(\mathrm{i}\lambda_a)}_{N\times N-\text{matrices}} \vartheta_a\right)$$

with $N = \frac{1}{2}(p+1)(q+1)(p+q+2)$.
We are particularly interested in the *characters*, which are constant on conjugation classes. Each $SU(3)$-matrix can be diagonalized. The eigenvalues are three complex numbers z_j, $j = 1, 2, 3$, of magnitude $|z_j| = 1$ which fulfill $z_1 \cdot z_2 \cdot z_3 = 1$.

$$SU(3) \ni U = U' \underbrace{\begin{pmatrix} \mathrm{e}^{\mathrm{i}\vartheta_1} & 0 & 0 \\ 0 & \mathrm{e}^{\mathrm{i}\vartheta_2} & 0 \\ 0 & 0 & \mathrm{e}^{-\mathrm{i}(\vartheta_1+\vartheta_2)} \end{pmatrix}}_{\text{generates a subgroup } U(1)\times U(1)} U'^{-1}, \quad \text{with } U' \in SU(3)$$

This subgroup $U(1) \times U(1) = S^1 \times S^1$ is the so-called *maximal torus* (the largest abelian subgroup) in $SU(3)$.



Irreducible characters of $SU(3)$-representations are complex-valued functions $\chi_{p,q}(\vartheta_1, \vartheta_2)$ of the two angles on the maximal torus. Since the exponent is

$$\begin{pmatrix} \vartheta_1 & 0 & 0 \\ 0 & \vartheta_2 & 0 \\ 0 & 0 & -\vartheta_1 - \vartheta_2 \end{pmatrix} = (\vartheta_1 - \vartheta_2)H_1 + \sqrt{3}(\vartheta_1 + \vartheta_2)H_2$$

with

$$H_1 = \frac{\lambda_3}{2} = \frac{1}{2}\begin{pmatrix} 1 & & \\ & -1 & \\ & & 0 \end{pmatrix} \quad \text{and} \quad H_2 = \frac{\lambda_8}{2} = \frac{1}{2\sqrt{3}}\begin{pmatrix} 1 & & \\ & 1 & \\ & & -2 \end{pmatrix}$$

we can construct the irreducible $SU(3)$-characters as follows:

$$\chi_{p,q}(\vartheta_1, \vartheta_2) = \sum_{\substack{\text{all weights} \\ \mu=(\mu_1,\mu_2) \\ \text{incl. multiplicity}}} \exp\left(\mathrm{i}(\vartheta_1 - \vartheta_2)\mu_1 + \mathrm{i}\sqrt{3}(\vartheta_1 + \vartheta_2)\mu_2\right)$$

1-representation $p = q = 0$:

$$\begin{aligned} &\mu = (0,0) \\ &\chi_{0,0}(\vartheta_1, \vartheta_2) = 1 \end{aligned}$$

3-representation $p = 1, q = 0$:

$$\mu \in \left\{ \left(\frac{1}{2}, \frac{\sqrt{3}}{6}\right), \left(-\frac{1}{2}, \frac{\sqrt{3}}{6}\right), \left(0, -\frac{\sqrt{3}}{3}\right) \right\}$$

We have the following exponents

$$\begin{aligned} &\frac{1}{2}(\vartheta_1 - \vartheta_2) + \frac{\sqrt{3}}{6}\sqrt{3}(\vartheta_1 + \vartheta_2) = \vartheta_1 \\ &-\frac{1}{2}(\vartheta_1 - \vartheta_2) + \frac{\sqrt{3}}{6}\sqrt{3}(\vartheta_1 + \vartheta_2) = \vartheta_2 \\ &-\frac{\sqrt{3}}{3}\sqrt{3}(\vartheta_1 + \vartheta_2) = -\vartheta_1 - \vartheta_2 \end{aligned}$$

and obtain

$$\chi_{1,0}(\vartheta_1, \vartheta_2) = \mathrm{e}^{\mathrm{i}\vartheta_1} + \mathrm{e}^{\mathrm{i}\vartheta_2} + \mathrm{e}^{-\mathrm{i}(\vartheta_1 + \vartheta_2)}$$

$\bar{\mathbf{3}}$-representation $p=0, q=1$:

$$\chi_{0,1}(\vartheta_1,\vartheta_2) = \mathrm{e}^{-\mathrm{i}\vartheta_1} + \mathrm{e}^{-\mathrm{i}\vartheta_2} + \mathrm{e}^{\mathrm{i}(\vartheta_1+\vartheta_2)} = \chi^*_{1,0}(\vartheta_1,\vartheta_2)$$

$\mathbf{8}$-representation $p=1, q=1$

$$\mu \in \left\{(0,0), \pm(1,0), \pm\left(\frac{1}{2},\frac{\sqrt{3}}{2}\right), \pm\left(-\frac{1}{2},\frac{\sqrt{3}}{2}\right)\right\}$$

We have the following exponents

$$0,\ \vartheta_1-\vartheta_2,\ \frac{1}{2}(\vartheta_1-\vartheta_2)+\frac{3}{2}(\vartheta_1+\vartheta_2) = 2\vartheta_1+\vartheta_2,\ -\frac{1}{2}(\vartheta_1-\vartheta_2)+\frac{3}{2}(\vartheta_1+\vartheta_2) = \vartheta_1+2\vartheta_2$$

and their negatives. Thus we obtain

$$\chi_{1,1}(\vartheta_1,\vartheta_2) = 2 + 2\cos(\vartheta_1-\vartheta_2) + 2\cos(2\vartheta_1+\vartheta_2) + 2\cos(\vartheta_1+2\vartheta_2)$$

There exists a closed determinant-formula for $\chi_{p,q}(\vartheta_1,\vartheta_2)$:

$$\chi_{p,q}(\vartheta_1,\vartheta_2) = \begin{vmatrix} z_1^{p+q+2} & z_1^{q+1} & 1 \\ z_2^{p+q+2} & z_2^{q+1} & 1 \\ z_3^{p+q+2} & z_3^{q+1} & 1 \end{vmatrix} \cdot \underbrace{\begin{vmatrix} z_1^2 & z_1 & 1 \\ z_2^2 & z_2 & 1 \\ z_3^2 & z_3 & 1 \end{vmatrix}}_{\text{Van-der-Monde-determinant}}{}^{-1}$$

with $z_{1,2} = \mathrm{e}^{\mathrm{i}\vartheta_{1,2}}$ and $z_3 = \frac{1}{z_1 z_2}$. In explicit form

$$\begin{aligned}\chi_{p,q}(\vartheta_1,\vartheta_2) = \Big\{ & z_1^{1-q} z_2^{3+p} + z_1^{1-p} z_2^{-p-q} + z_1^{4+p+q} z_2^{3+q} - z_1^{-p-q} z_2^{1-p} \\ & - z_1^{3+p} z_2^{1-q} - z_1^{3+q} z_2^{4+p+q}\Big\} \cdot \left\{(z_1-z_2)(z_1 z_2^2-1)(z_1^2 z_2-1)\right\}^{-1}\end{aligned}$$

this is a *Laurent polynomial* in z_1 and z_2 with positive and negative exponents and *non-negative integer* coefficients. Denote the numerator of this expression by $\Phi_{p,q}(z_1,z_2)$. On the other hand the denominator is given by

$$(z_1-z_2)(z_1 z_2^2-1)(z_1^2 z_2-1) = -2\mathrm{i}\mathrm{e}^{-2\mathrm{i}(\vartheta_1+\vartheta_2)}\left[\sin(\vartheta_1+2\vartheta_2) - \sin(2\vartheta_1+\vartheta_2) + \sin(\vartheta_1-\vartheta_2)\right]$$

The irreducible characters form a orthonormalized set of functions (on the group $SU(3)$):

$$\int\limits_{SU(3)} \mathrm{d}U \, \chi^*_{p',q'}(U)\, \chi_{p,q}(U) = \delta_{p'p}\delta_{q'q}$$

$$= \frac{1}{6\pi^2} \int\limits_0^{2\pi} \mathrm{d}\vartheta_1 \int\limits_0^{2\pi} \mathrm{d}\vartheta_2 \, [\sin(\vartheta_1 + 2\vartheta_2) - \sin(2\vartheta_1 + \vartheta_2) + \sin(\vartheta_1 - \vartheta_2)]^2$$

$$\cdot\, \chi^*_{p',q'}(\vartheta_1, \vartheta_2)\, \chi_{p,q}(\vartheta_1, \vartheta_2)$$

$$= \frac{1}{24\pi^2} \int\limits_0^{2\pi} \mathrm{d}\vartheta_1 \int\limits_0^{2\pi} \mathrm{d}\vartheta_2 \, \Phi_{p,q}(z_1, z_2)\, \Phi^*_{p',q'}(z_1, z_2)$$

$$= \frac{1}{24\pi^2} \oint\limits_{|z_1|=1} \frac{\mathrm{d}z_1}{\mathrm{i}z_1} \oint\limits_{|z_2|=1} \frac{\mathrm{d}z_2}{\mathrm{i}z_2} \, \Phi_{p,q}(z_1, z_2)\, \underbrace{\Phi_{p',q'}\left(\frac{1}{z_1}, \frac{1}{z_2}\right)}_{\Phi^* \text{ on unit circle}}$$

$$= \frac{1}{6}\left[\text{constant term in } \Phi_{p,q}(z_1, z_2)\, \Phi_{p',q'}\left(\frac{1}{z_1}, \frac{1}{z_2}\right)\right]$$

$$= \frac{1}{6} \cdot 6 \cdot \delta_{pp'}\delta_{qq'} = \delta_{pp'}\delta_{qq'} \;\checkmark$$

The constant term in $\Phi_{p,q}(z_1, z_2)\, \Phi_{p',q'}\left(\frac{1}{z_1}, \frac{1}{z_2}\right)$ is given by

$$\delta_{qq'}\delta_{pp'} \cdot 6 - \underbrace{\delta_{p-p',1+q'}\delta_{p-p',-1-q} - \delta_{q'-q,-1-p'}\delta_{q'-q,1+p} - \delta_{q'-q,1+p}\delta_{q'-q,-1-p'} - \delta_{p-p',-1-q}\delta_{p-p',1+q'}}_{\text{vanishes due to conditions on the exponents}}$$

$$\left.\begin{array}{l} p - p' = \underbrace{1 + q'}_{\geq 1} = \underbrace{-1 - q}_{\leq -1} \\ q' - q = 1 + p = -1 - p' \end{array}\right\} \text{impossible}$$

How does the Weyl-group S_3 arise? The diagonalizeation of a $SU(3)$-matrix yields:

$$U = U' \begin{pmatrix} z_1 & 0 & 0 \\ 0 & z_2 & 0 \\ 0 & 0 & z_3 \end{pmatrix} U'^{-1}, \quad \text{with } U, U' \in SU(3)$$

where $z_1 \cdot z_2 \cdot z_3 = 1$. The order of the eigenvalues z_1, z_2, z_3 is undetermined. For arbitrary permutation of z_1, z_2, z_3 one stays in the same conjugation class of $SU(3)$. Therefore the character is a totally symmetric function of the eigenvalues

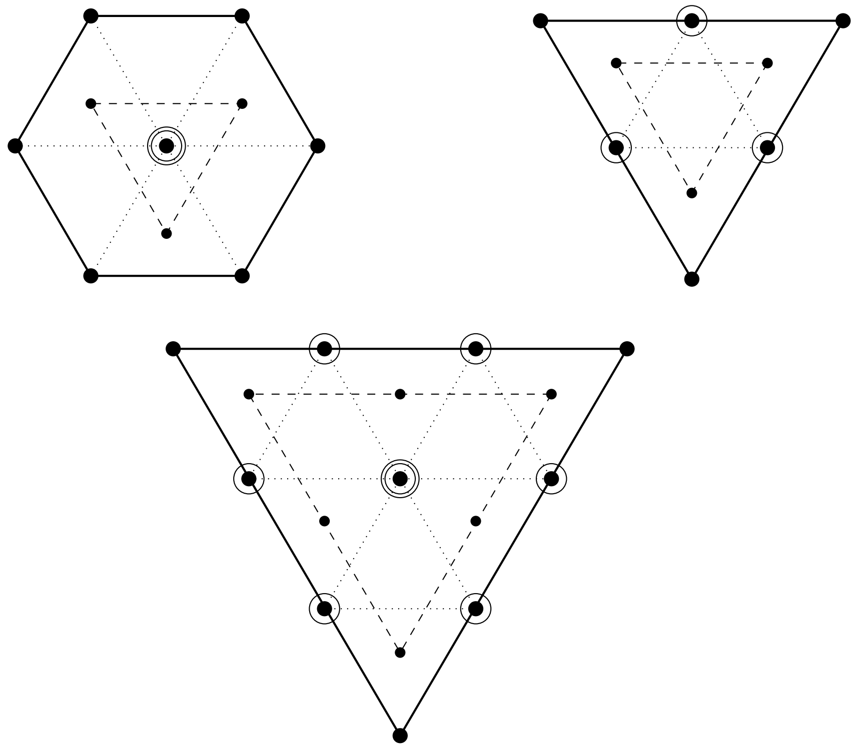
$$\chi(z_1, z_2, z_3) = \chi\big(z_{\sigma(1)}, z_{\sigma(2)}, z_{\sigma(3)}\big) \quad \text{for all } \sigma \in S_3$$

The condition $z_1 \cdot z_2 \cdot z_3 = 1$ remains unaffected by permutations. For those reasons $sl(3,\mathbb{C})$-weight diagrams possess a S_3-symmetry.
We expect that the Weyl-group of $SU(n)$ is the permutation group S_n with $n!$ elements.

Reduction of tensor products:

A possible method is the addition of weight diagrams:

$$
\begin{aligned}
3 \otimes 3 &= 6 \oplus \bar{3} \\
3 \otimes \bar{3} &= 8 \oplus 1 \\
6 \otimes 3 &= 10 \oplus 8
\end{aligned}
$$



General decomposition of tensor products of $sl(3,\mathbb{C})$-irreps.:
We use the Dynkin-label $p, q \in \mathbb{N}_0$ to denote the irreps.

$$V(p_1,q_1)\otimes V(p_2,q_2) = \bigoplus_{i,j} S(p_1-i,p_2-j;q_1-j,q_2-i)$$
$$\text{with } 0 \le i \le \min(p_1,q_2) \text{ and } 0 \le j \le \min(q_1,p_2)$$

The intermediate vector spaces $S(\mu_1,\mu_2;\nu_1,\nu_2)$ are decomposed into the direct sum of irreps. of $sl(3,\mathbb{C})$ as follows

$$\begin{aligned} S(\mu_1,\mu_2;\nu_1,\nu_2) =& V(\mu_1+\mu_2,\nu_1+\nu_2) \\ &\oplus \bigoplus V(\mu_1+\mu_2-2i,\nu_1+\nu_2+i) \quad \text{with } 1 \le i \le \min(\mu_1,\mu_2) \\ &\oplus \bigoplus V(\mu_1+\mu_2+j,\nu_1+\nu_2-2j) \quad \text{with } 1 \le j \le \min(\nu_1,\nu_2) \end{aligned}$$

If $\min(\mu_1,\mu_2)$ or $\min(\nu_1,\nu_2)$ vanishes these terms are absent.

Example: $\mathbf{8}\otimes\mathbf{8}$:

$$\begin{aligned} V(1,1)\otimes V(1,1) &= S(1,1;1,1)\oplus S(1,0;0,1)\oplus S(0,1;1,0)\oplus S(0,0;0,0) \\ S(1,1;1,1) &= V(2,2)\oplus V(0,3)\oplus V(3,0) \\ S(1,0;0,1) &= V(1,1) = S(0,1;1,0) \\ S(0,0;0,0) &= V(0,0) \end{aligned}$$

$$\begin{aligned} 8\otimes 8 &= 27\oplus 10\oplus \overline{10}\oplus 8\oplus 8\oplus 1 \\ &= (27\oplus 8\oplus 1)_{\text{sym.}}\oplus(10\oplus\overline{10}\oplus 8)_{\text{antisym.}} \end{aligned}$$

Example: $\mathbf{10}\otimes\mathbf{8}$:

$$V(3,0)\otimes V(1,1) = V(4,1)\oplus V(2,2)\oplus V(3,0)\oplus V(1,1)$$

$$10\otimes 8 = 35\oplus 27\oplus 10\oplus 8$$

Triality rule:

$$V(p_1,q_1)\otimes V(p_2,q_2) = V(p_1+p_2,q_1+q_2)\oplus\bigoplus V(r,s)$$

where

$$r-s=(p_1+p_2-q_1-q_2)\ \mathrm{mod}\ 3$$

Consider the representation of the center of $SU(3)$: $Z(SU(3)) = \{\zeta,\zeta^2,1\}$. The action of $\zeta = \mathrm{e}^{\frac{2\pi\mathrm{i}}{3}}$ must be the same on the right and left hand side, thus

$$\zeta^{p_1-q_1}\zeta^{p_2-q_2} = \zeta^{r-s}$$

which implies $r-s=(p_1+p_2-q_1-q_2)\ \mathrm{mod}\ 3$. ✓

Young-tableaus:

We assign a *tableau* with $p+q$ boxes in the first row and q boxes in the second row to the irrep. $V(p,q)$ of $SU(3)$:

$$\begin{array}{|c|c|c|c|c|c|c|c|c|c|}\hline & & & & & & & & & \\ \hline & & & & & & \multicolumn{4}{c}{} \\ \cline{1-6} \end{array}$$

The dimension is calculated via the quotient where the entries in the denominator tableau are the so-called *hook-lengths* of each box:

$$\dim = \frac{\begin{array}{|c|c|c|c|c|c|c|c|c|c|}\hline 3 & 4 & 5 & & & & & & & {\scriptstyle p+q+2} \\ \hline 2 & 3 & 4 & & & {\scriptstyle q+1} & \multicolumn{4}{c}{} \\ \cline{1-6} \end{array}}{\begin{array}{|c|c|c|c|c|c|c|c|c|c|}\hline {\scriptstyle p+q+1} & {\scriptstyle p+q} & & & {\scriptstyle p+3} & {\scriptstyle p+2} & p & & 2 & 1 \\ \hline q & & & & 2 & 1 & \multicolumn{4}{c}{} \\ \cline{1-6} \end{array}}$$

$$= \frac{(p+q+2)!(q+1)!}{2!(p+q+1)!(p+1)^{-1}q!} = \frac{1}{2}(p+q+2)(p+1)(q+1)$$

for $SU(3)$. (There are analogous rules for $SU(N)$)

Reduction of tensor product repre3sentations via coupling of two tableaus:

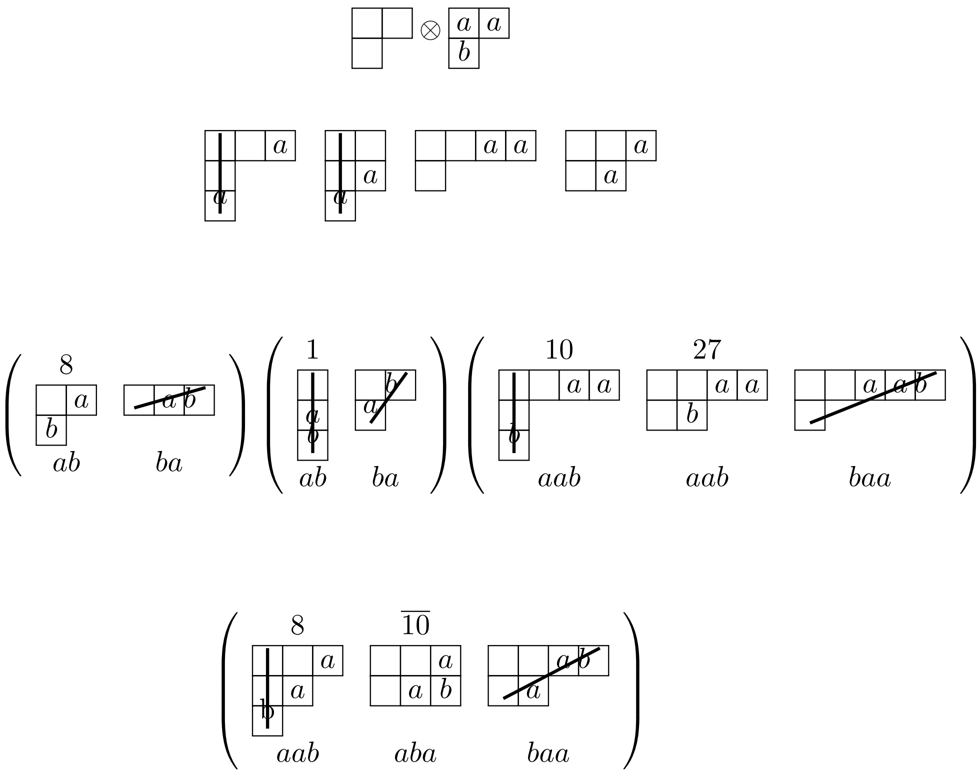
i) Denote the boxes in the first row of the second tableau with $\boxed{a}$ and the ones in the second row with $\boxed{b}$.

ii) Add the $\boxed{a}$s to the right end of the first tableau, whereby only one $\boxed{a}$ may appear in the same column and the length of a row must not grow from top to bottom. Repeat this procedure for the $\boxed{b}$s applying the same rules. The length of a column must be ≤ 3.

iii) Read of the added symbols (a, b) for each new tableau from right to left starting in the first row, then second row and finally in the third row. Write down this sequence from left to right.

A sequence is *allowed*, if there are not less $\boxed{a}$s then $\boxed{b}$s left of each symbol. Tableaus with *not allowed* sequences are to be *dropped*.
A column of length 3 on the left edge may be dropped. If this generates an empty tableau, it encodes the (trivial) **1**-representation.

Example 8 ⊗ 8:



Exercise: Show that

$$\mathbf{10} \otimes \mathbf{8} = \mathbf{35} \oplus \mathbf{27} \oplus \mathbf{10} \oplus \mathbf{8}$$
$$V(3,0) \otimes V(1,1) = V(4,1) \oplus V(2,2) \oplus V(3,0) \oplus V(1,1)$$

3.8 Outlook: Representations of $sl(4, \mathbb{C})$ resp. $SU(4)$

The Lie-algebra $sl(4, \mathbb{C})$ has the complex dimension 15. Its (maximal abelian) Cartan-algebra h is of dimension three

$$h = \mathbb{C}H_1 \oplus \mathbb{C}H_2 \oplus \mathbb{C}H_3$$

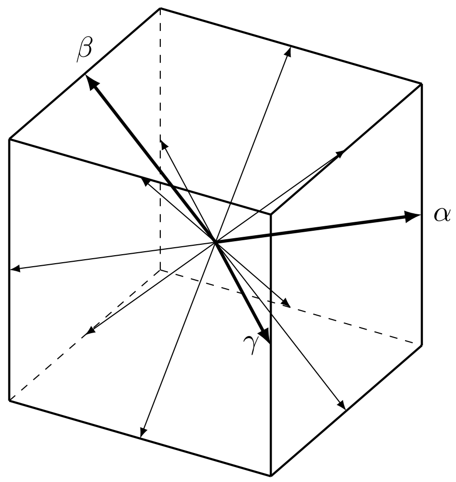
with

$$\begin{aligned} H_1 &= \frac{1}{2}\mathrm{diag}(1, -1, 0, 0) \\ H_2 &= \frac{1}{2\sqrt{3}}\mathrm{diag}(1, 1, -2, 0) \\ H_3 &= \frac{1}{2\sqrt{6}}\mathrm{diag}(1, 1, 1, -3) \end{aligned}$$

subject to the orthonormalization condition

$$tr\,(H_i H_j) = \frac{1}{2}\delta_{ij}$$

There are in total twelve roots $\pm\alpha_j$, $j = 1, \ldots, 6$. They all have length 1 and point from the center of a cube with edge length $\sqrt{2}$ to the midpoints of the twelve edges of the cube.



There are three simple roots

$$\alpha = (1, 0, 0), \quad \beta = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}, 0\right), \quad \gamma = \left(0, -\frac{\sqrt{3}}{3}, \frac{\sqrt{6}}{3}\right)$$

satisfying $\alpha\cdot\beta = -\frac{1}{2}$, $\beta\cdot\gamma = -\frac{1}{2}$, $\alpha\cdot\gamma = 0$, such that every other root can be written as a linear combination of these with integer coefficients with uniform sign.
The dominant weights of $sl(4,\mathbb{C})$ irreps. are given by

$$\begin{aligned}
\mu_{\max} &= p\hat{\alpha} + q\hat{\beta} + r\hat{\gamma}, \quad \text{with } p,q,r \in \mathbb{N}_0 \\
\hat{\alpha} &= \left(\frac{1}{2}, \frac{\sqrt{3}}{6}, \frac{\sqrt{6}}{12}\right) = \frac{1}{4}(3\alpha + 2\beta + \gamma) \\
\hat{\beta} &= \left(0, \frac{\sqrt{3}}{3}, \frac{\sqrt{6}}{6}\right) = \frac{1}{2}(\alpha + 2\beta + \gamma) \\
\hat{\gamma} &= \left(0, 0, \frac{\sqrt{6}}{4}\right) = \frac{1}{4}(\alpha + 2\beta + 3\gamma)
\end{aligned}$$

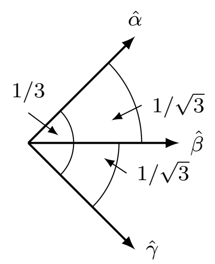


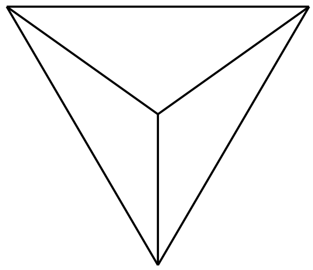
Figure 3.5: Weyl-chamber of $sl(4,\mathbb{C})$ spanned by $\hat{\alpha}, \hat{\beta}$ and $\hat{\gamma}$.

The weight diagrams are S_4-symmetric polyhedrons with three (different) side lengths (p,q,r).
The dimension (symmetric only under the exchange of p and r) is given by:

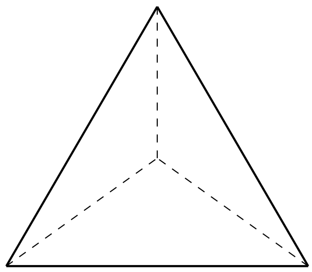
$$V(p,q,r) = \frac{1}{12}(p+1)(q+1)(r+1)(p+q+2)(q+r+2)(p+q+r+3)$$

There are three fundamental representations:

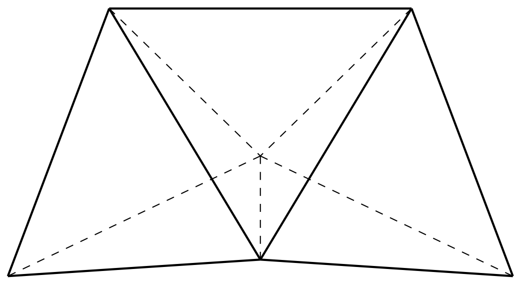
$$\begin{aligned}
&V(1,0,0) = \mathbb{C}^4 \quad \text{(4-dim. defining repr.)} \\
&V(0,1,0) = \mathbb{C}^6 = Alt_2\big(\mathbb{C}^4\big) \quad \text{(rank 2 antisym. tensors)} \\
&V(0,0,1) = \mathbb{C}^4 = Alt_3\big(\mathbb{C}^4\big) \quad \text{(complex conjugated of } V(1,0,0))
\end{aligned}$$



4
tetrahedron



4^*



$6 = 6^*$

Young-tableaus for $sl(4, \mathbb{C})$ representations:

$$\begin{array}{lllllllllllll}
\square & \square & \square & \square & \square & \square & \square & \square & \square & \square & \square & \square & \square \leftarrow p+q+r \text{ boxes} \\
\square & \square & \square & \square & \square & \square & \square & \square & \square & \square \leftarrow q+r \text{ boxes} & & & \\
\square & \square & \square & \square & \square & \square \leftarrow r \text{ boxes} & & & & & & &
\end{array}$$

Again, the entries in the denominator tableau are the hook-lengths of each box and the dimension is given by:

$$dim = \frac{\begin{array}{|c|c|c|c|c|c|c|c|c|c|c|c|}
4 & 5 & & & & & & & & & & \leftarrow p{+}q{+}r{+}3 \\
3 & 4 & & & & & \leftarrow q{+}r{+}2 & & & & & \\
2 & 3 & & & \leftarrow r{+}1 & & & & & & &
\end{array}}{\begin{array}{r|c|c|c|c|c|c|c|c|c|c|c|c|}
p{+}q{+}r{+}2 \rightarrow & & & & p{+}q{+}4 & p{+}q{+}3 & p{+}q{+}1 & p{+}2 & p & & & & 1 \\
q{+}r{+}1 \rightarrow & & & & q{+}3 & q{+}2 & q & 1 & & & & & \\
r \rightarrow & r & & & 2 & 1 & & & & & & &
\end{array}}$$

$$= \frac{(p+q+r+3)!(q+r+2)!(r+1)!}{3!2!(p+q+r+2)!(p+1)^{-1}(p+q+2)^{-1}(q+r+1)!(q+1)^{-1}r!}$$

$$= \frac{1}{12}(p+q+r+3)(p+q+2)(q+r+2)(p+1)(q+1)(r+1)$$

Coupling of tableaus:

- Boxes in the third row are denoted by $\boxed{c}$. The column length must not exceed 4. A column of length 4 encodes the trivial **1**-representation.
- An allowed series has the property: $\#\boxed{a} \geq \#\boxed{b} \geq \#\boxed{c}$

Example:

$$6 \otimes 6 = 20 \oplus 15 \oplus 1$$
$$(0,1,0) \otimes (0,1,0) = (0,2,0) \oplus \underbrace{(1,0,1)}_{\text{adjoint repr.}} \oplus (0,0,0)$$

General results for the decomposition of tensor products:

$$\chi_{\mu_1} \cdot \chi_{\mu_2} = \sum_{\mu} m(\mu_1, \mu_2, \mu)\, \chi_\mu$$

where μ_1, μ_2, μ are dominant weights and $m(\mu_1, \mu_2, \mu)$ is the multiplicity.

Theorem of Steinberg:

$$m(\mu_1, \mu_2, \mu) = \sum_{v,w \in W} \underbrace{\det(v \cdot w)}_{\pm 1} P\left[v(\mu_1 + \rho) + w(\mu_2 + \rho) - \mu - 2\rho\right]$$

where $\rho = \frac{1}{2} \sum \alpha_i$ is the half sum of the positive roots and the so-called *constant-function*

$$P[\mu] = \#\{(n_1, \ldots, n_k)\} \quad \text{with } \mu = \sum_{j=1}^{k} n_j \alpha_j \text{ and } n_j \in \mathbb{N}_0$$

counts the number of decompositions of μ as a non-negative linear combination of the positive roots α_j.

3.9 Finite-dimensional representations of the proper, orthochronus Lorentz group

We recall the isomorphisms

$$SO^+(1,3) = Sl(2,\mathbb{C})\,/\{\pm 1\}$$
$$so(1,3) = sl(2,\mathbb{C}) = \mathbb{C} \otimes su(2)$$

Irreps. of $sl(2,\mathbb{C})$ can be obtained by taking the irreps. $V(j)$ of $SU(2)$ and allowing for three *complex* parameters $\boldsymbol{w} \in \mathbb{C}^3$ instead of the three real ones:

$$D_j\left(\underbrace{\exp\left(\frac{\mathrm{i}}{2}\boldsymbol{\sigma} \cdot \boldsymbol{w}\right)}_{\in Sl(2,\mathbb{C})}\right) = \exp\left(\mathrm{i} \underbrace{\boldsymbol{J}}_{(2j+1)\times(2j+1)-\text{matrices}} \cdot\, \boldsymbol{w}\right) \quad \text{(not unitary)}$$

These would correspond to holomorphic (complex differentiable) representations. The holomorphic structure is of no interest here and thus we can take the complex conjugated representation as well:

$$\tilde{D}_j\left(\exp\left(\frac{\mathrm{i}}{2}\boldsymbol{\sigma}\cdot\boldsymbol{w}\right)\right) = [\exp(\mathrm{i}\boldsymbol{J}\cdot\boldsymbol{w})]^*$$

The finite dimensional irreps. of $Sl(2,\mathbb{C})$ have the representation spaces $V(j)\otimes\tilde{V}(k)$ of dimension $(2j+1)(2k+1)$, $j,k\in\frac{1}{2}\mathbb{N}_0$.

$$\Psi = \begin{pmatrix}\psi_{11} & \cdots & \psi_{1b}\\ \vdots & \ddots & \vdots\\ \psi_{a1} & \cdots & \psi_{ab}\end{pmatrix}, \quad \text{with } a = 2j+1,\ b = 2k+1$$

The quantity Ψ transforms under $Sl(2,\mathbb{C})$ as:

$$\Psi \overset{A\in sl(2,\mathbb{C})}{\longrightarrow} D_j(A)\,\Psi D_k(A)^\dagger$$

If $\pm A$ yield the *same* transformation (hence $\pm 1\in Sl(2,\mathbb{C})$ yields *id*), we obtain a representation of the Lorentz-group $SO^+(1,3)$

$$(-1)^{2j}(-1)^{2k} = 1 \Leftrightarrow j+k\in\mathbb{N}_0 \text{ integer}$$

$(j,k)=(0,0)$: trivial representation: *Lorentz scalar*
$\left(\frac{1}{2},0\right),\left(0,\frac{1}{2}\right)$: left-, right-handed *Weyl-spinor*
$\left(\frac{1}{2},0\right)\oplus\left(0,\frac{1}{2}\right)$: 4-component *Dirac-spinor* (reducible w.r.t. $sl(2,\mathbb{C})$, but irreducible if parity is taken into account)
$\left(\frac{1}{2},\frac{1}{2}\right)$: *Lorentz 4-vector*

$$X = \begin{pmatrix}x^0+x^3 & x^1-\mathrm{i}x^2\\ x^1+\mathrm{i}x^2 & x^0-x^3\end{pmatrix} \longrightarrow AXA^\dagger$$

In physical literature one often finds *van der Waerden's multi-spinors* with $2j\in\mathbb{N}_0$ "un-dotted" and $2k\in\mathbb{N}_0$ "dotted" indices

$$\psi_{\alpha_1,\ldots,\alpha_{2j},\dot{\beta}_1,\ldots,\dot{\beta}_{2k}}$$

Here the "un-dotted" indices transform with $A\in Sl(2,\mathbb{C})$ and the "dotted" indices transform with $A^*\in Sl(2,\mathbb{C})$.